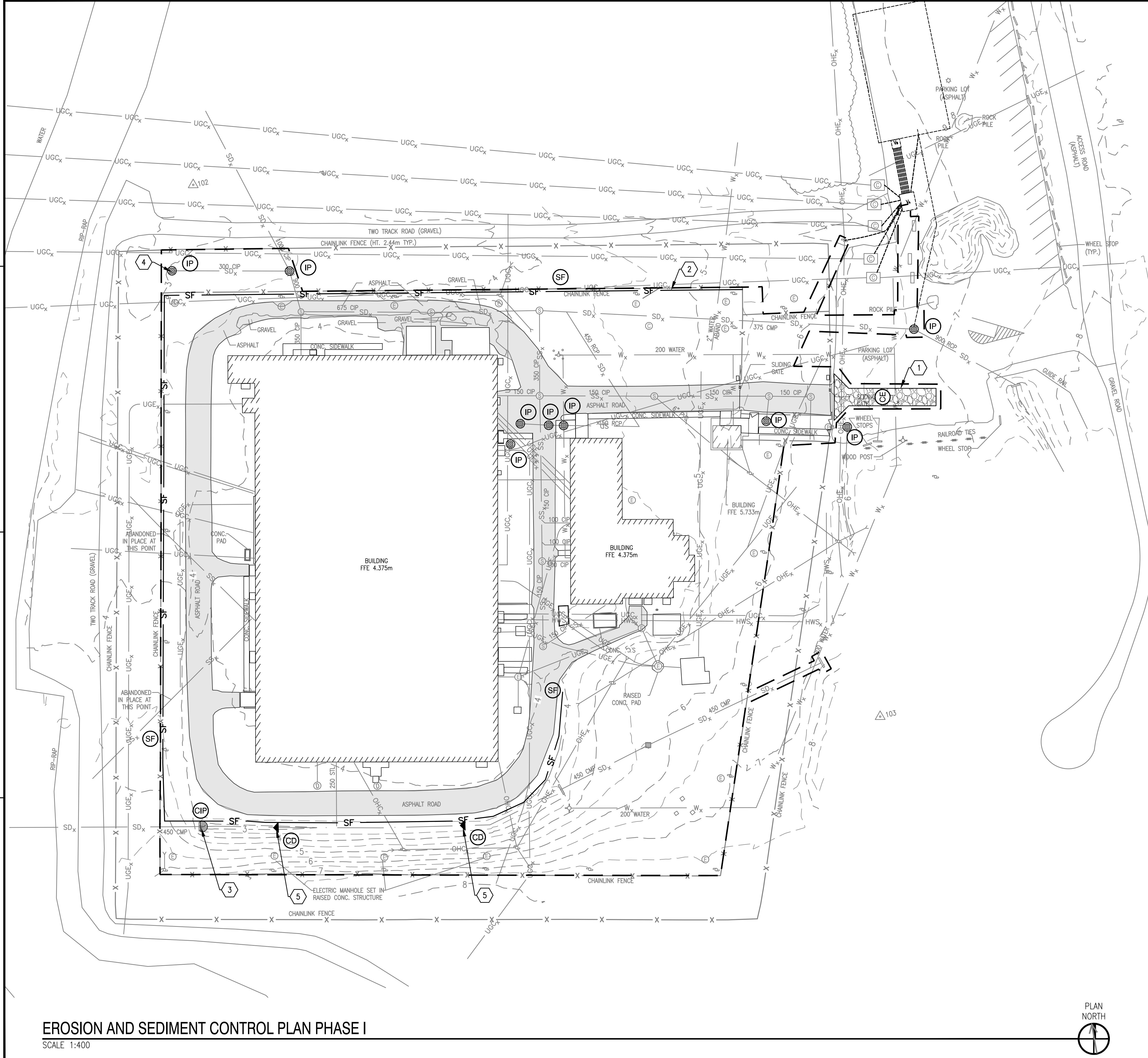


FILE NAME: Y:\DDONUS\MultipleSta\2026_1783264-P1360_MMP\B_Design\DWG\CE101.dwg LAYOUT NAME: CE101 EROSION AND SEDIMENT CONTROL PLAN PHASE I PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: mayam.durant



EROSION AND SEDIMENT CONTROL PLAN PHASE I
SCALE 1:400



GENERAL NOTES

1. PRIOR TO COMMENCEMENT OF OPERATIONS, CONFIRM PERMITS HAVE BEEN APPROVED.
2. COORDINATE WITH CONTRACTING OFFICER FOR LOCATION OF LAYDOWN AND STOCKPILE AREAS.
3. PLACE EXCAVATED MATERIAL ON THE UPHILL SIDE OF UTILITY TRENCHES AND FOUNDATION EXCAVATION PITS.
4. REFER TO SHEETS C-001 & C-002 FOR CIVIL LEGEND, NOTES & ABBREVIATIONS.
5. REFER TO SHEET B-101 FOR GEOTECHNICAL INFORMATION.
6. REFER TO SHEET C-001 FOR SURVEY CONTROL MONUMENT INFORMATION.
7. REFER TO DEMOLITION PACKAGE PREPARED BY ALLIANCE FOR DEMOLITION.
8. REFER TO SITE ELECTRICAL PLANS FOR POWER AND COMMUNICATION LINE WORK. PLACE PERIMETER SEDIMENT CONTROLS AS APPROPRIATE.



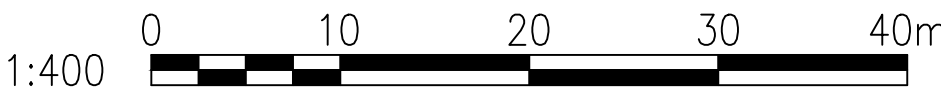
CONSTRUCTION KEYNOTES

1. PROVIDE CONSTRUCTION ENTRANCE.
2. PROVIDE SILT FENCE.
3. PROVIDE CULVERT INLET PROTECTION.
4. PROVIDE INLET PROTECTION (TYP).
5. PROVIDE CHECK DAM (TYP).

EROSION AND SEDIMENT CONTROL LEGEND

(SF)	SILT FENCE	SF
(IP)	INLET PROTECTION	
(CE)	CONSTRUCTION ENTRANCE	
(CD)	CHECK DAM	
(CIP)	CULVERT INLET PROTECTION	
LIMIT OF DISTURBANCE = 16,150 M2 (4.0 AC)		

GRAPHIC SCALE



APPROVED		DATE	APPR
Win B Forbes		2026.01.28	
FOR COMMANDER NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		16:55:54-05:00	
ACTIVITY			
D. EVANS VIA EMAIL			
SATISFACTORY TO DATE 01/15/2026			
DES	MAD	DRW	DTK
PM/DM	MJD/ATP		
BRANCH MANAGER	JMW		
DES PROJ DIR	WBF		
FIRE PROTECTION ENGINEER	DSN		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		VARIOUS LOCATIONS	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		VARIOUS LOCATIONS	
PLANNING, DESIGN AND CONSTRUCTION		VARIOUS LOCATIONS	
JOINT MARITIME FACILITY		VARIOUS LOCATIONS	
EROSION AND SEDIMENT CONTROL PLAN		VARIOUS LOCATIONS	
PHASE I		VARIOUS LOCATIONS	
SCALE: AS NOTED			
PROJECT NO.: 1838720			
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO. 14157083			
SHEET 21 OF 170			
CE101			

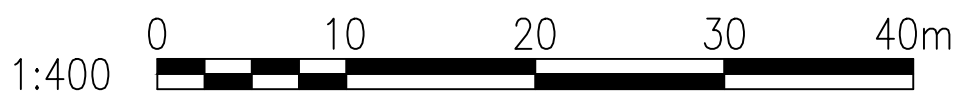
FILE NAME: V:\DDO\USV\MultipleSta\2026_1783264-P1360_MMP\B_Design\DWG\US_CIV\F1360-1783264-CE102.dwg LAYOUT NAME: CE102 EROSION AND SEDIMENT CONTROL PLAN PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: maryam.durant

EROSION AND SEDIMENT CONTROL PLAN PHASE II

SCALE 1:400



GRAPHIC SCALE



GENERAL NOTES

1. PRIOR TO COMMENCEMENT OF OPERATIONS, CONFIRM PERMITS HAVE BEEN APPROVED.
2. COORDINATE WITH CONTRACTING OFFICER FOR LOCATION OF LAYDOWN AND STOCKPILE AREAS.
3. PLACE EXCAVATED MATERIAL ON THE UPHILL SIDE OF UTILITY TRENCHES AND FOUNDATION EXCAVATION PITS.
4. REFER TO SHEETS C-001 & C-002 FOR CIVIL LEGEND, NOTES & ABBREVIATIONS.
5. REFER TO SHEET B-101 FOR GEOTECHNICAL INFORMATION.
6. REFER TO SHEET C-001 FOR SURVEY CONTROL MONUMENT INFORMATION.
7. REFER TO DEMOLITION PACKAGE PREPARED BY ALLIANCE FOR DEMOLITION.
8. REFER TO SITE ELECTRICAL PLANS FOR POWER AND COMMUNICATION LINE WORK. PLACE PERIMETER SEDIMENT CONTROLS AS APPROPRIATE.

CONSTRUCTION KEYNOTES

1. INSPECT AND REPAIR CONSTRUCTION ENTRANCE PROVIDED DURING PHASE I.
2. INSPECT AND REPAIR SILT FENCE, INLET PROTECTION AND CULVERT INLET PROTECTION PROVIDED DURING PHASE I.
3. INSPECT AND REPAIR CHECK DAM PROVIDED DURING PHASE I.
4. REMOVE PORTION OF SILT FENCE PRIOR TO GRADING OPERATIONS IN THIS AREA.
5. PROVIDE CHECK DAM FOR SWALE.
6. PROVIDE INLET PROTECTION.
7. PROVIDE OUTLET PROTECTION.

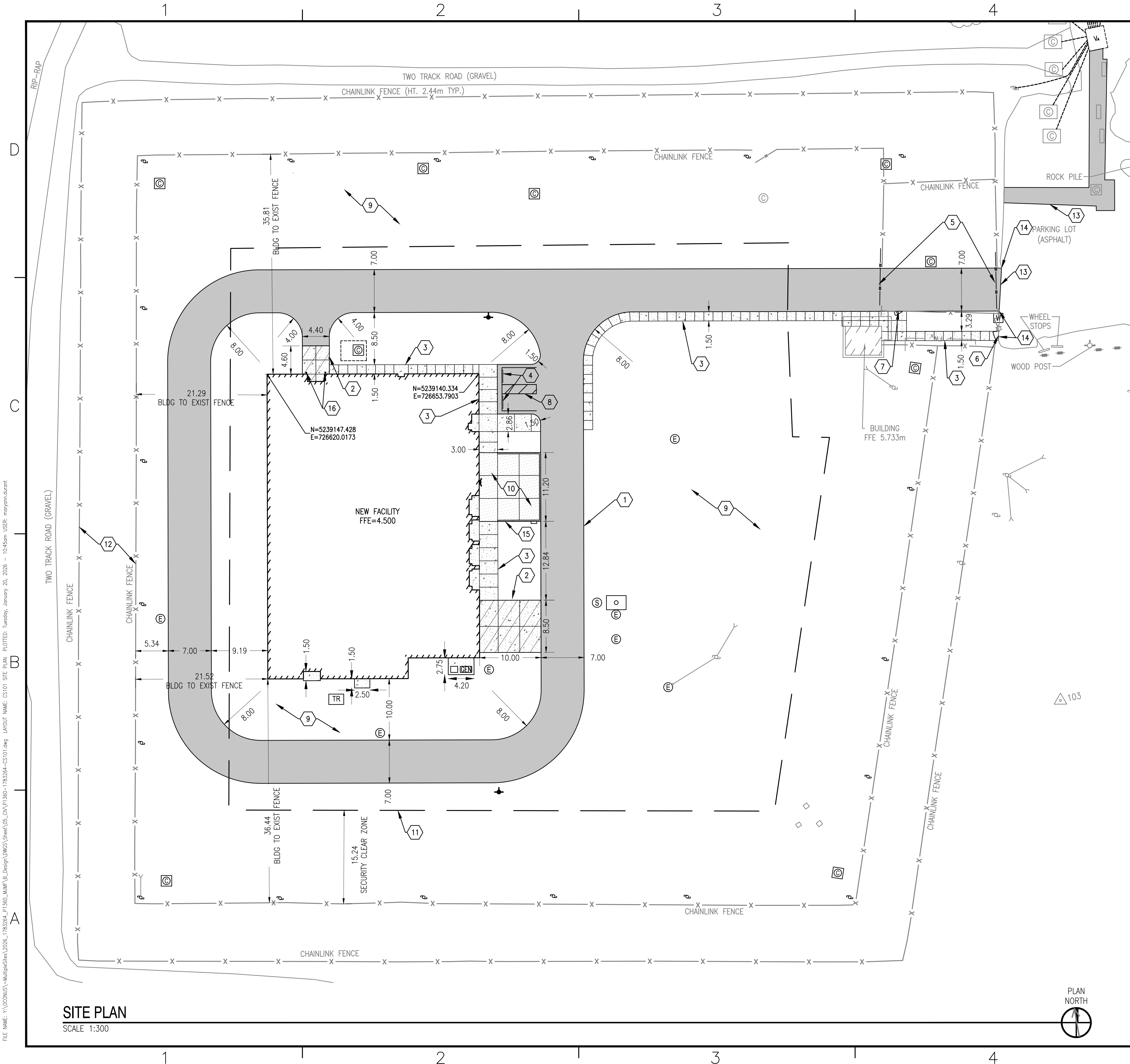
EROSION AND SEDIMENT CONTROL LEGEND

(SF)	SILT FENCE	SF
(IP)	INLET PROTECTION	
(CE)	CONSTRUCTION ENTRANCE	
(CD)	CHECK DAM	
(CIP)	CULVERT INLET PROTECTION	
(OP)	CULVERT OUTLET PROTECTION	
— — — — — LIMIT OF DISTURBANCE = 16,150 M2 (4.0 AC)		

APPROVED	Digitally signed by signature Win B Forbes Date: 2026.01.28 16:55:54-05'00'	DATE	APPR
ACTIVITY	D. EVANS VIA EMAIL	DATE	APPR
SATISFACTORY TO	DATE 01/15/2026	DATE	APPR
DES	MAD	DRW	DTK
PM/DM	MJD/ATP	CHK	JWN
BRANCH MANAGER	JMW	DATE	DATE
DES PROJ DIR	WBF	DATE	DATE
FIRE PROTECTION ENGINEER	DSN	DATE	DATE
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION	PLANNING, DESIGN AND CONSTRUCTION	PLANNING, DESIGN AND CONSTRUCTION	PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS	VARIOUS LOCATIONS	VARIOUS LOCATIONS	VARIOUS LOCATIONS
JOINT MARITIME FACILITY	JOINT MARITIME FACILITY	JOINT MARITIME FACILITY	JOINT MARITIME FACILITY
EROSION AND SEDIMENT CONTROL PLAN	EROSION AND SEDIMENT CONTROL PLAN	EROSION AND SEDIMENT CONTROL PLAN	EROSION AND SEDIMENT CONTROL PLAN
PHASE II	PHASE II	PHASE II	PHASE II
SCALE:	AS NOTED	SCALE:	AS NOTED
PROJECT NO.:	1838720	PROJECT NO.:	1838720
CONSTR. CONTR. NO.		CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14157084	NAVFAC DRAWING NO.	14157084
SHEET	22 OF 170	SHEET	22 OF 170
CE102	CE102	CE102	CE102

NAVFAC METRIC DRAWING REVISION: 01 OCTOBER 2018

FILE NAME: Y:\DDONUS\Multiple\2026_1783264-P1360_MMF\B_Design\DWG\Sheet\05_CIV\F1360-1783264-CS101.dwg LAYOUT NAME: CS101 SITE PLAN PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: mayam.durant



SITE PLAN
SCALE 1:300

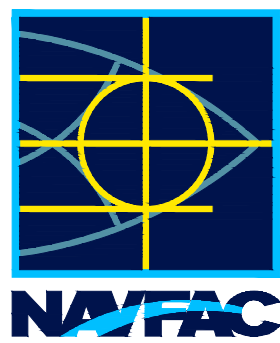
GENERAL NOTES

1. REFER TO SHEETS C-001 & C-002 FOR CIVIL LEGEND, NOTES & ABBREVIATIONS.
2. REFER TO SHEET B-101 FOR GEOTECHNICAL INFORMATION.
3. REFER TO SHEET C-001 FOR SURVEY CONTROL MONUMENT INFORMATION. ANY BASE CONTROL BENCHMARK OR PROPERTY LINE MONUMENTATION DISTURBED DURING CONSTRUCTION MUST BE RE-ESTABLISHED BY A PROFESSIONAL LICENSED SURVEYOR AND SET BY THE CONTRACT, AT NO ADDITIONAL COST TO THE GOVERNMENT.
4. REFER TO DEMOLITION PACKAGE PREPARED BY ALLIANCE FOR DEMOLITION.
5. REFER TO SHEET CS401 FOR CONCRETE JOINTING PLAN.
6. SEE ARCHITECTURAL DRAWINGS FOR EXACT BUILDING DIMENSIONS AND DETAILS CONTIGUOUS TO THE BUILDING, INCLUDING RAMPS, BUILDING ENTRANCES, STAIRWAY, UTILITY PENETRATIONS, CONCRETE DOOR PADS, ETC.



CONSTRUCTION KEYNOTES

1. ASPHALT PAVEMENT. REFER TO DETAIL B1/C-504.
2. HEAVY DUTY PORTLAND CEMENT CONCRETE. REFER TO DETAIL C1/C-504.
3. CONCRETE SIDEWALK. REFER TO DETAIL A2/C-503.
4. CONCRETE PARKING BUMPER. REFER TO DETAIL B2/C-503.
5. PROVIDE 6.5M SLIDING SECURITY GATE. REFER TO DETAIL A1/C-510.
6. PROVIDE A PEDESTRIAN SECURITY SWING GATE AND CYPHER LOCK AT THIS LOCATION ONLY. REFER TO DETAIL A1/C-511.
7. PROVIDE A PEDESTRIAN SECURITY SWING GATE. REFER TO DETAIL A1/C-511.
8. PARKING AREA. REFER TO DETAIL C2/C-503.
9. LANDSCAPE PLANTING AREA. REFER TO SHEET LS101.
10. MECHANICAL YARD CONCRETE PAVEMENT. REFER TO STRUCTURAL PLANS FOR SECTION DETAIL.
11. SECURITY CLEAR ZONE.
12. REPLACE BARBED WIRE AT TOP OF FENCE. REFER TO DETAILS A3/C-511 AND B3/C-511.
13. SAWCUT EXIST PAVEMENT. REFER TO DETAIL C4/C-503.
14. TIE TO EXIST PAVEMENT. CENTERLINE OF PAVEMENT TO MATCH CENTERLINE OF GATE.
15. MECHANICAL YARD ENCLOSURE. REFER TO LANDSCAPE PLANS FOR DETAIL.
16. BOLLARD. REFER TO DETAIL A1/C-503.



APPROVED	Digitally signed by signature Date: 2026.01.28 FOR COMMANDER NAVFAC 165554-0500
ACTIVITY	D. EVANS VIA EMAIL
SUBSTANTIAL TO DATE	01/15/2026
DESIGNER	MAD
DRW	DTK
CHK	JWN
PM/DM	MJD/ATP
BRANCH MANAGER	JMW
DESIGN PROJ DIR	WBF
FIRE PROTECTION ENGINEER	DSN

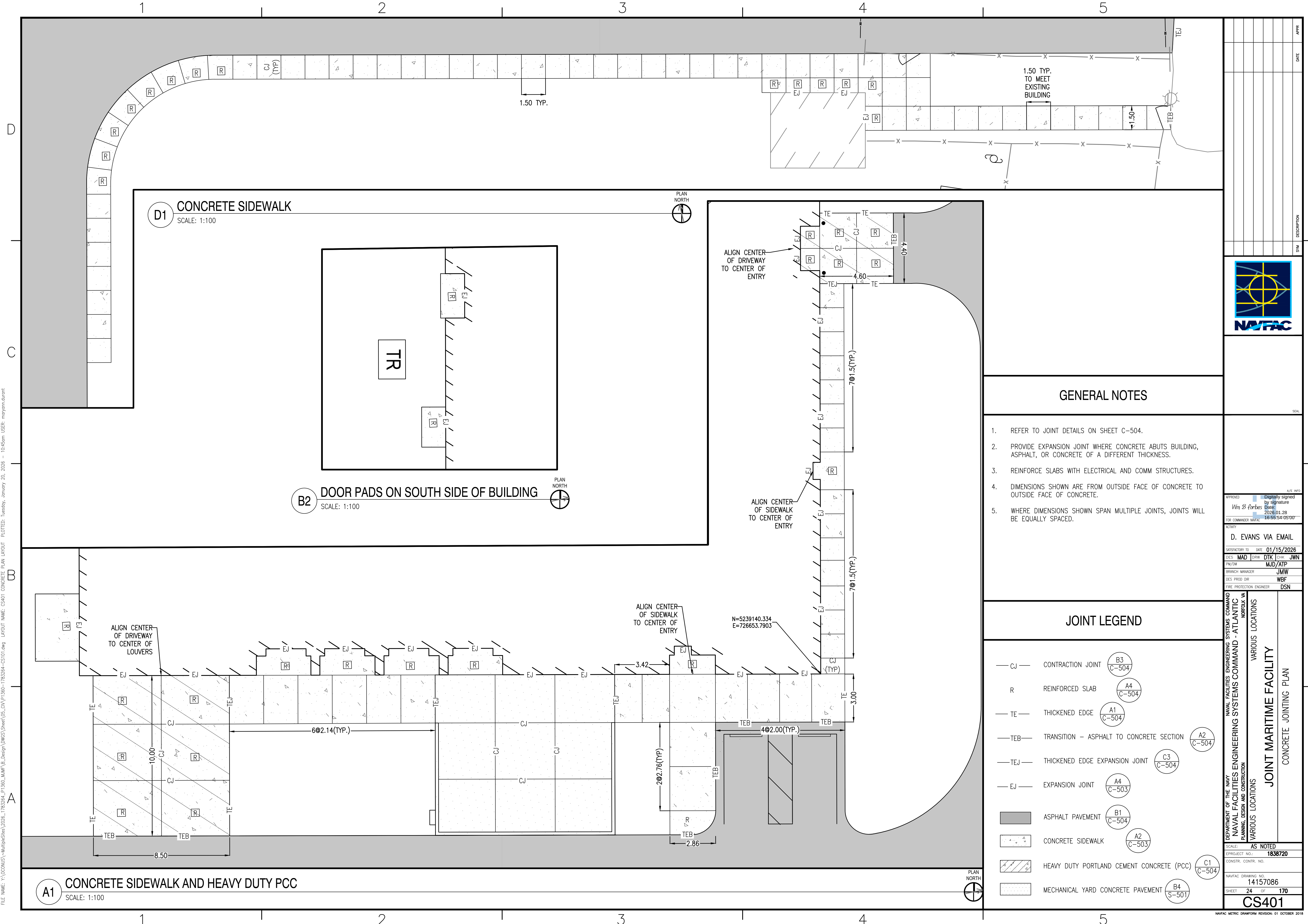
DEPARTMENT OF THE NAVY	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION	VARIOUS LOCATIONS	VARIOUS LOCATIONS	VARIOUS LOCATIONS
JOINT MARITIME FACILITY			
SITE PLAN			

GRAPHIC SCALE



SCALE:	AS NOTED
PROJECT NO.:	1838720
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14157085
SHEET	23 OF 170
CS101	

FILE NAME: V:\DDDD\Navy\Multiple\2026_1783264-P1360_MMP\B_Design\DWG\Sheet\05_CIV\P1360-1783264-CS101.dwg LAYOUT NAME: CS401 CONCRETE PLAN LAYOUT PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: mayam.durant



GENERAL NOTES

- REFER TO JOINT DETAILS ON SHEET C-504.
- PROVIDE EXPANSION JOINT WHERE CONCRETE ABUTS BUILDING, ASPHALT, OR CONCRETE OF A DIFFERENT THICKNESS.
- REINFORCE SLABS WITH ELECTRICAL AND COMM STRUCTURES.
- DIMENSIONS SHOWN ARE FROM OUTSIDE FACE OF CONCRETE TO OUTSIDE FACE OF CONCRETE.
- WHERE DIMENSIONS SHOWN SPAN MULTIPLE JOINTS, JOINTS WILL BE EQUALLY SPACED.

JOINT LEGEND

— CJ —	CONTRACTION JOINT	(B3 C-504)
R	REINFORCED SLAB	(A4 C-504)
— TE —	THICKENED EDGE	(A1 C-504)
— TEB —	TRANSITION - ASPHALT TO CONCRETE SECTION	(A2 C-504)
— TEJ —	THICKENED EDGE EXPANSION JOINT	(C3 C-504)
— EJ —	EXPANSION JOINT	(A4 C-503)
	ASPHALT PAVEMENT	(B1 C-504)
	CONCRETE SIDEWALK	(A2 C-503)
	HEAVY DUTY PORTLAND CEMENT CONCRETE (PCC)	(C1 C-504)
	MECHANICAL YARD CONCRETE PAVEMENT	(B4 S-501)

APPR

DATE

DESCRIPTION

SW

SEAL

APPROVED

Digitally signed by signature

Win B forbes

Date: 2026.01.28 16:55:54-05'00'

FOR COMMANDER NAVFAC

ACTIVITY

D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES MAD DRW DTK CHK JWN

PM/DM MJD/ATP

BRANCH MANAGER JMW

DES PROJ DIR WBF

FIRE PROTECTION ENGINEER DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC

PLANNING, DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

JOINT MARITIME FACILITY

CONCRETE JOINTING PLAN

SCALE: AS NOTED

PROJECT NO.: 1838720

CONSTR. CONTR. NO.

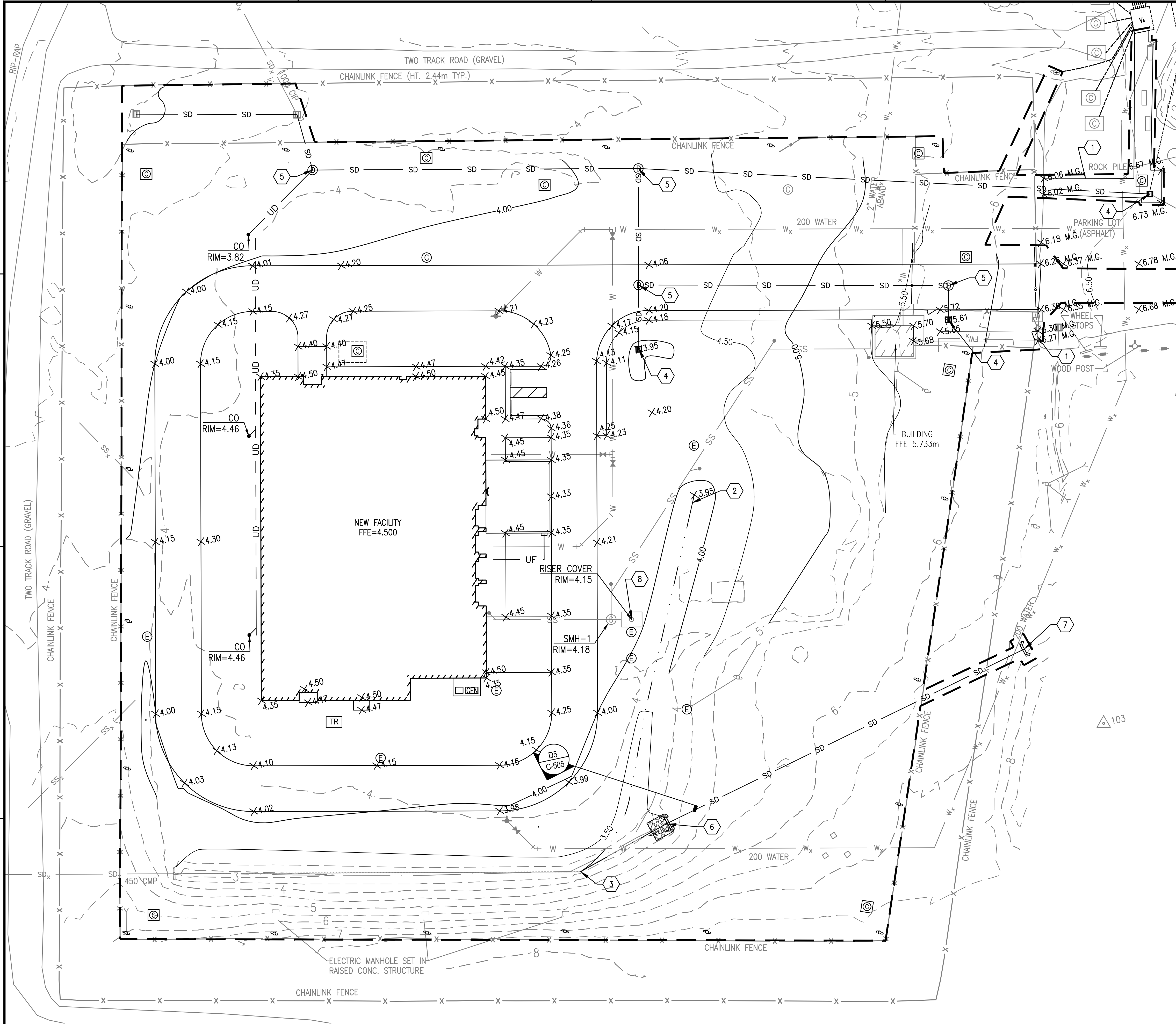
NAVFAC DRAWING NO. 14157086

SHEET 24 OF 170

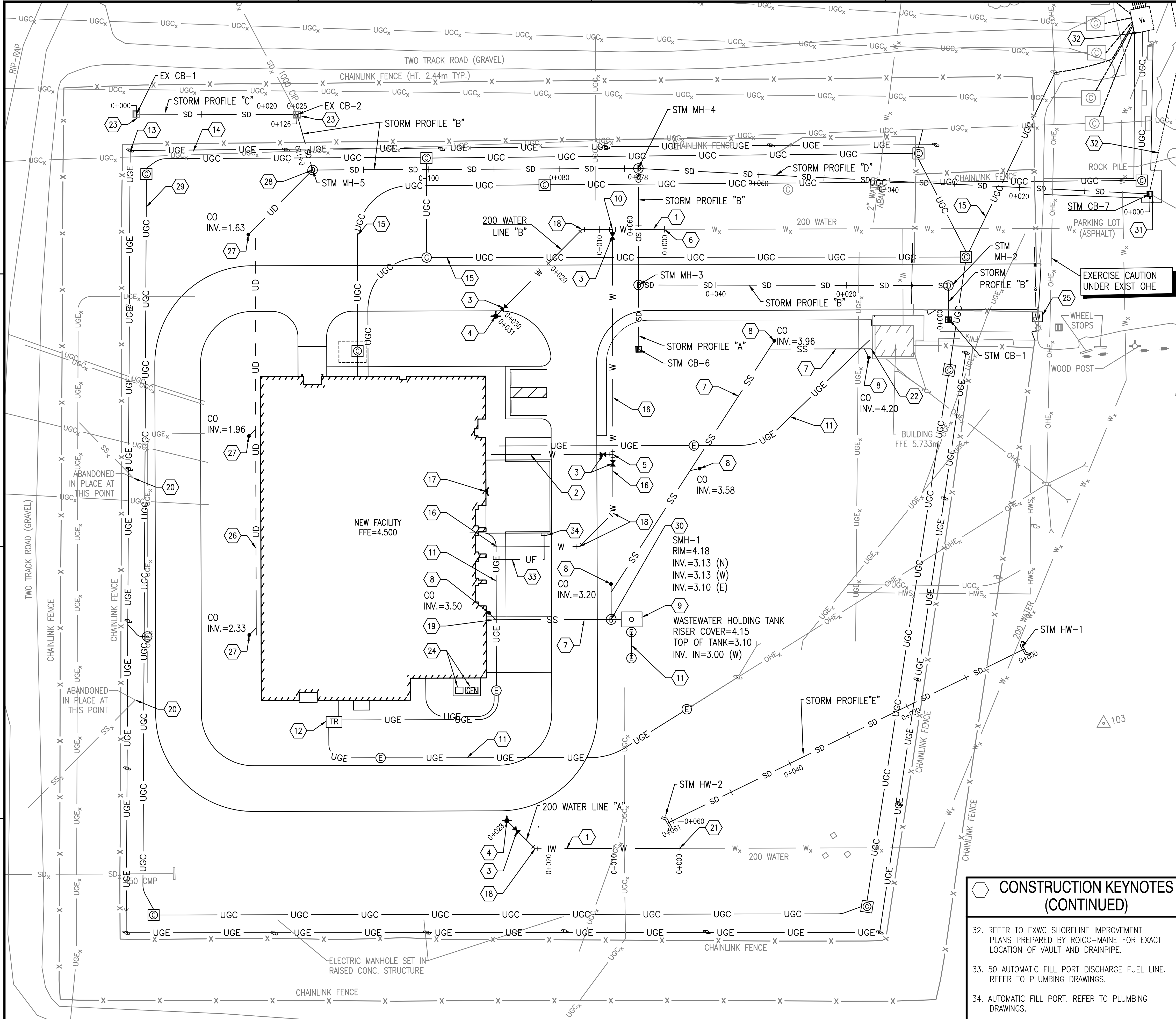
CS401

NAVFAC METRIC DRAWING REVISION: 01 OCTOBER 2018

FILE NAME: V:\DDONIS\MultipleSheets\2026_1783264-P1360_MMP\B_Design\DWG\Sheets\05_CIV\F1360-1783264-CG101.dwg LAYOUT NAME: CG101 GRADING PLAN PLOTTED: Tuesday, January 20, 2026 - 10:46am USER: moryam.turant



FILE NAME: V:\CDONUGS\MultipleSta\2026_1783264-P1360-MMF1B_Design\DWGS\Sheet\UG_CIV\F1360-1783264-CU101.dwg LAYOUT NAME: CU101 UTILITY PLAN PLOTTED: Tuesday, January 20, 2026 - 10:46am USER: mayam.durant



UTILITY PLAN
SCALE 1:300



GENERAL NOTES

1. REFER TO SHEETS C-001 & C-002 FOR CIVIL LEGEND, NOTES & ABBREVIATIONS.
2. REFER TO SHEET B-101 FOR GEOTECHNICAL INFORMATION.
3. REFER TO SHEET C-001 FOR SURVEY CONTROL MONUMENT INFORMATION.
6. REFER TO SHEET CU201 FOR WATER LINE PROFILES.
7. REFER TO PLUMBING PLANS FOR CONTINUATION OF UTILITIES IN THE BUILDING.
8. REFER TO ELECTRICAL PLANS FOR POWER, COMMUNICATION, CCTV, AND SITE LIGHTING.
7. REFER TO DEMOLITION PACKAGE PREPARED BY ALLIANCE FOR DEMOLITION.
8. WATER METER AND RPZ ARE INSIDE BUILDING. REFER TO PLUMBING PLANS FOR DETAILS.

CONSTRUCTION KEYNOTES

1. 200 WATER MAIN (FIRE).
2. 50 WATER MAIN (DOMESTIC).
3. SLIDING TYPE VALVE BOX. REFER TO DETAIL A3/C-507.
4. FIRE HYDRANT ASSEMBLY. REFER TO DETAIL A1/C-508.
5. 150 X 50 WATER SERVICE CONNECTION. REFER TO DETAIL A1/C-507.
6. REMOVE EXIST FIRE HYDRANT. CONNECT TO EXIST 200 WATER MAIN.
7. 100 SANITARY SEWER LATERAL @ 1.04% MIN SLOPE.
8. 100 SANITARY SEWER CLEANOUT. REFER TO DETAIL A5/C-507.
9. CONTRACTOR IS RESPONSIBLE FOR THE DELEGATED DESIGN OF THE WASTEWATER HOLDING TANK (10,000L). REFER TO DETAIL A1/C-509.
10. 200 X 150 WATER TEE.
11. UNDERGROUND ELECTRICAL LINE. REFER TO ELECTRICAL DRAWINGS.
12. ELECTRICAL TRANSFORMER. REFER TO ELECTRICAL DRAWINGS.
13. SITE LIGHTING AND POLE (TYP). REFER TO ELECTRICAL DRAWINGS.
14. UNDERGROUND SITE LIGHTING LINE. REFER TO ELECTRICAL DRAWINGS.
15. UNDERGROUND COMMUNICATION LINE. REFER TO ELECTRICAL DRAWINGS.
16. 150 WATER MAIN (FIRE).
17. WALL MOUNTED FDC ON BUILDING. REFER TO FIRE PROTECTION DRAWINGS FOR EXACT LOCATION.
18. 45 DEGREE BEND
19. 100 SAN SEWER
INV ELEV @ BLDG = 3.35
20. CAP EXIST UTILITY LINE THAT WAS ABANDONED IN PLACE DURING DEMOLITION.
21. EXIST FIRE HYDRANT WAS REMOVED IN 2025. REMOVE EXIST PLUG AND CONNECT TO EXIST 200 WATER MAIN.
22. REROUTE AND TIE EXIST SAN SEWER SERVICE LATERAL FROM EXIST GUARD SHACK BUILDING TO 100 SAN SEWER TO HOLDING TANK. CONTRACTOR TO FIELD VERIFY LOCATION AND INV ELEV TO TIE-IN. IMMEDIATELY CONTACT NAVFAC CIVIL ENGINEER AND THE CONTRACTING OFFICER IF 1.04% MIN SLOPE CANNOT BE ACHIEVED.
23. GROUT AROUND PIPES AND OPENINGS IN EXIST CATCH BASIN.
24. ELECTRICAL GENERATOR EQUIPMENT. REFER TO ELECTRICAL DRAWINGS.
25. KNOX GATE AND KEY SWITCH BOX. CONTRACTOR TO COORDINATE EXACT LOCATION WITH THE CONTRACTING OFFICER.
26. 150 UNDERDRAIN PIPE. REFER TO DETAIL A2/S-301 IN STRUCTURAL DRAWINGS.
27. 150 UNDERDRAIN CLEANOUT.
28. TIE 150 UNDERDRAIN PIPE TO STM MH-5 @ INV IN (SW) = 1.49
29. UNDERGROUND CCTV LINE. REFER TO ELECTRICAL DRAWINGS.
30. SANITARY SEWER MANHOLE
31. TIE EXIST STORM PIPE (SE) AND 100 DRAINPIPE (NE) TO STM CB-7. GROUT AROUND PIPES AND OPENINGS.

CONSTRUCTION KEYNOTES
(CONTINUED)

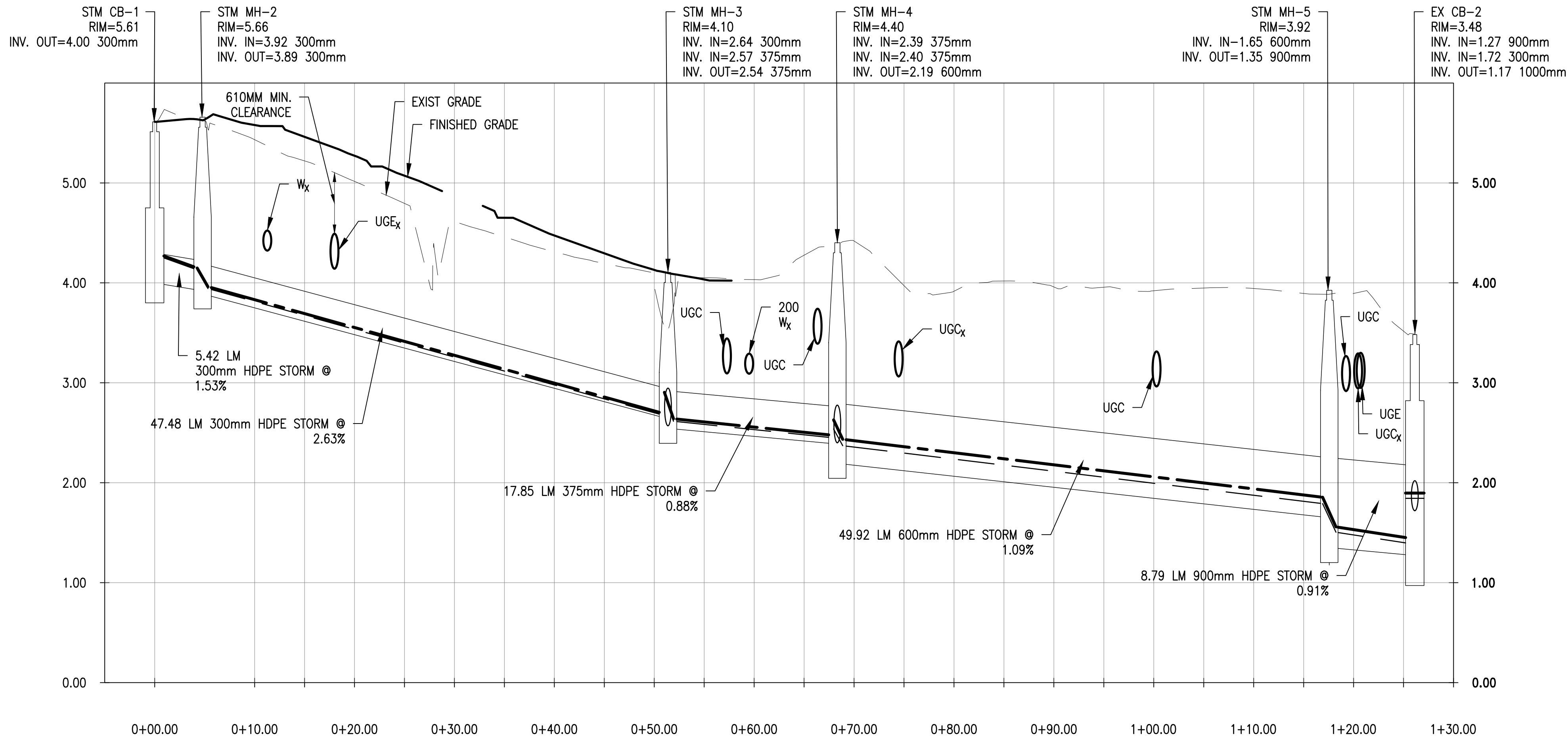
32. REFER TO EXWC SHORELINE IMPROVEMENT PLANS PREPARED BY ROICC-MAINE FOR EXACT LOCATION OF VAULT AND DRAINPIPE.
33. 50 AUTOMATIC FILL PORT DISCHARGE FUEL LINE. REFER TO PLUMBING DRAWINGS.
34. AUTOMATIC FILL PORT. REFER TO PLUMBING DRAWINGS.

GRAPHIC SCALE

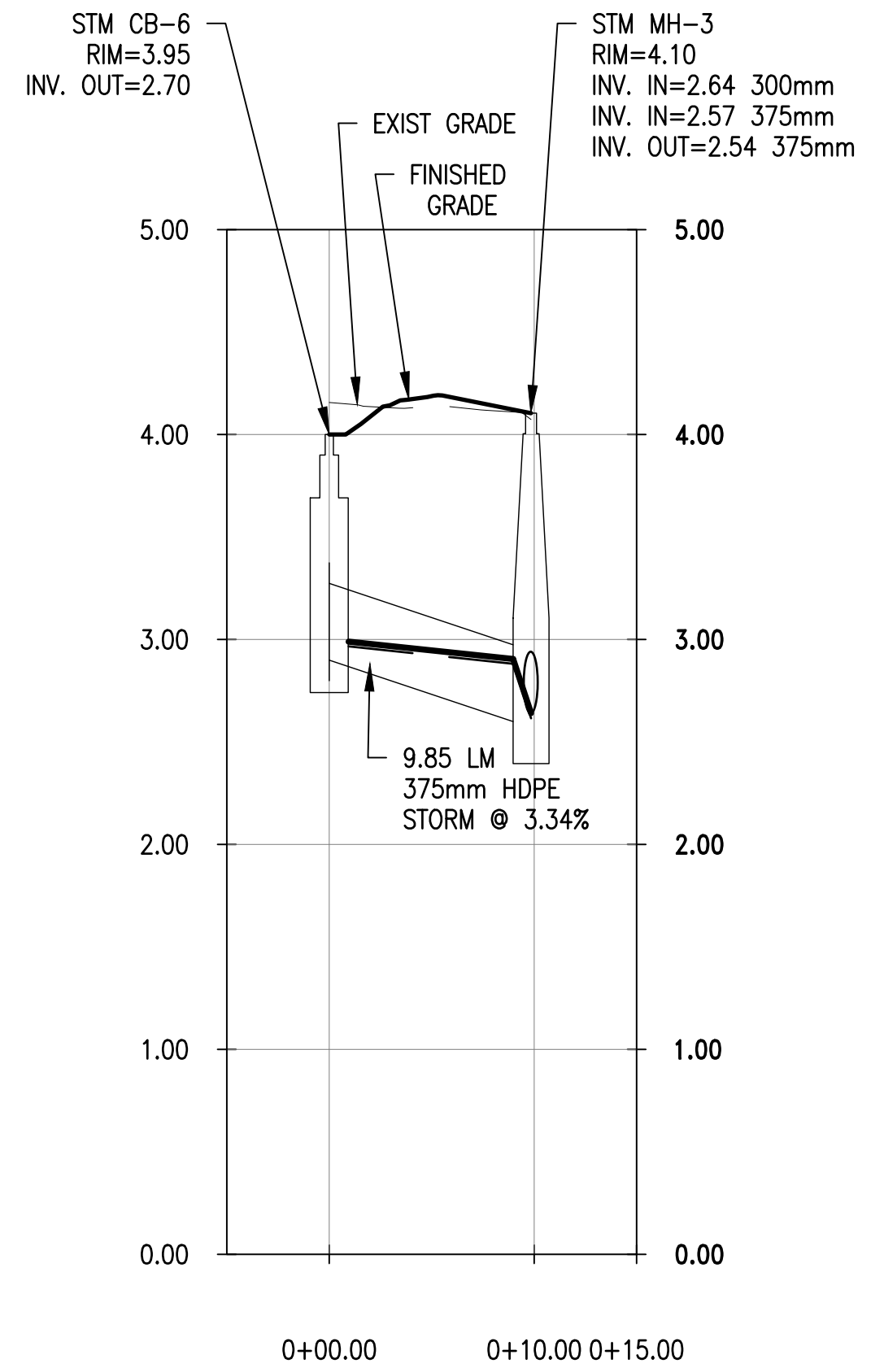


APPROVED		DATE		APPR	
Win B Forbes		2026.01.28		16:55:54-05'00"	
FOR COMMANDER NAVFAC		FIRE PROTECTION ENGINEER		DSN	
D. EVANS VIA EMAIL		01/15/2026			
DES	MAD	DRW	DTK	CHK	JWN
PM/DM	MJD/ATP				
BRANCH MANAGER	JMW				
DES PROJ DIR	WBF				
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND					
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC					
PLANNING, DESIGN AND CONSTRUCTION					
VARIOUS LOCATIONS					
VARIOUS LOCATIONS					
JOINT MARITIME FACILITY					
UTILITY PLAN					
SCALE: AS NOTED					
PROJECT NO.: 1838720					
CONSTR. CONTR. NO.					
NAVFAC DRAWING NO. 14157088					
SHEET 26 OF 170					
CU101					

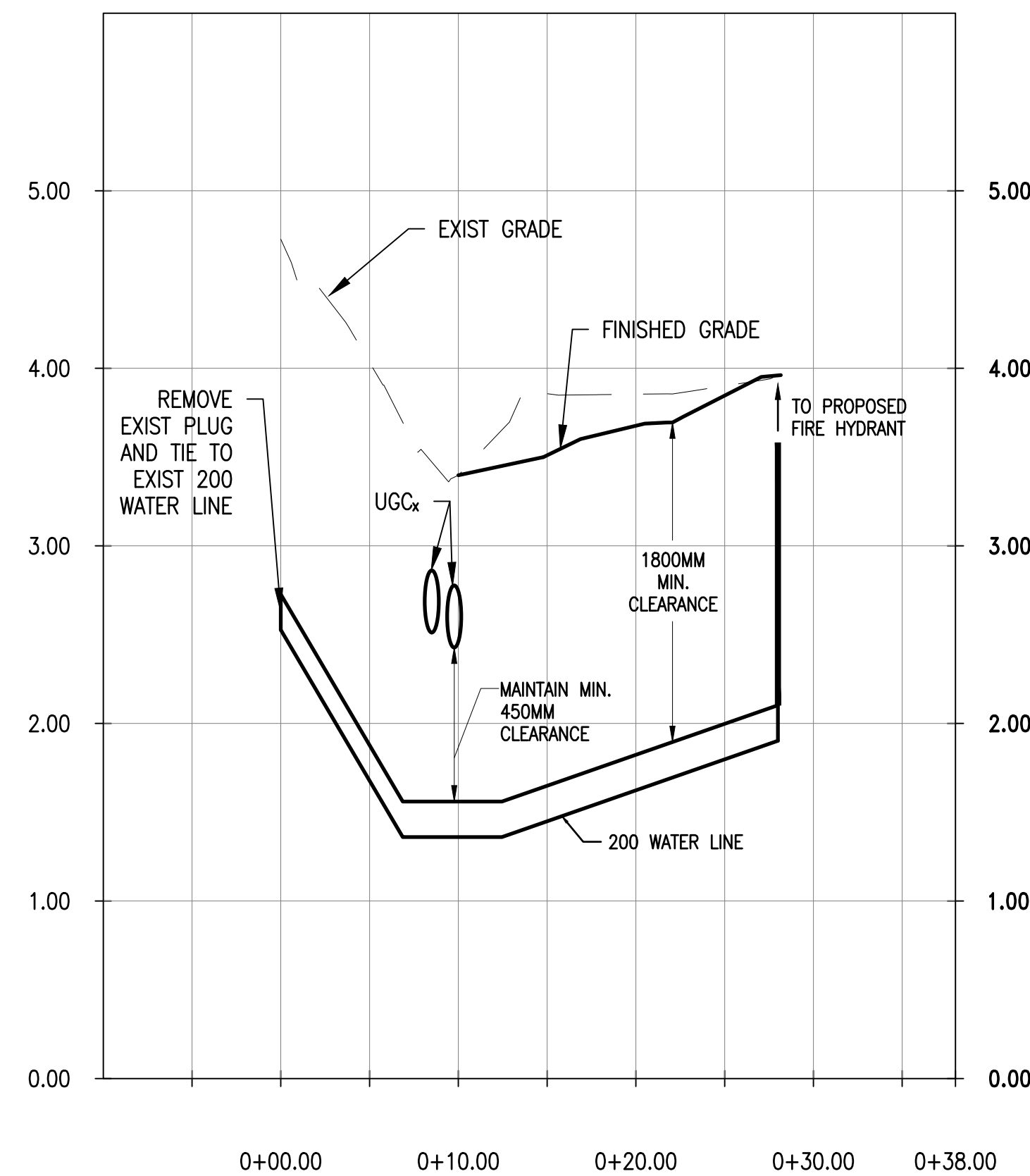
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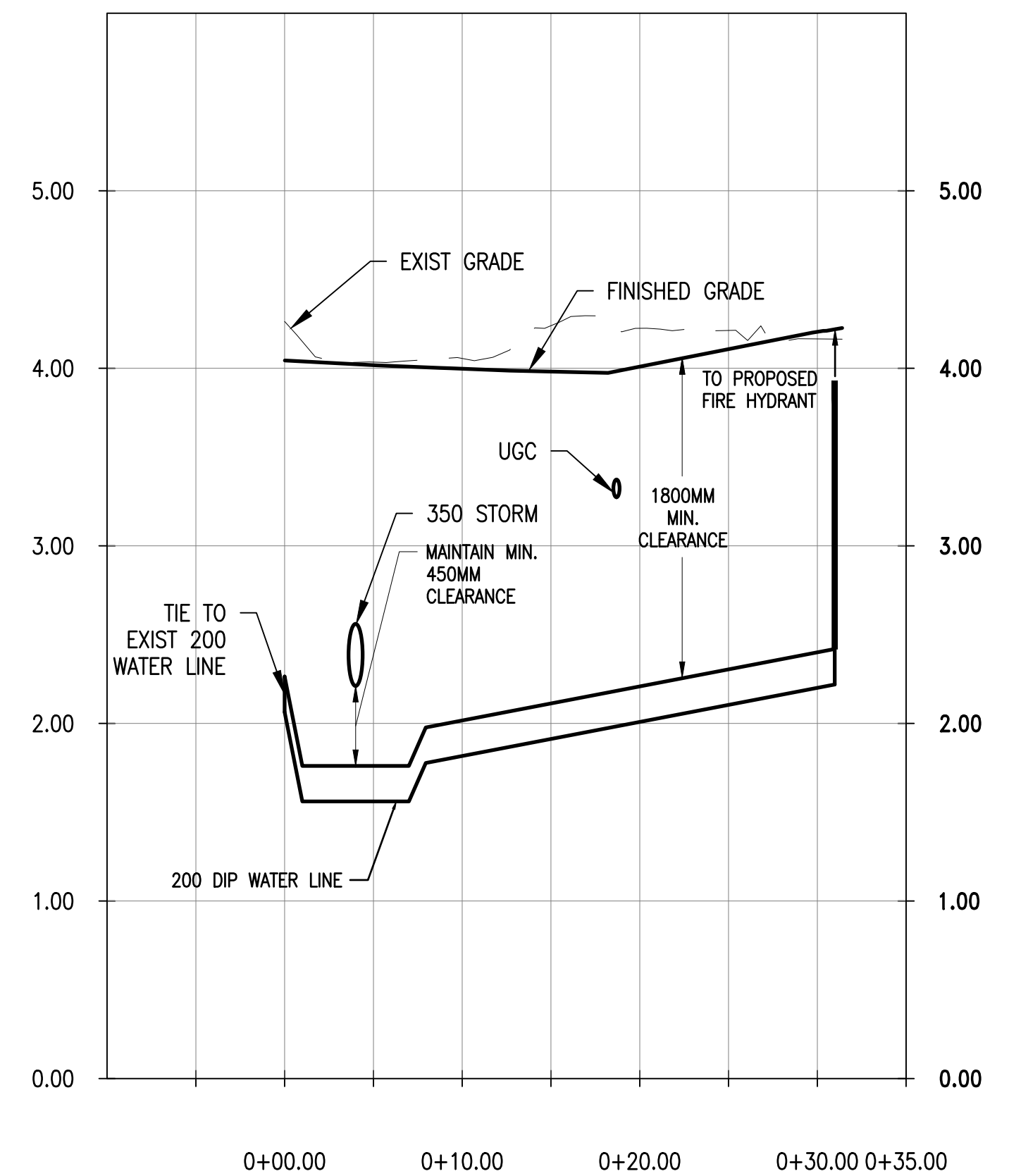
C3 STORM PROFILE "B"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL



C5 STORM PROFILE "A"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL



A3 200 WATER LINE "A"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL



A5 200 WATER LINE "B"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL



APPROVED: Digitally signed by signature
Win B forbes
Date: 2026.01.28
16:55:54-05'00'

ACTIVITY

D. EVANS VIA EMAIL

SATISFACTORY TO DATE: 01/15/2026

DES: MAD DRW: DTK CHK: JWN

PM/DM: MJD/ATP

BRANCH MANAGER: JMW

DES PROD DIR: WBF

FIRE PROTECTION ENGINEER: DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC

PLANNING, DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

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VARIOUS LOCATIONS

VARIOUS LOCATIONS

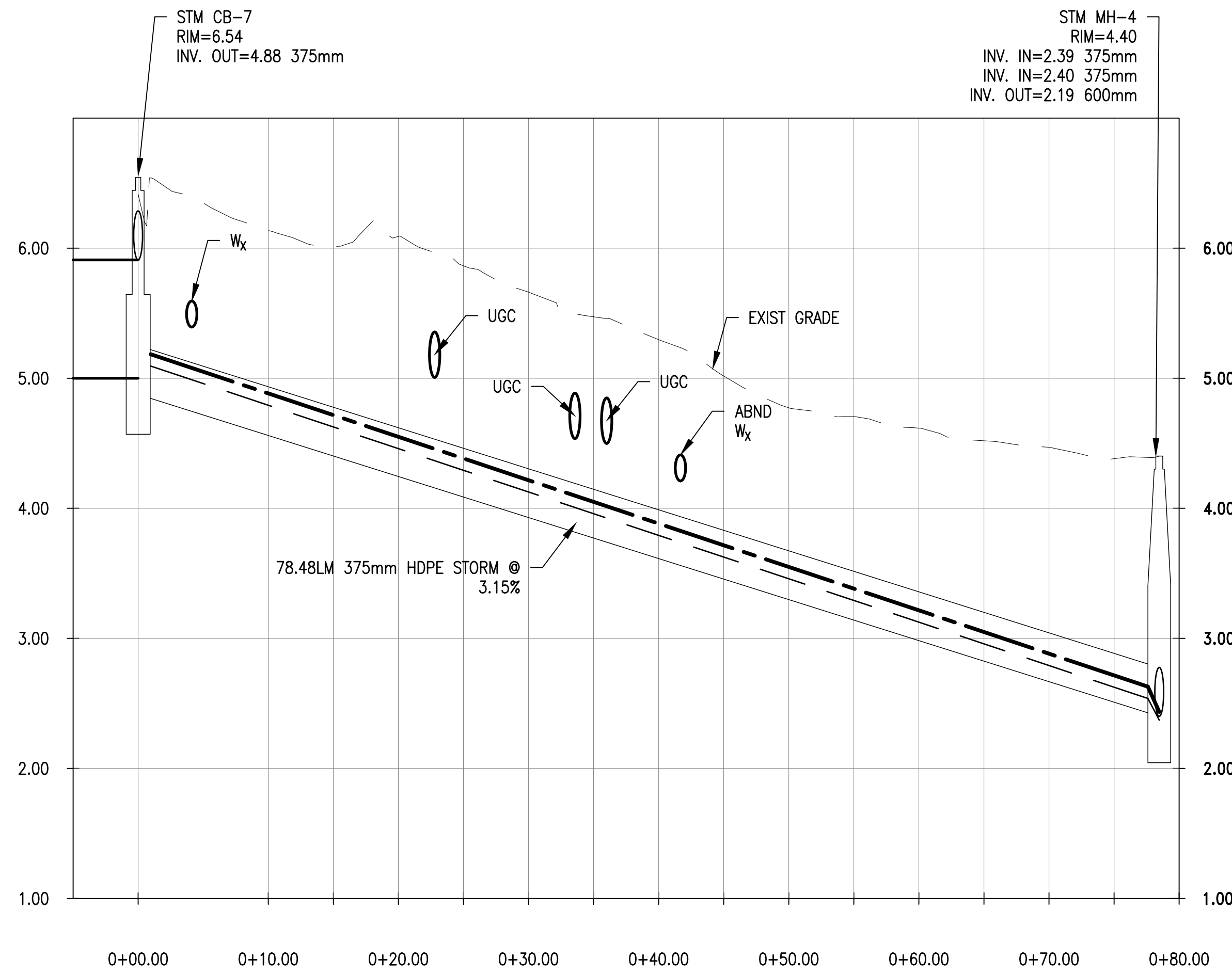
VARIOUS LOCATIONS

VARIOUS LOCATIONS

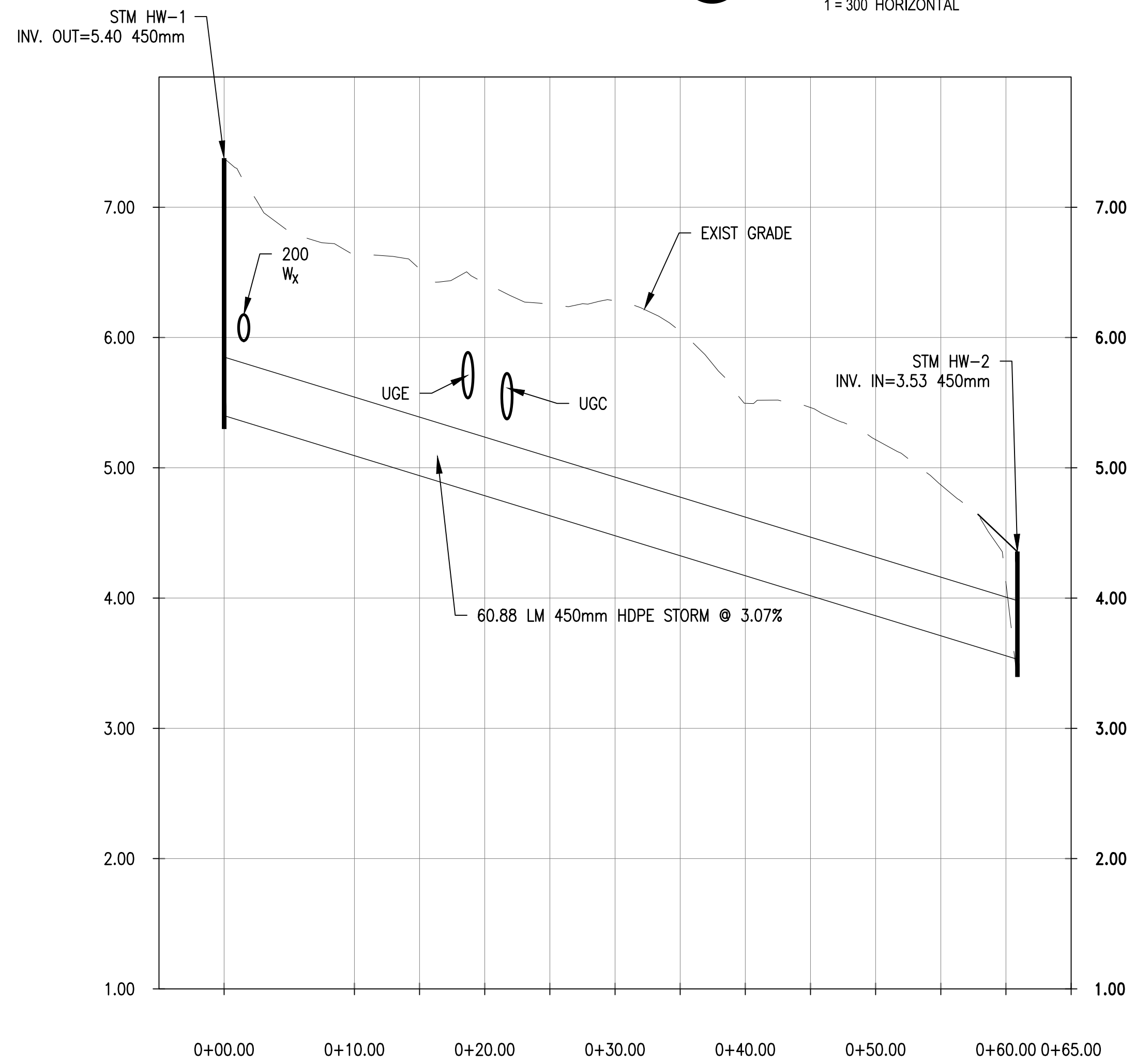
VARIOUS LOCATIONS

VARIOUS LOCATIONS

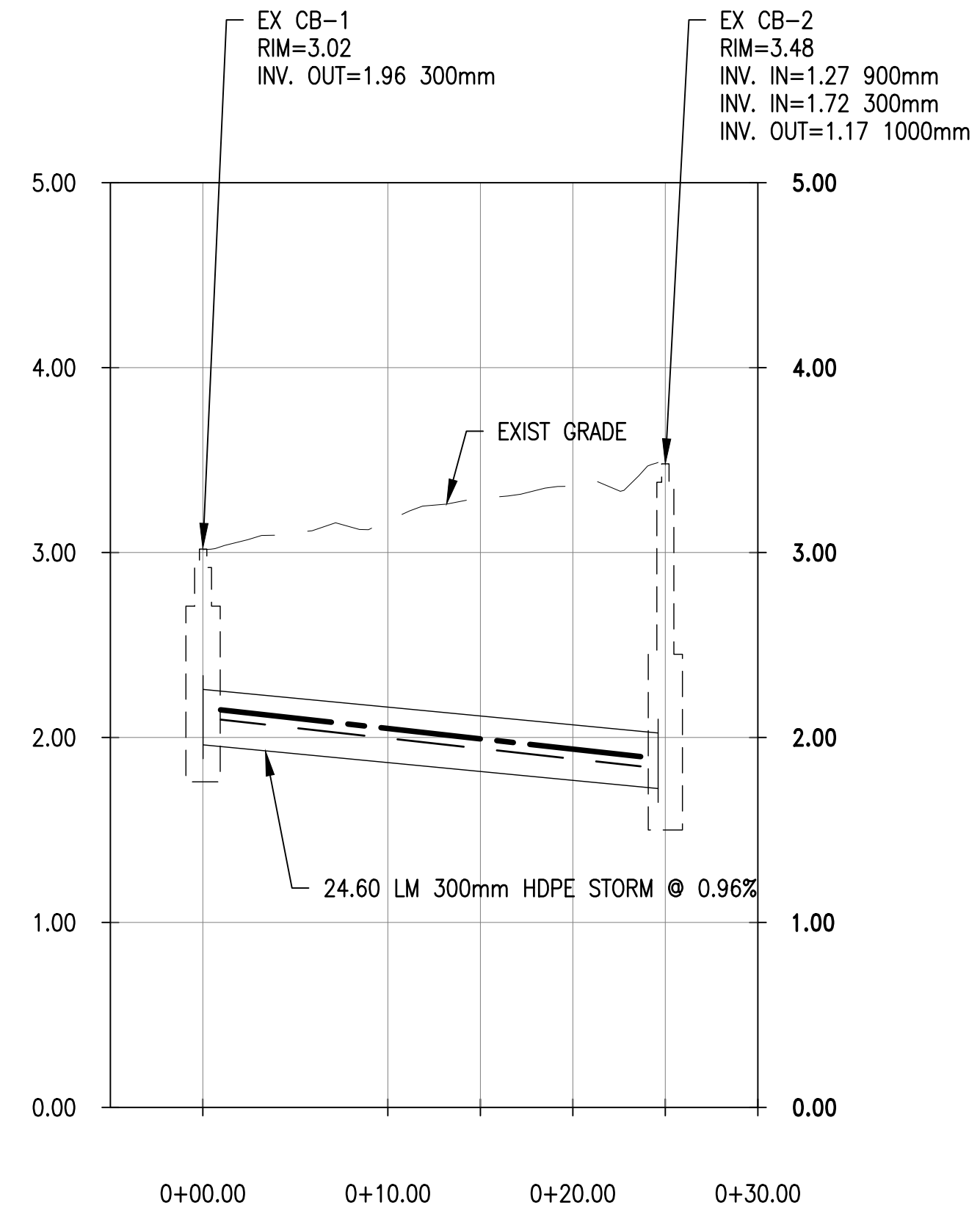
FILE NAME: V:\DDONUS\Multipipe\2026-1783264-P1360_MMP\B_Design\DWG\Sheet\05_CIV\PI360-P1360-1783264-CU101.dwg LAYOUT NAME: CU002 STORM SEWER PROFILES PLOTTED: Tuesday, January 20, 2026 - 10:46am USER: mayam.durant



A2 STORM PROFILE "D"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL

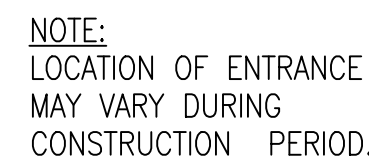


A4 STORM PROFILE "E"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL



C5 STORM PROFILE "C"
SCALE: 1 = 30 VERTICAL
1 = 300 HORIZONTAL

APPR	DATE	SYN	DESCRIPTION	SEAL
APPROVED: FOR COMMANDER NAVFAC				
ACTIVITY: D. EVANS VIA EMAIL				
SATISFACTORY TO: DATE: 01/15/2026				
DES	MAD	DRW	DTK	CHK
PM/DM	MJD/ATP			
BRANCH MANAGER	JMW			
DES PROJ DIR	WBF			
FIRE PROTECTION ENGINEER	DSN			
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PLANNING, DESIGN AND CONSTRUCTION NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC NORFOLK VA VARIOUS LOCATIONS VARIOUS LOCATIONS JOINT MARITIME FACILITY STORM SEWER PROFILES				
SCALE: AS NOTED				
PROJECT NO.: 1838720				
CONSTR. CONTR. NO.				
NAVFAC DRAWING NO.: 14157090				
SHEET 28 OF 170				
CU002				



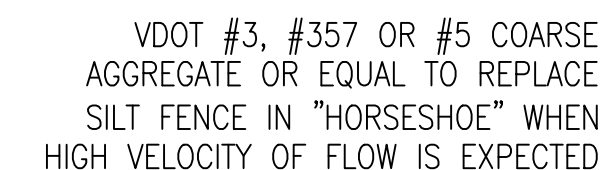
C1 CONSTRUCT
SCALE: NOT TO SCALE



C2 TEMPORARY
SCALE: NOT TO SCALE



A1 DROP INLET
SCALE: NOT TO SCALE



A4 CULVERT INLET
SCALE: NOT TO SCALE

PHASE I:
PHASE I OPERATIONS WILL INCLUDE THE CONSTRUCTION NEEDED TO DIVERT MAJOR OFF-SITE CLEAN WATER AND ON-SITE DRAINAGE FLOWS AROUND THE WORK AREA AND CONTROL EROSION AND SILTATION ASSOCIATED WITH INITIAL CLEARING AND GRADING OPERATIONS. AS REQUIRED, PHASE I CONTROLS SHALL BE ESTABLISHED IN A ONE (1) STEP PROCESS:

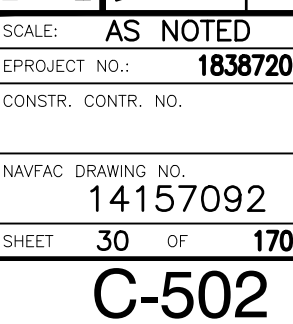
1. CONTRACTOR AND CONTRACTING OFFICER SHALL ATTEND A PRE-CONSTRUCTION MEETING PRIOR TO STARTING ANY LAND DISTURBING ACTIVITIES.
2. INSTALL CONSTRUCTION ENTRANCE (CE) AS SHOWN ON THE PLAN. PROVIDE THE CONSTRUCTION ENTRANCE AND ESTABLISH A PROTECTED STAGING AND EQUIPMENT PARKING AREA.
3. INSTALL THE REMAINING PERIMETER CONTROLS SUCH AS SILT FENCE (SF), CULVERT INLET PROTECTION (CIP), CHECK DAMS (CD), AND INLET PROTECTION (IP) AS SHOWN ON THE PLAN.
4. CONTACT THE CONTRACTING OFFICER FOR AN E&S PERIMETER INSPECTION PRIOR TO STARTING THE CLEARING AND GRUBBING OF REMAINING SITE.
5. CLEARING AND GRUBBING OR REMOVAL OF EXISTING VEGETATION MUST BE LIMITED WITHIN THE LIMITS OF DISTURBANCE, AS SHOWN ON THE PLAN.

PHASE II:
THE PHASE II SEDIMENT CONTROL MEASURES ARE INTENDED FOR THE FINAL STAGES OF SITE CONSTRUCTION. THESE MEASURES INCLUDE INLET PROTECTION (IP) WHICH SHALL BE INSTALLED BEFORE STORMWATER IS ALLOWED TO ENTER NEW INLETS. PHASE I CONTROL MEASURES, WHICH ARE NOT IN CONFLICT WITH FINAL CONSTRUCTION AND PROVIDE EFFECTIVE CONTROL, SHALL REMAIN IN PLACE FOR FINAL DEVELOPMENT. THE CONTRACTOR SHALL PROVIDE ADDITIONAL CONTROL MEASURES NEEDED DURING THE CONSTRUCTION PROCESS, AT NO ADDITIONAL COST TO THE GOVERNMENT, TO INSURE THAT SEDIMENT IS PREVENTED FROM POLLUTING OFF-SITE AREAS, STREAMS AND/OR PROTECTED ON-SITE AREAS.

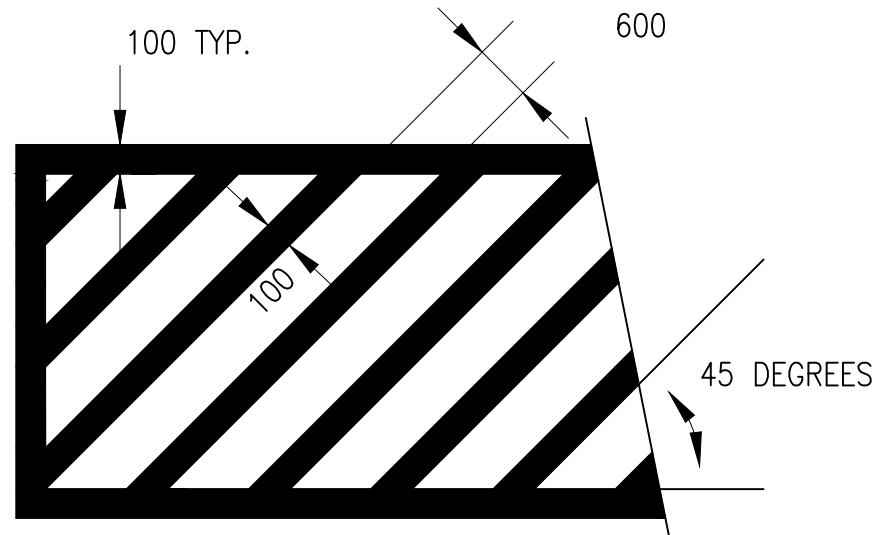
THE FOLLOWING SEQUENCE OF CONSTRUCTION IS RECOMMENDED:

1. ROUGH GRADE THE SITE. REMOVE PORTION OF SILT FENCE SHOWN ON THE PLAN PRIOR TO GRADING OPERATIONS IN AREA SOUTH OF THE ASPHALT ROAD.
2. INSTALL THE STORM SEWER SYSTEM, UTILITIES AND OTHER SITE FEATURES. INSTALL OUTLET PROTECTION. INSTALL INLET PROTECTION ON NEW STORM CATCH BASINS.
3. CONSTRUCT THE PAVEMENTS AND PERMANENTLY STABILIZE SITE BY PLANTING AND SEEDING.
4. REMOVE EROSION CONTROL MEASURES ONLY AFTER INSPECTION AND APPROVAL BY THE CONTRACTING OFFICER.

[illegible]



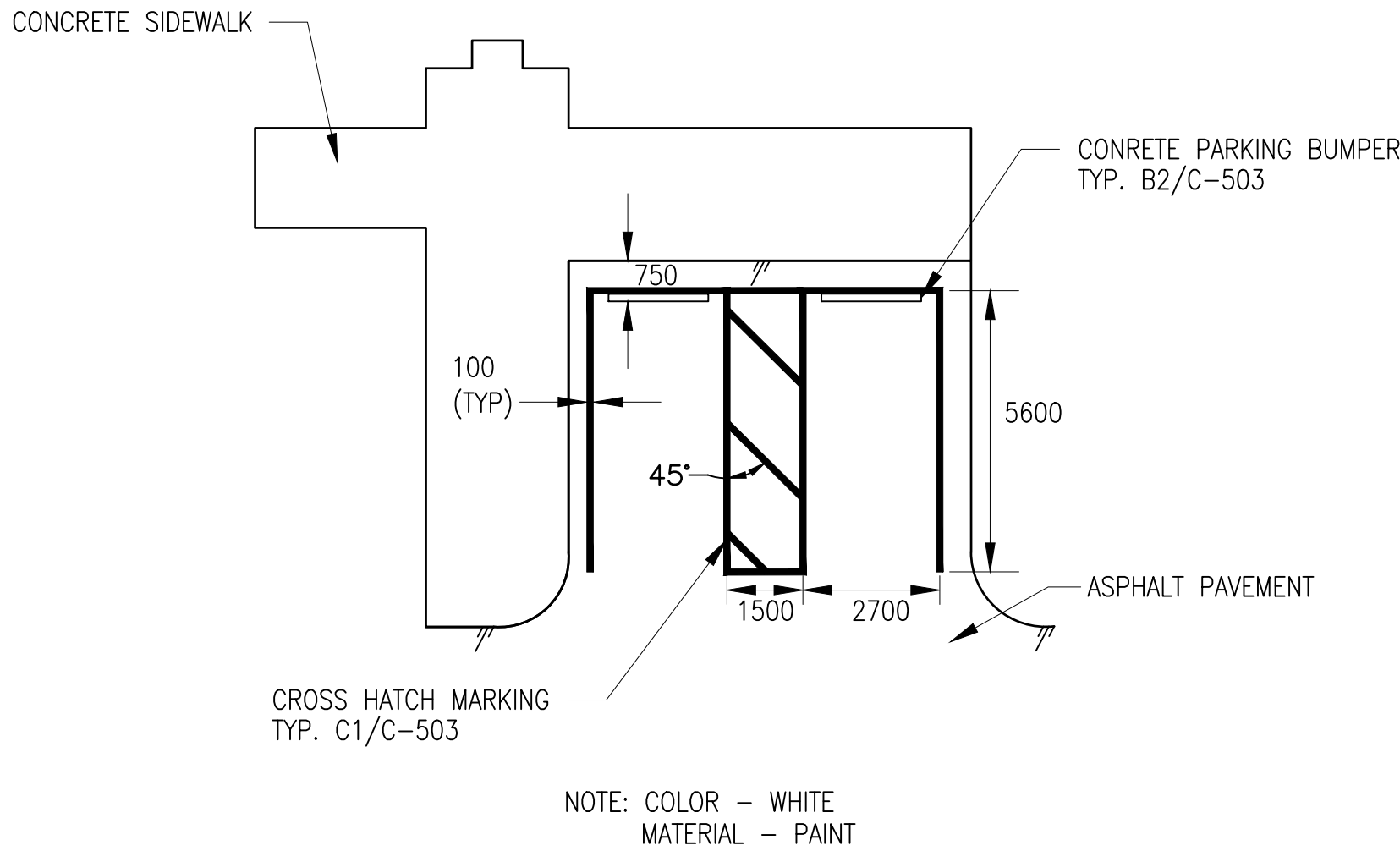
FILE NAME: V:\DDO\US\MultipleSite\2026_1783264-P1360_MMP\B_Design\DWG\Sheet\05_CIV\F1360-1783264-C-501.dwg LAYOUT NAME: C-503 DETAILS - SITE PLOTTED: Tuesday, January 20, 2026 - 10:46am USER: mayam.durant



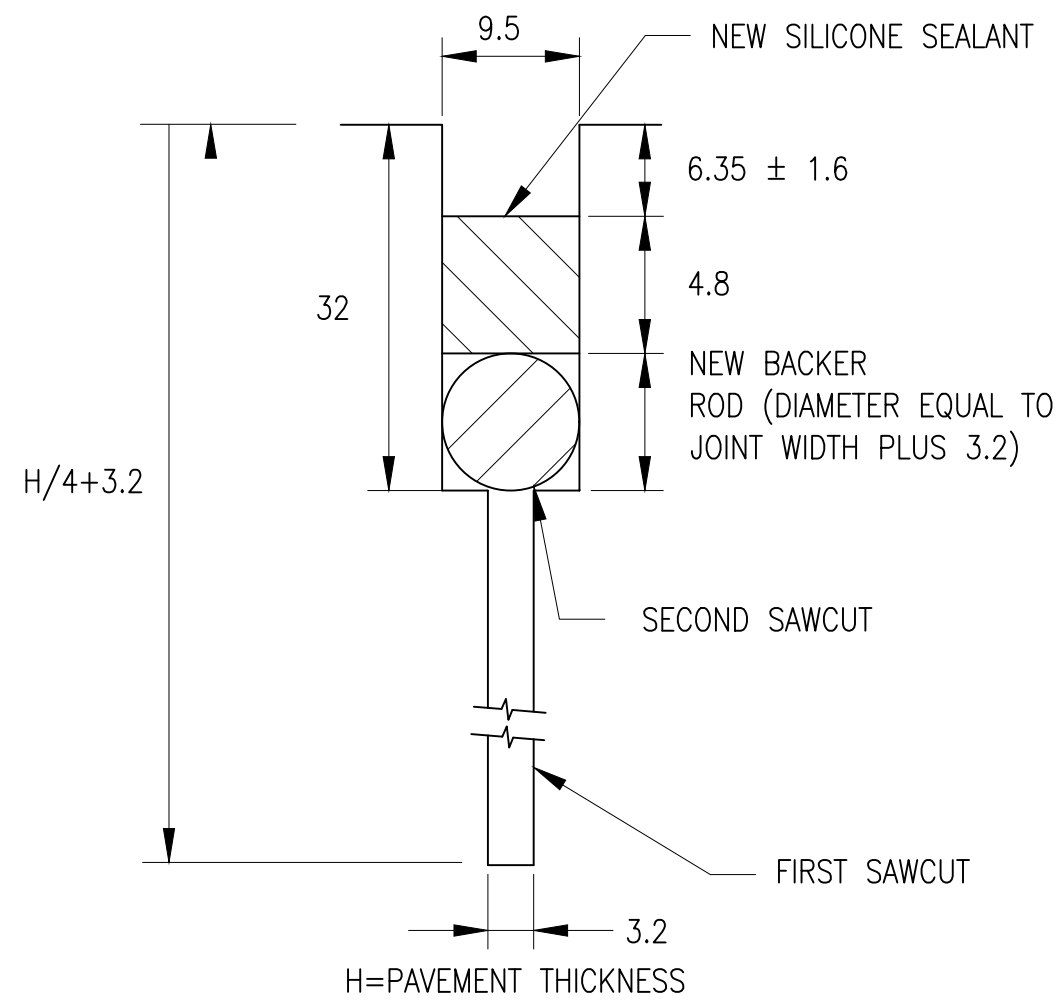
COLORS : WHITE ON REGULAR SPACES AND END ISLANDS.
YELLOW ON H/C SPACES AND H/C AISLE.
OTHER COLORS AS NOTED.

MATERIAL: PAINT

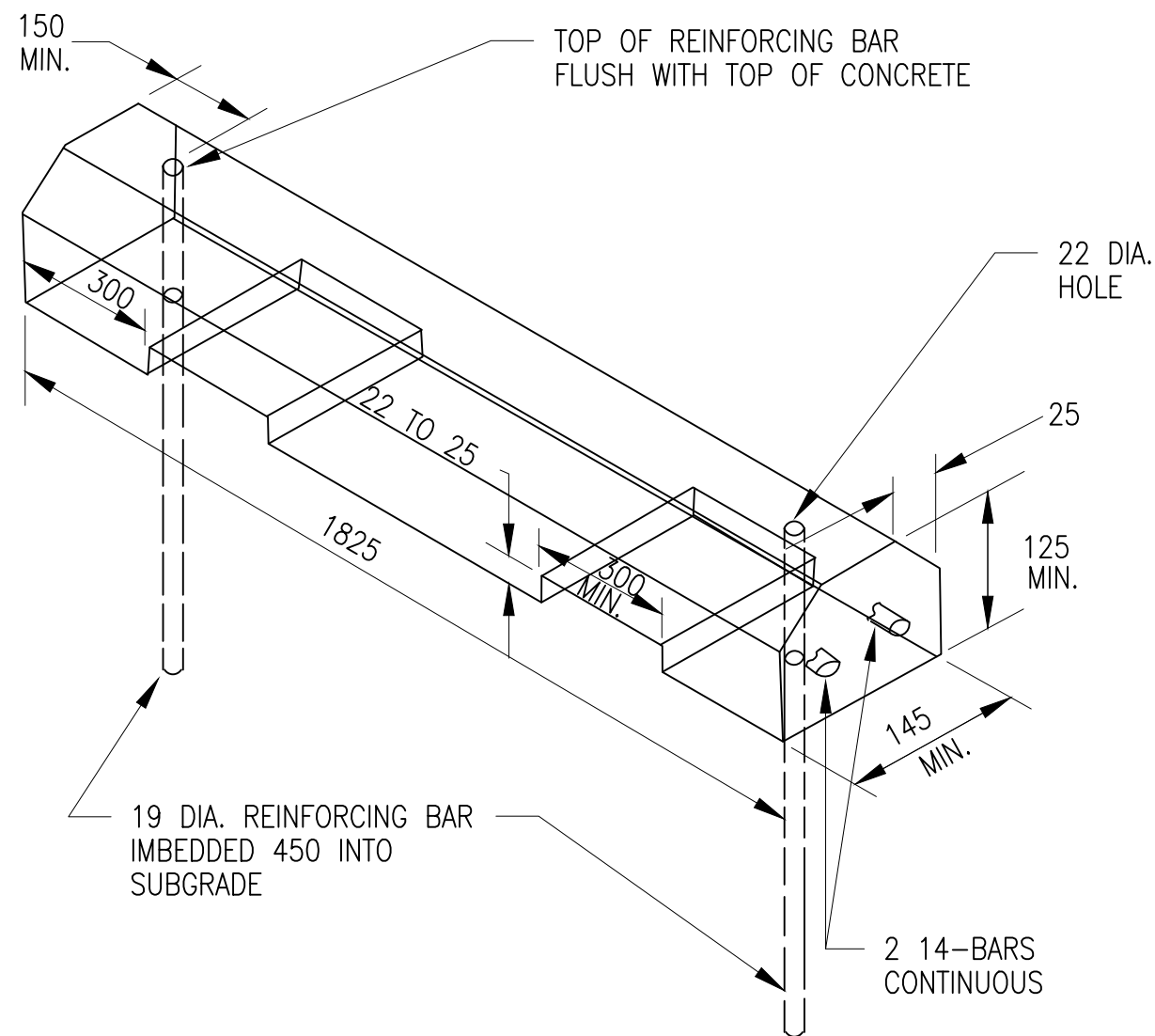
C1 CROSS HATCH MARKING
SCALE: NOT TO SCALE



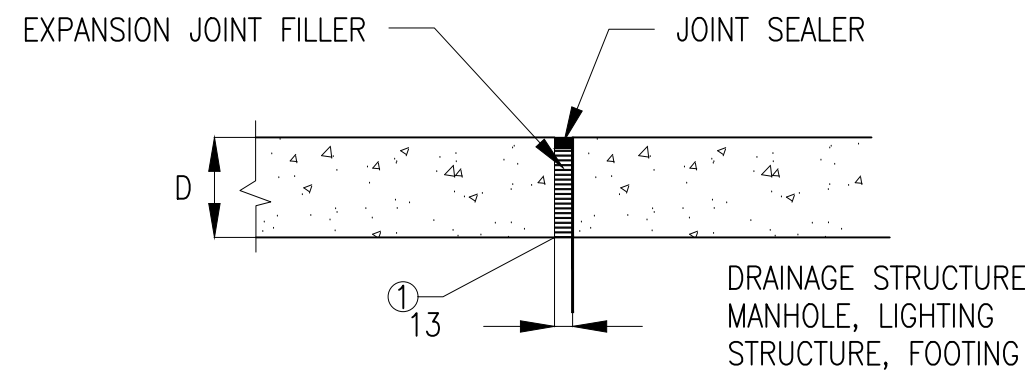
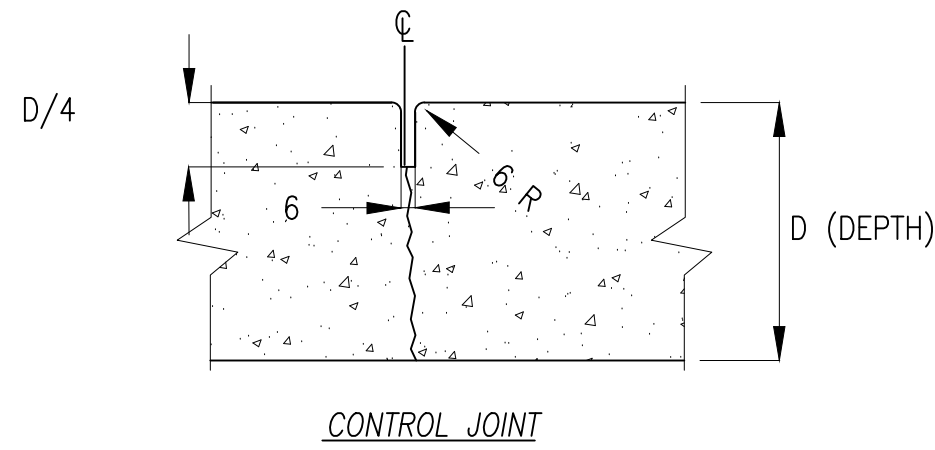
C2 PARKING AREA
SCALE: NOT TO SCALE



C4 SAWED CONTRACTION OR CONSTRUCTION JOINTS
SCALE: NOT TO SCALE

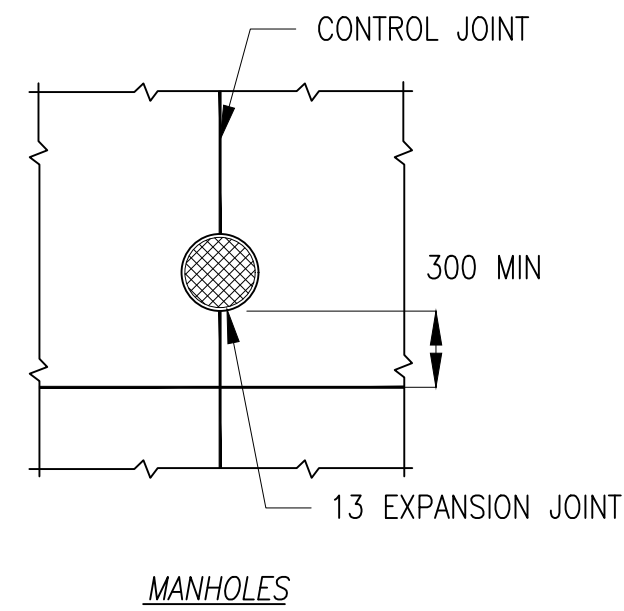


B2 CONCRETE PARKING BUMPER
SCALE: NOT TO SCALE



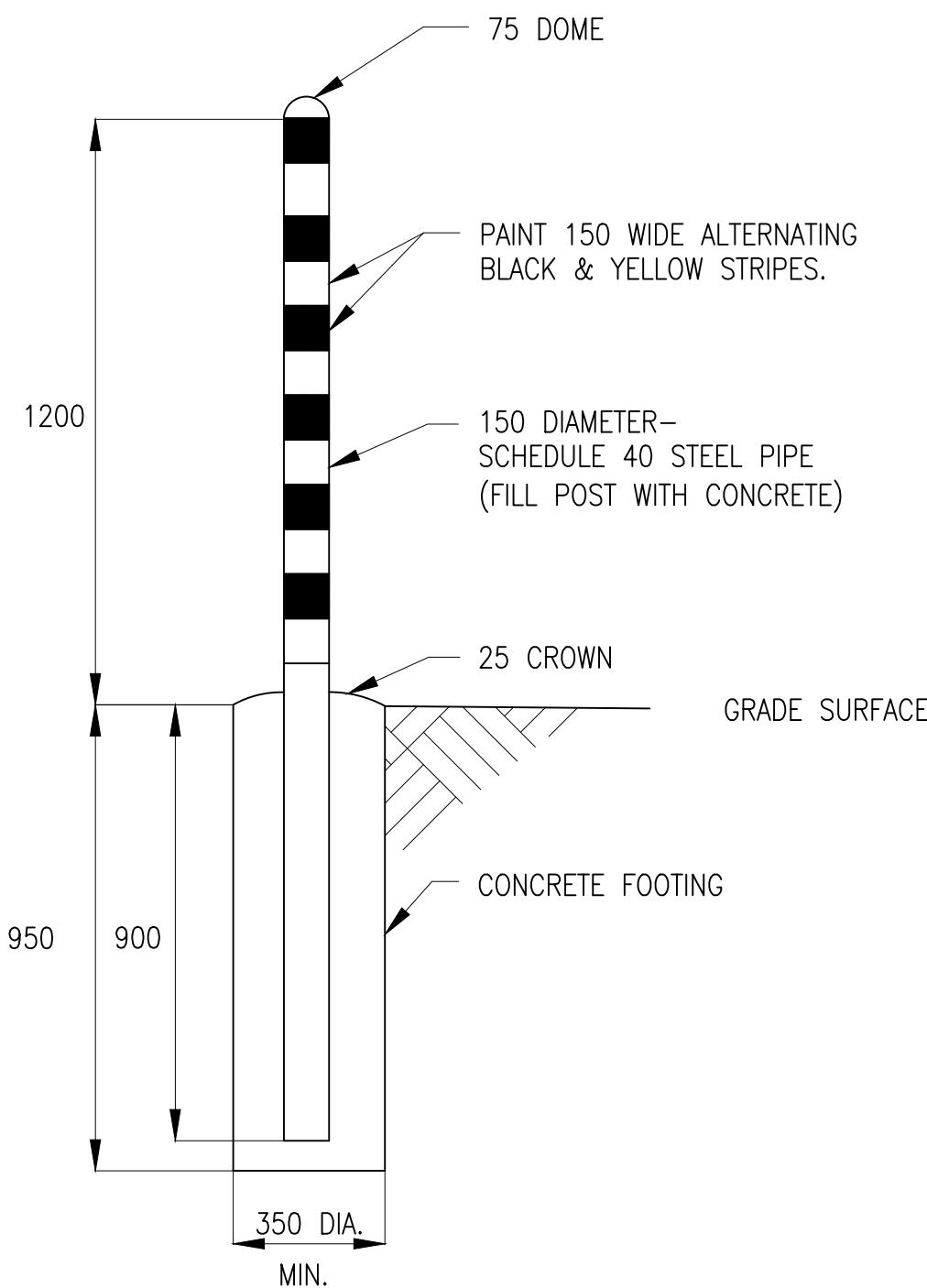
NOTE:
① 9.5 JOINT FILLER FOR SIDEWALKS, RAMPS, AND DRIVEWAYS

EXPANSION JOINT DETAIL

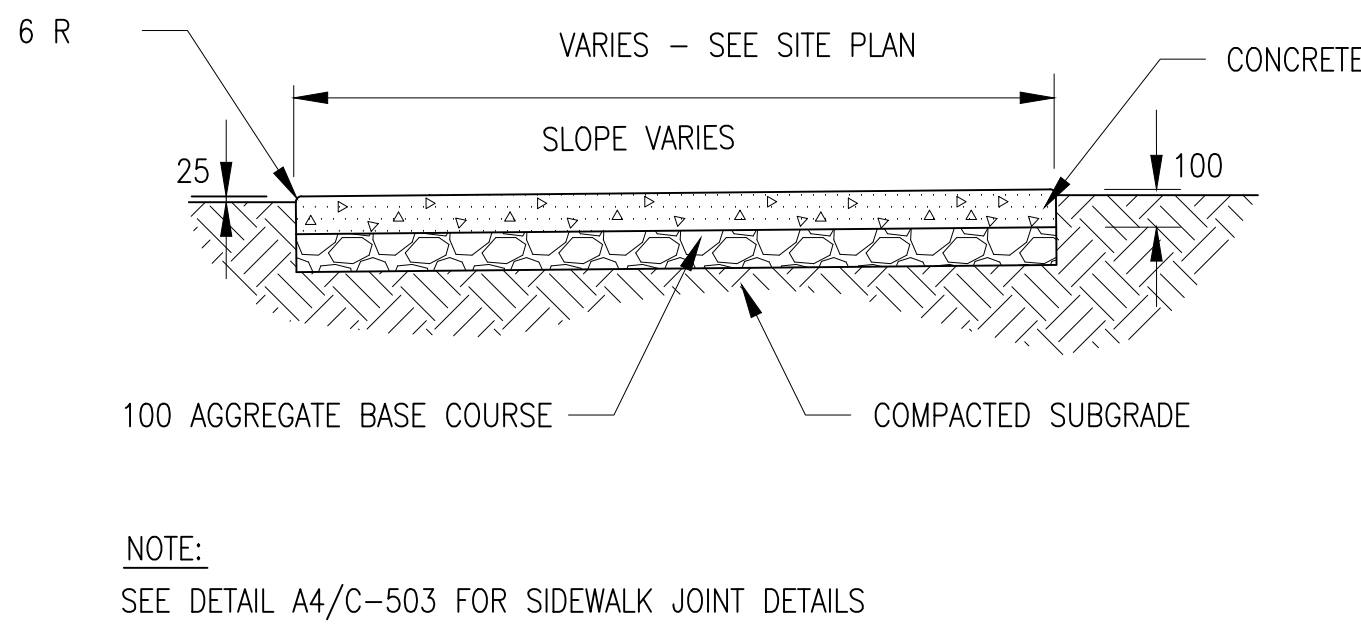


NOTES:
1. EXPANSION JOINTS MAY BE USED IN LIEU OF CONTROL JOINTS.

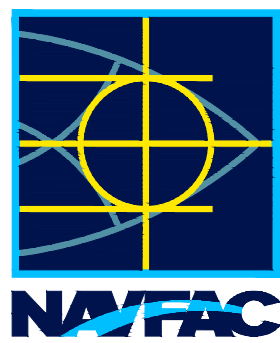
A4 TYPICAL SIDEWALK CONCRETE JOINT
SCALE: NOT TO SCALE



A1 BOLLARD
SCALE: NOT TO SCALE



A2 CONCRETE SIDEWALK
SCALE: NOT TO SCALE



APPROVED: Digitally signed by signature
Win B Forbes
Date: 2026.01.28
16:55:54-05'00'

ACTIVITY

D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES: MAD DRW: DTK CHK: JWN

PM/DM: MJD/ATP

BRANCH MANAGER: JMW

DES PROJ DIR: WBF

FIRE PROTECTION ENGINEER: DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

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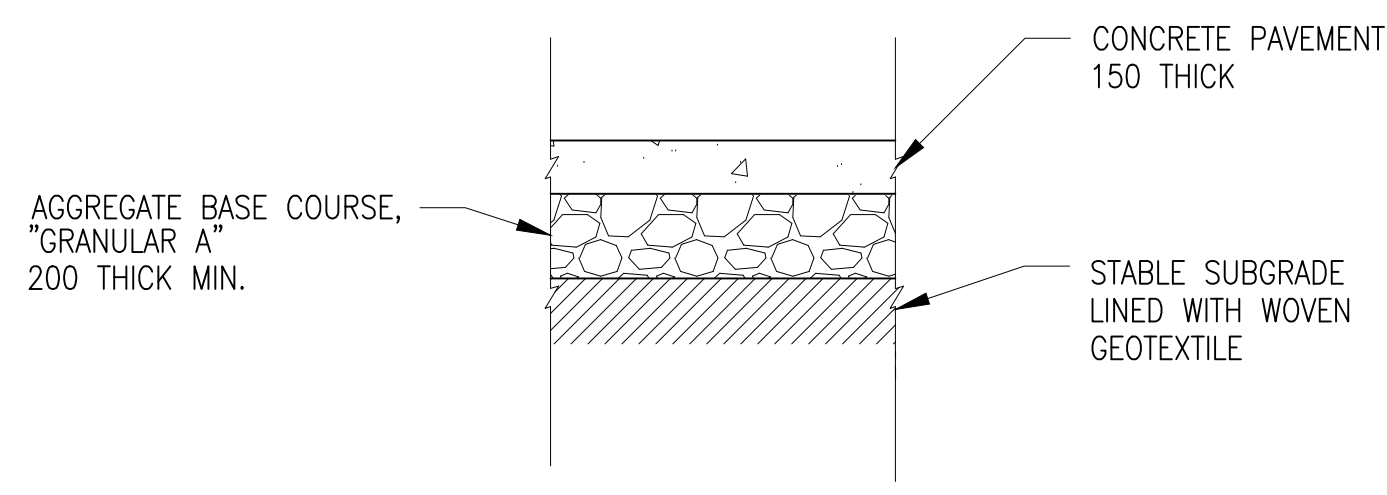
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D

C

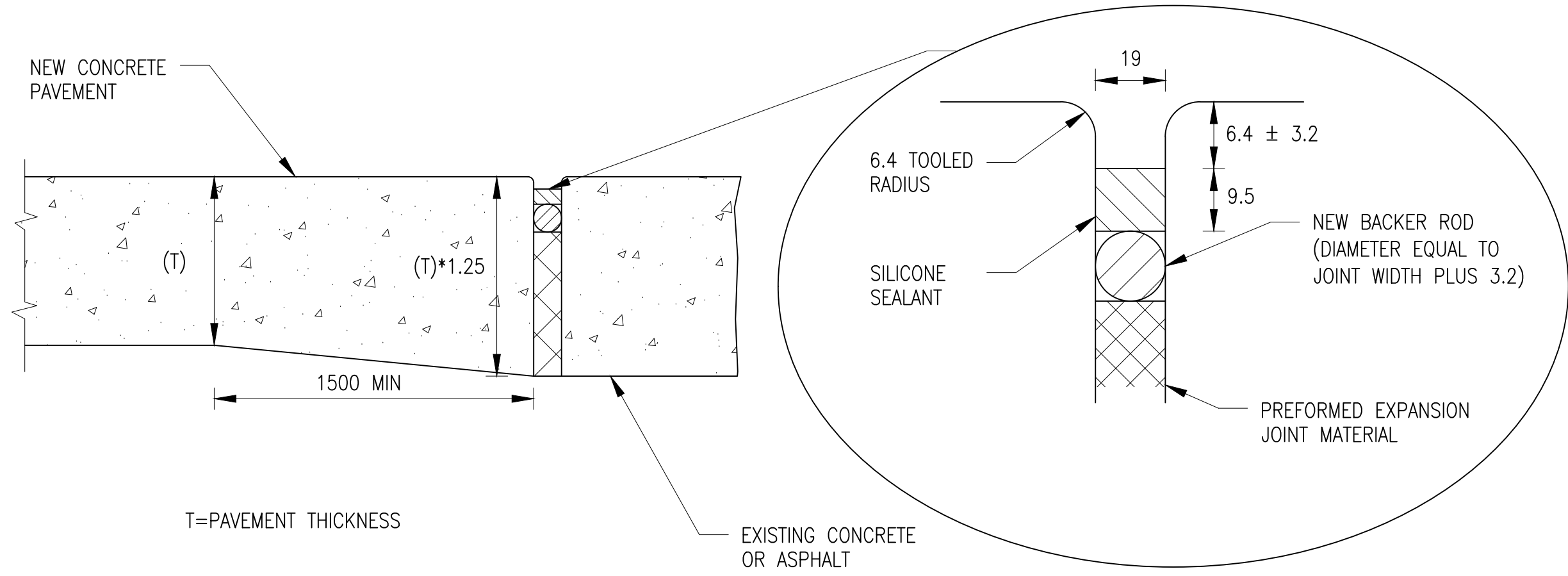
B

A

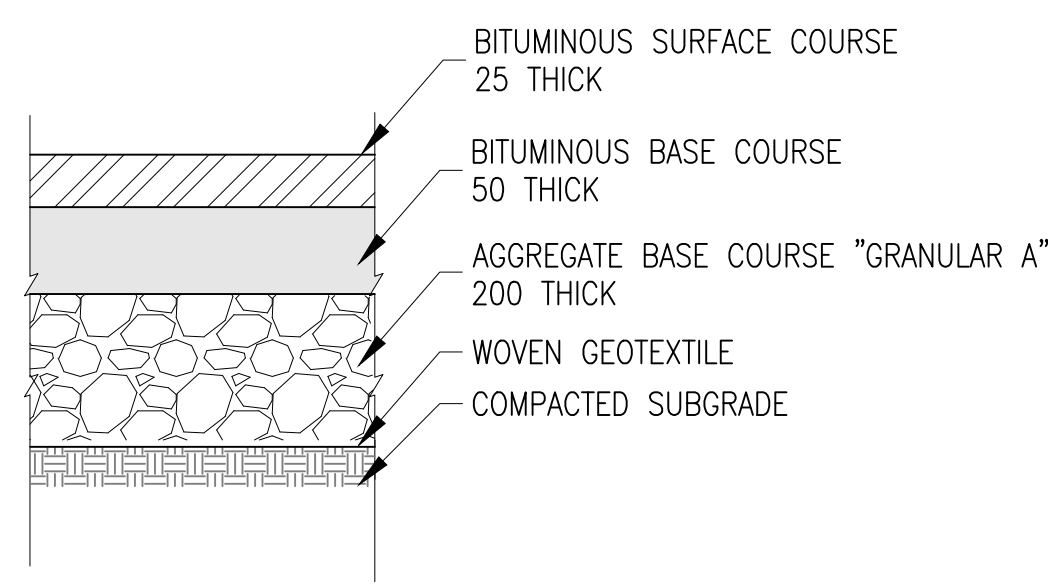


NOTE:
SEE DETAIL A4/C-502 FOR REINFORCEMENT DETAIL.

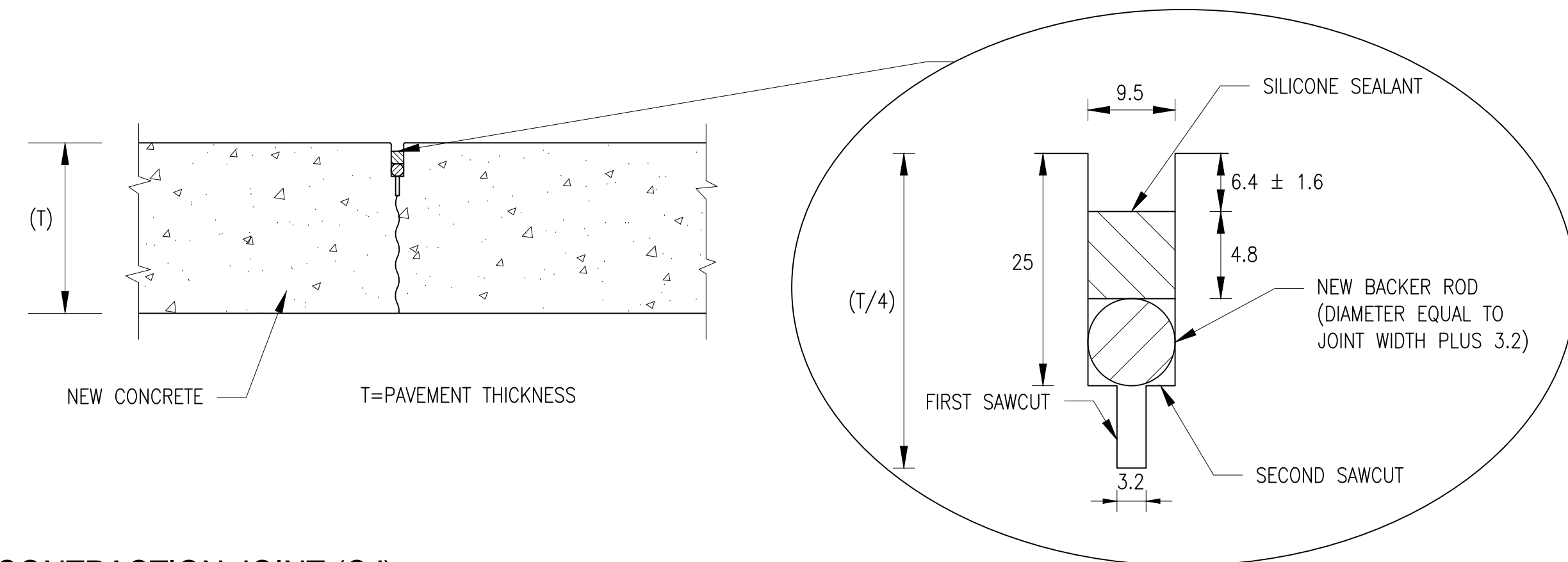
C1 HEAVY DUTY PORTLAND CEMENT CONCRETE (PCC)
SCALE: NOT TO SCALE



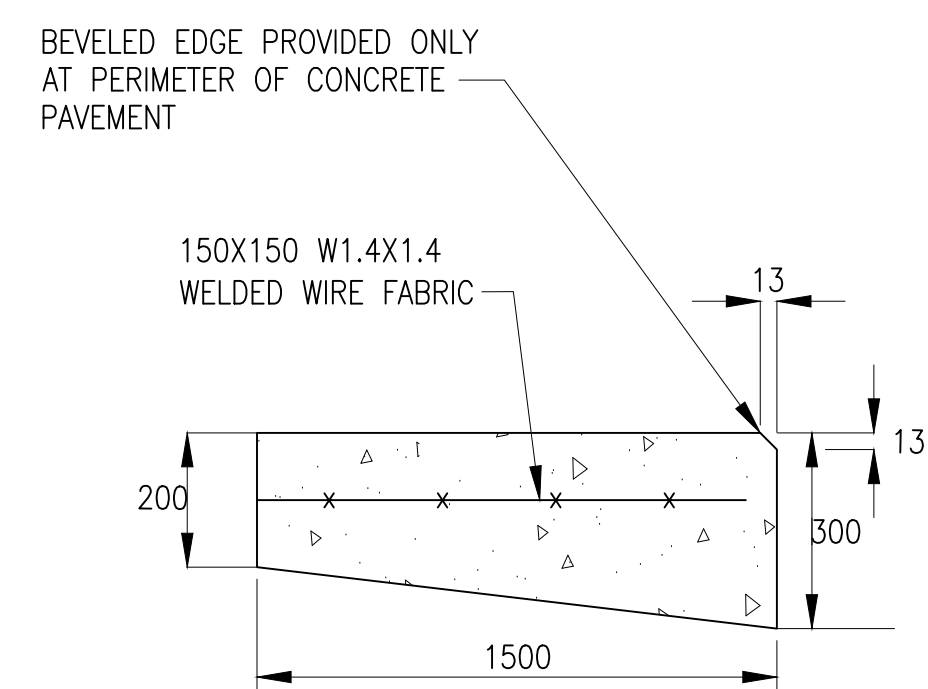
C3 THICKENED EDGE EXPANSION JOINT (TEJ)
SCALE: NOT TO SCALE



B1 ASPHALT PAVEMENT
SCALE: NOT TO SCALE

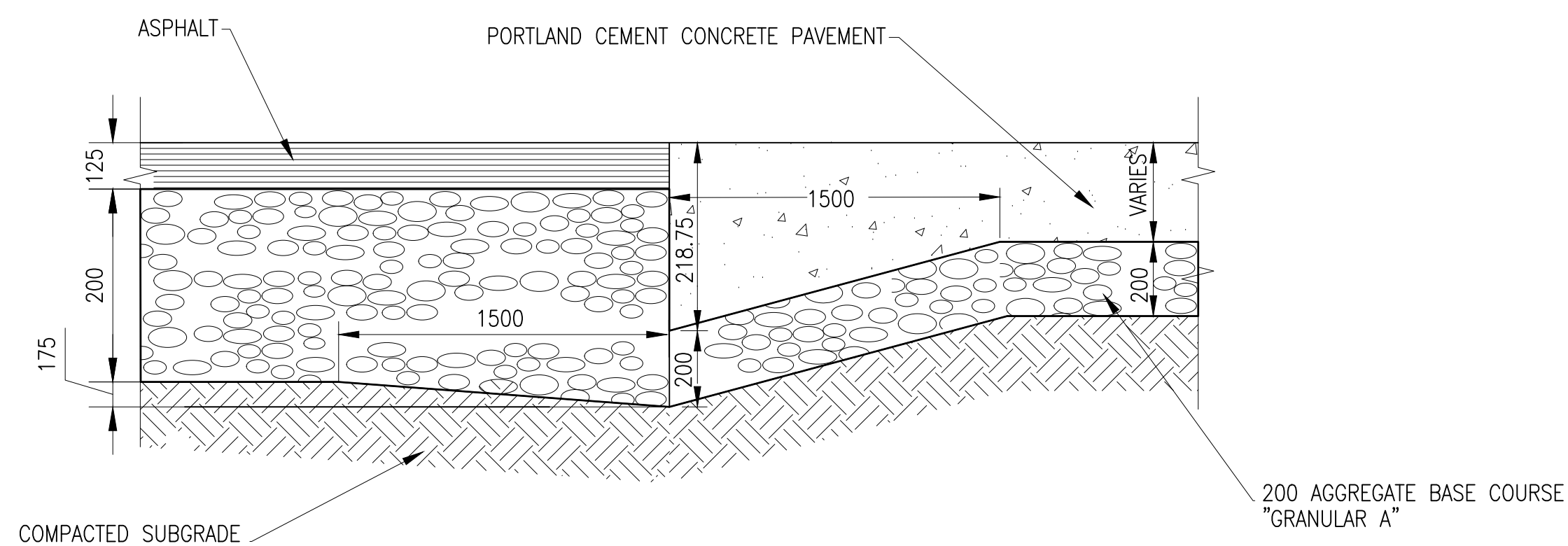


B3 CONTRACTION JOINT (CJ)
SCALE: NOT TO SCALE

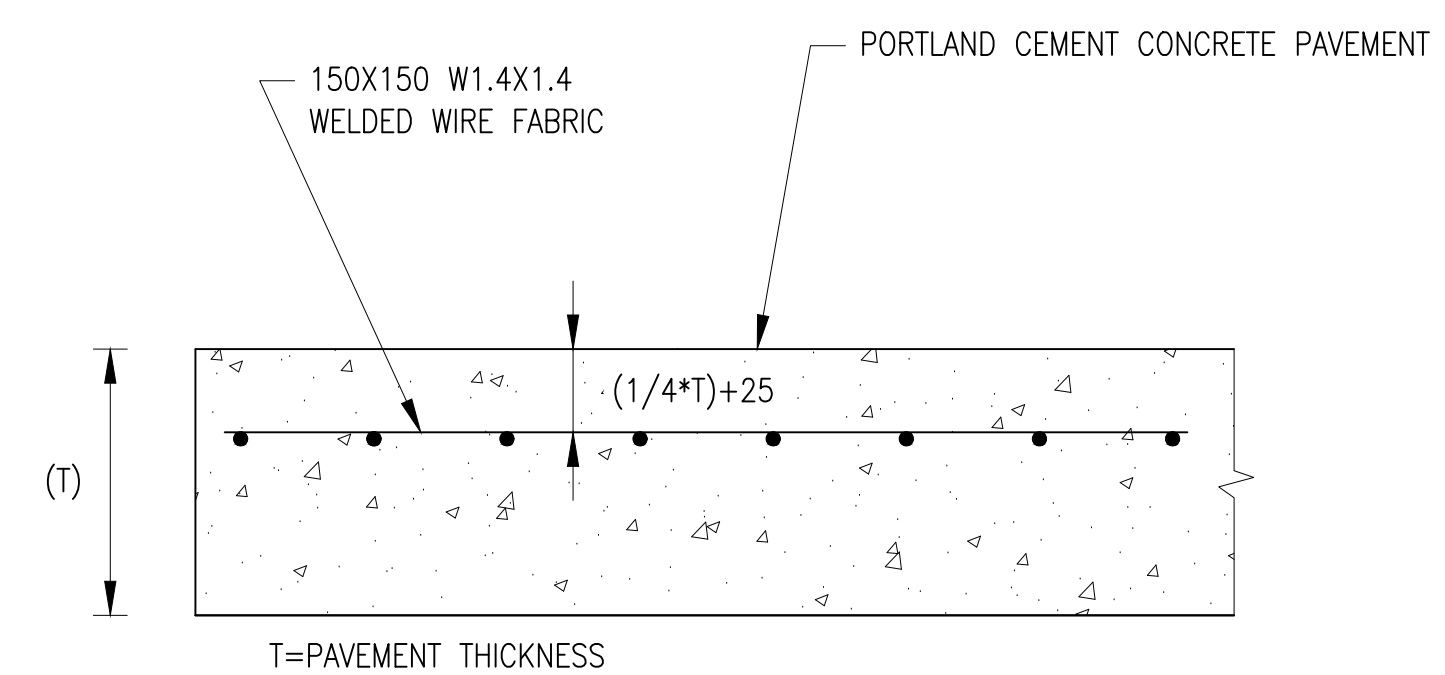


NOTE:
PROVIDE EXPANSION JOINT MATERIAL WHERE SLAB EDGE ABUTS BUILDINGS, FOUNDATIONS, WALLS, MANHOLES, AND OTHER FIXED OBJECTS.

A1 THICKENED EDGE (TE)
SCALE: NOT TO SCALE

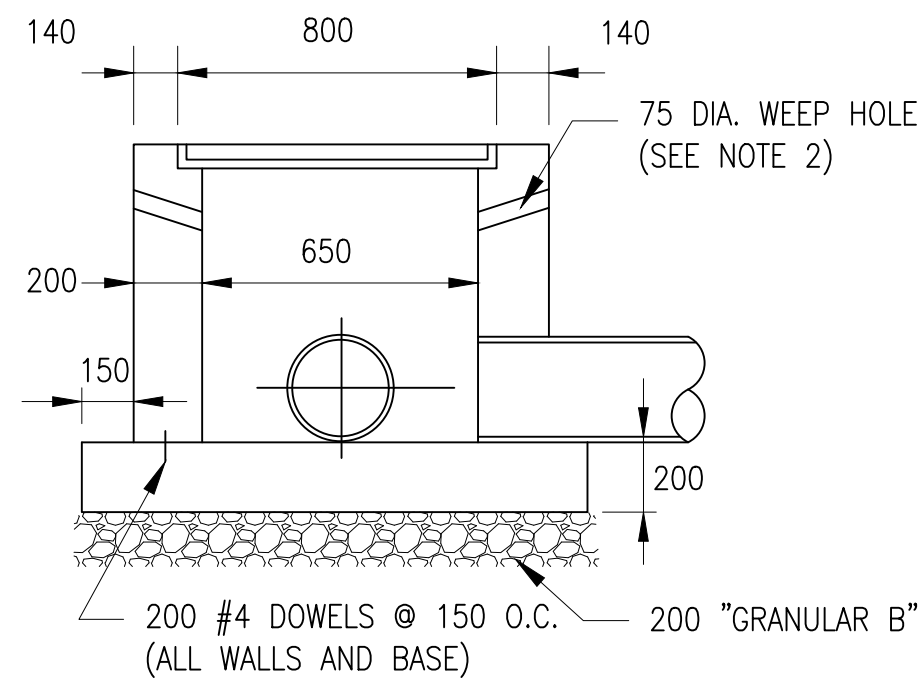


A2 TRANSITION - ASPHALT TO CONCRETE SECTION (TEB)
SCALE: NOT TO SCALE

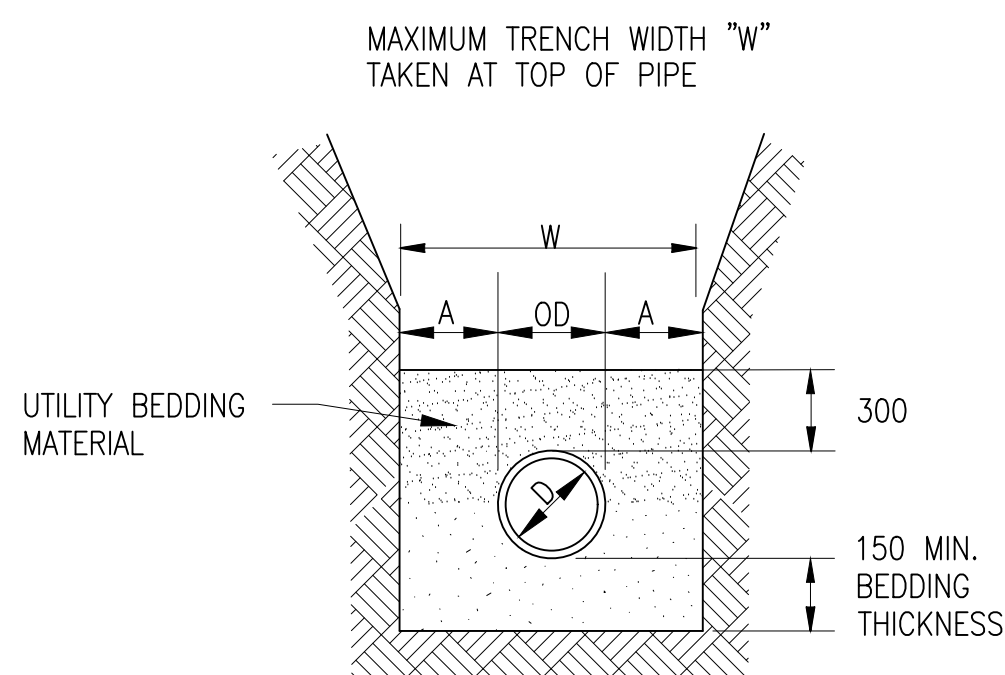


A4 REINFORCED SLAB **R**
SCALE: NOT TO SCALE

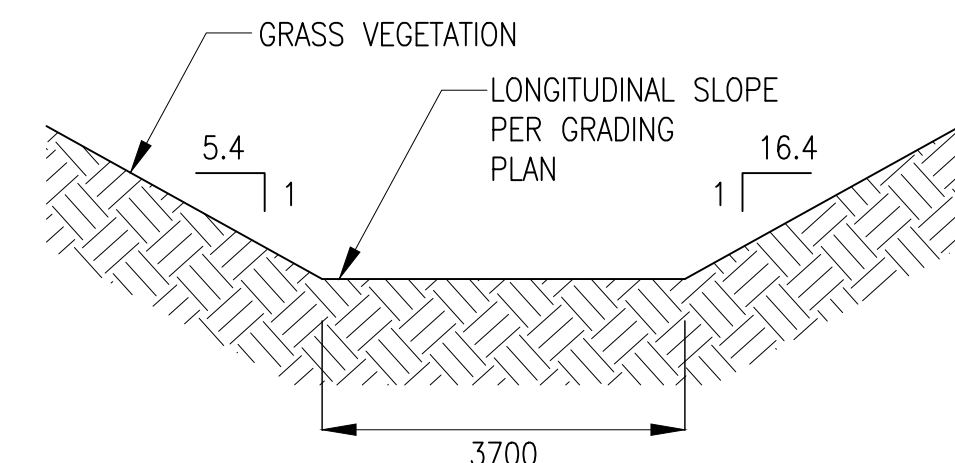
APPROVED	DATE	APPROVED
Win B Forbes	2026.01.28	
FOR COMMANDER NAFAC: 16:55:54-05:00		
ACTIVITY		
D. EVANS VIA EMAIL		
SATISFACTORY TO DATE: 01/15/2026		
DESIGN	DRW	CHK
MAD	DTK	JWN
PM/DM	MJD/ATP	
BRANCH MANAGER	JMW	
DESIGN PROD DIR	WBF	
FIRE PROTECTION ENGINEER	DSN	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		
PLANNING, DESIGN AND CONSTRUCTION		
VARIOUS LOCATIONS		
VARIOUS LOCATIONS		
JOINT MARITIME FACILITY		
DETAILS - SITE		
SCALE: AS NOTED		
PROJECT NO.: 1838720		
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO. 14157094		
SHEET 32 OF 170		
C-504		
NAVFAC METRIC DRAWING REVISION: 01 OCTOBER 2018		



SECTION A-A



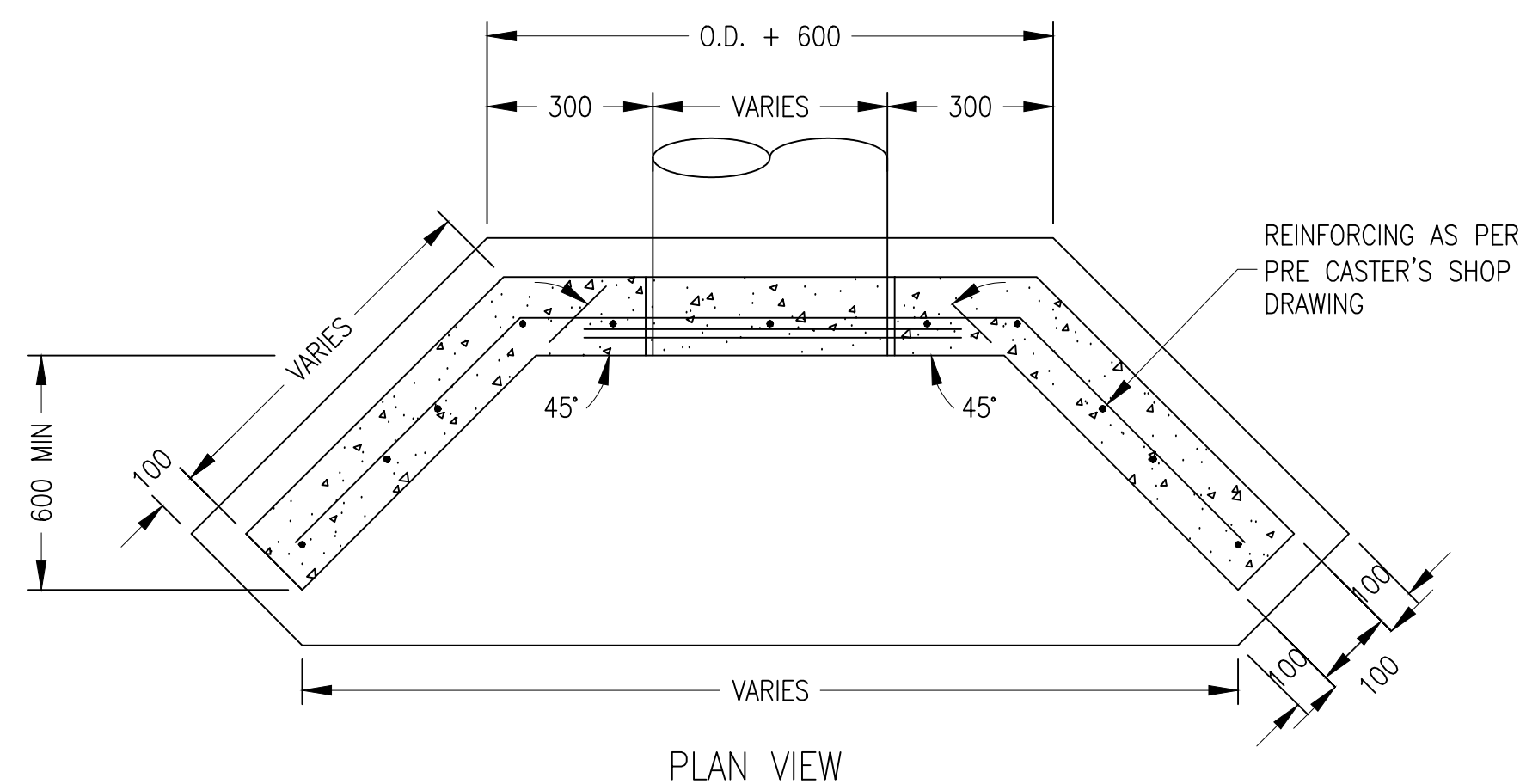
PIPE DIA "D"	MAXIMUM "A"
150 TO 375	200
400 TO 525	250
600 TO 750	300
825 TO 1050	375
1200 & LARGER	450



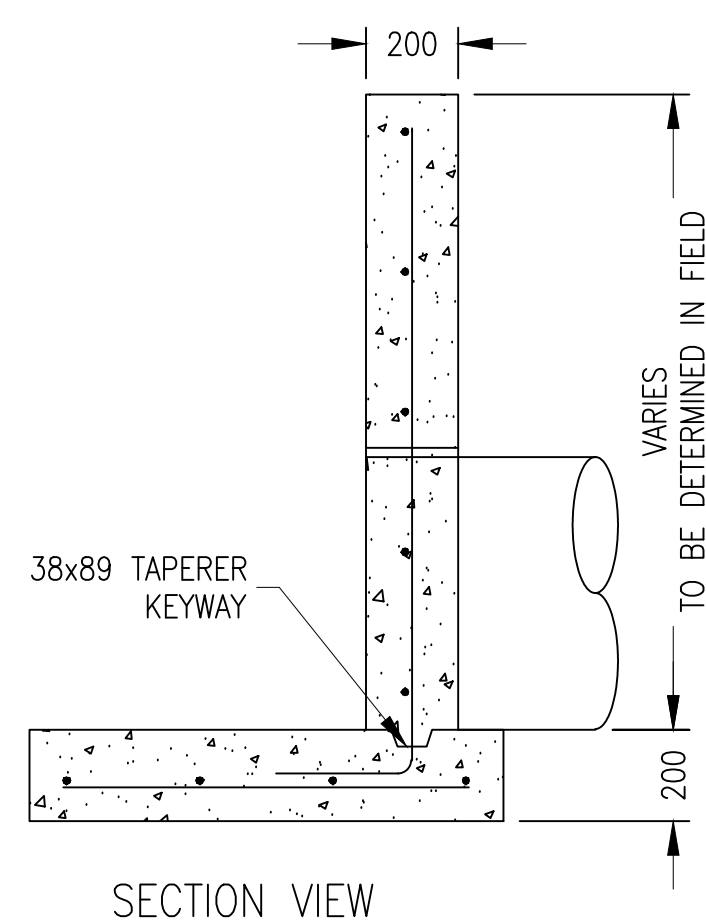
TRAPEZOIDAL SWALE SECTION

DROP INLET

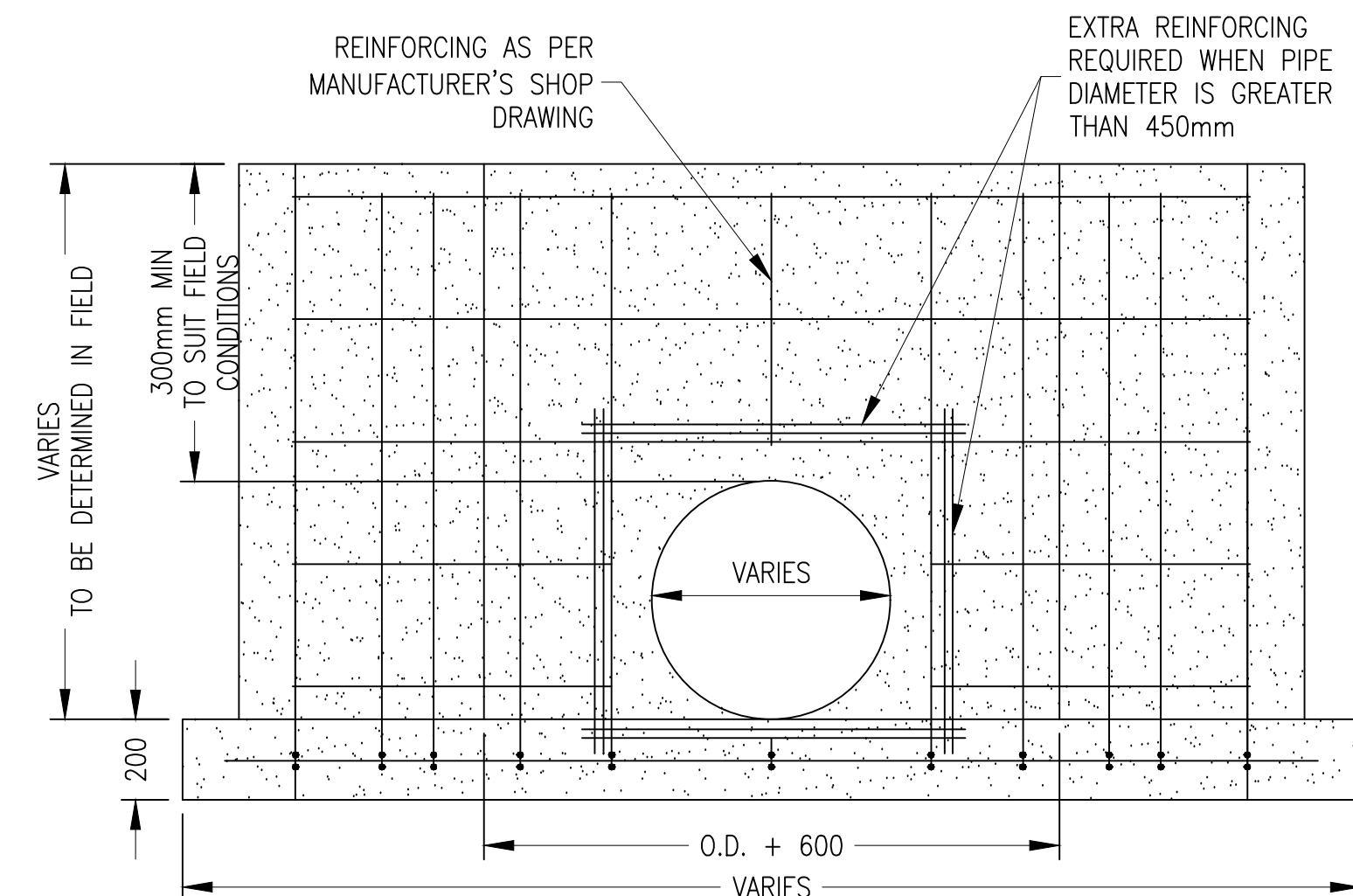
SCALE: NOT TO SCALE



PLAN VIEW



SECTION VIEW



ELEVATION VIEW

NOTES:

1. 30MPa CONCRETE.
2. COVER TO REINFORCING BARS: 75mm $\pm$ 20mm.
3. GRANULAR "B" BACKFILL TO BE PLACED TO 300mm MIN THICKNESS ON ALL SIDES.
4. HAND RAIL REFER TO DRAWING 4170 TO BE PLACED ON WALLS WITH A HEIGHT EQUAL TO OR GREATER THAN 1200mm.
5. ALL EXPOSED CORNERS ON CONCRETE WORK SHALL BE CHAMFERED 25mm.
6. DEBRIS RACK TO BE USED FOR PIPES 600mm AND LARGER. FOR PIPES 750mm AND LARGER, A 600mm SQUARE HINGED ACCESS OPENING C/W HASP & CLASP SHALL BE INSTALLED IN THE TOP OF THE DEBRIS RACK.
7. HEADWALL SHALL BE SITED AT A GRADE SUCH THAT THE ULTIMATE OVERFLOW ROUTE IS OVER THE HEADWALL AND NOT THE UPSTREAM BANKS.
8. HEADWALLS WITH PIPE OPENING LARGER THAN 1800mm ARE REQUIRED TO BE STRUCTURALLY DESIGNED BY LICENSED PROFESSIONAL ENGINEER.
9. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.
10. DEBRIS RACK DIMENSIONED TO MATCH

PRECAST CONCRETE HEADWALL

SCALE: NOT TO SCALE

DRAWING NUMBER 04060

REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

TRENCH CROSS SECTION FOR PIPE

SCALE: NOT TO SCALE

PIPE SIZE (mm)	MINIMUM ALLOWABLE DEFLECTION ANGLE					
	1200 MH	1500 MH	1800 MH	2100 MH	2400 MH	3000 MH
200	90	90	90	90	90	90
250	90	90	90	90	90	90
300	90	90	90	90	90	90
375	90	90	90	90	90	90
450	90	90	90	90	90	90
525	110	90	90	90	90	90
600	115	90	90	90	90	90
750	N/A	110	90	90	90	90
900	N/A	135	110	90	90	90
1050	N/A	N/A	115	110	90	90

MIN. ALLOWABLE DEFLECTION
ANGLES FOR PVC PIPE

PIPE SIZE (mm)	1200 MH MIN ANGLE	1500 MH MIN ANGLE	1800 MH MIN ANGLE	2100 MH MIN ANGLE	2400 MH MIN ANGLE
200	90	90	90	90	90
250	90	90	90	90	90
300	90	90	90	90	90
375	90	90	90	90	90
450	90	90	90	90	90
525	95	90	90	90	90
600	110	90	90	90	90
750	N/A	N/A	N/A	95	90
900	N/A	N/A	N/A	110	90
1050	N/A	N/A	N/A	105	95

MH TYPE	MIN INVERT DRO
STRAIGHT RUN to 45° TURN	50mm
45° TURN +	150mm

NOTES:

1. THE SUMP IS MEASURED FROM THE LOWEST INVERT.
2. GRANULAR BACKFILL SHALL BE PLACED TO A MINIMUM OF 300mm ALL AROUND THE MAINTENANCE HOLE.
3. PRECAST CONCRETE COMPONENTS SHALL BE ACCORDING TO MANUFACTURERS SHOP DRAWING.
4. STRUCTURES EXCEEDING 5.0m IN DEPTH SHALL INCLUDE SAFETY PLATFORM ACCORDING TO STANDARD DRAWING 04270.
5. FOR ADJUSTMENT UNIT AND FRAME INSTALLATION, SEE STANDARD DRAWING 04250.
6. ALL DIMENSIONS ARE NOMINAL AND IN MILLIMETERS UNLESS OTHERWISE SHOWN.
7. ALL CONCRETE TO BE 30 MPa TO MEET CSA A23.1, CLASS C-2 EXPOSURE.
8. IF A TRANSITION SLAB IS REQUIRED THE DETAIL SHALL BE PROVIDED BY THE PRECASTER

9. BENCHING
 -TO BE 30MPa CONCRETE AND START AT 2/3 THE HEIGHT OF THE PIPE AND SLOPE UPWARDS AT 4:1.
 -BENCHING AND CHANNEL TO BE GIVEN STEEL TROWEL FINISH.
 -SLOPES SHALL BE MAINTAINED FROM THE OUTLET HOLE OPENING FOR TOP OF BENCHING.
 -THERE SHALL BE NO GAP BETWEEN THE PIPE AND THE BENCHING.

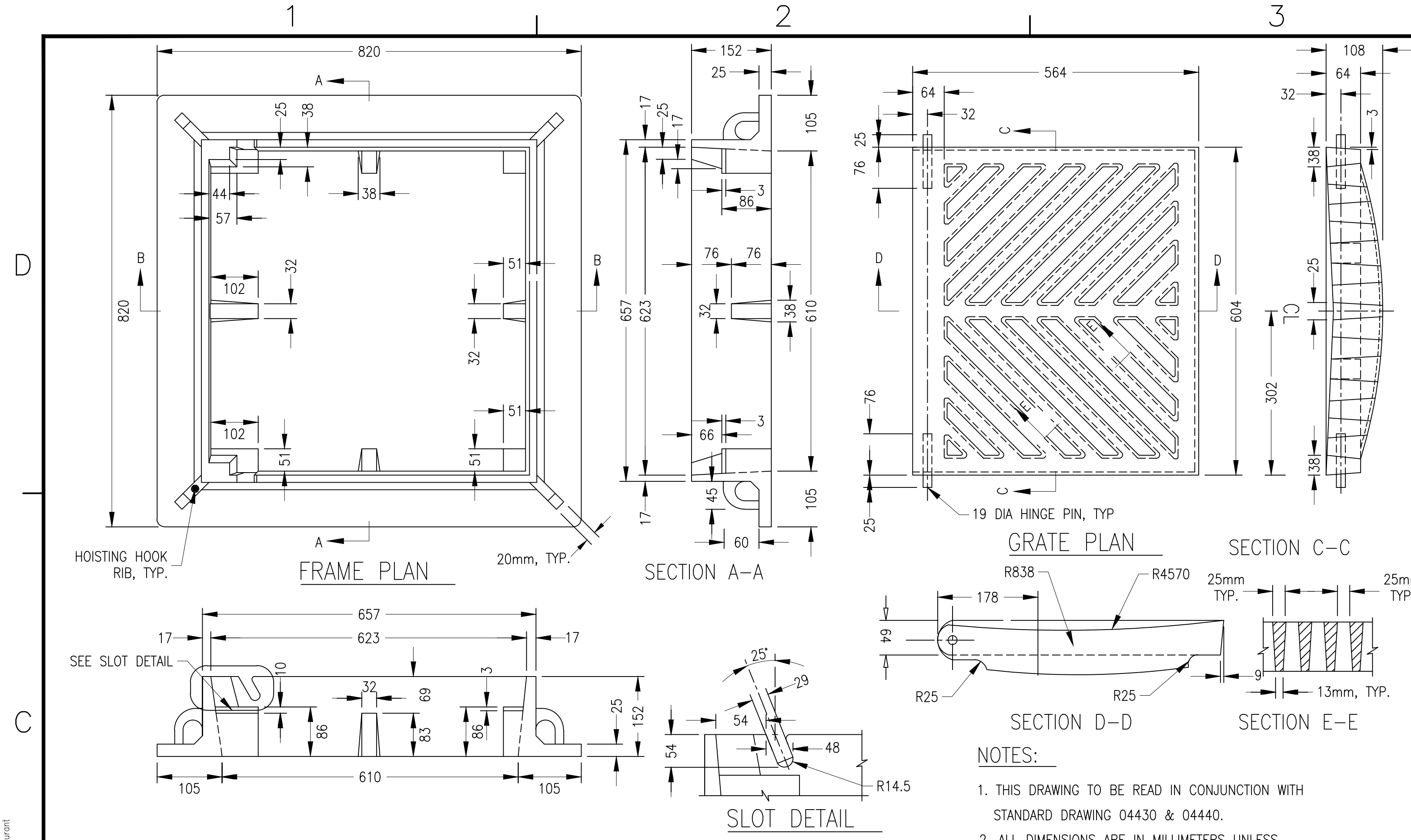
4.2 PRECAST MANHOLE DETAIL

SCALE: NOT TO SCALE

DRAWING NUMBER 04190

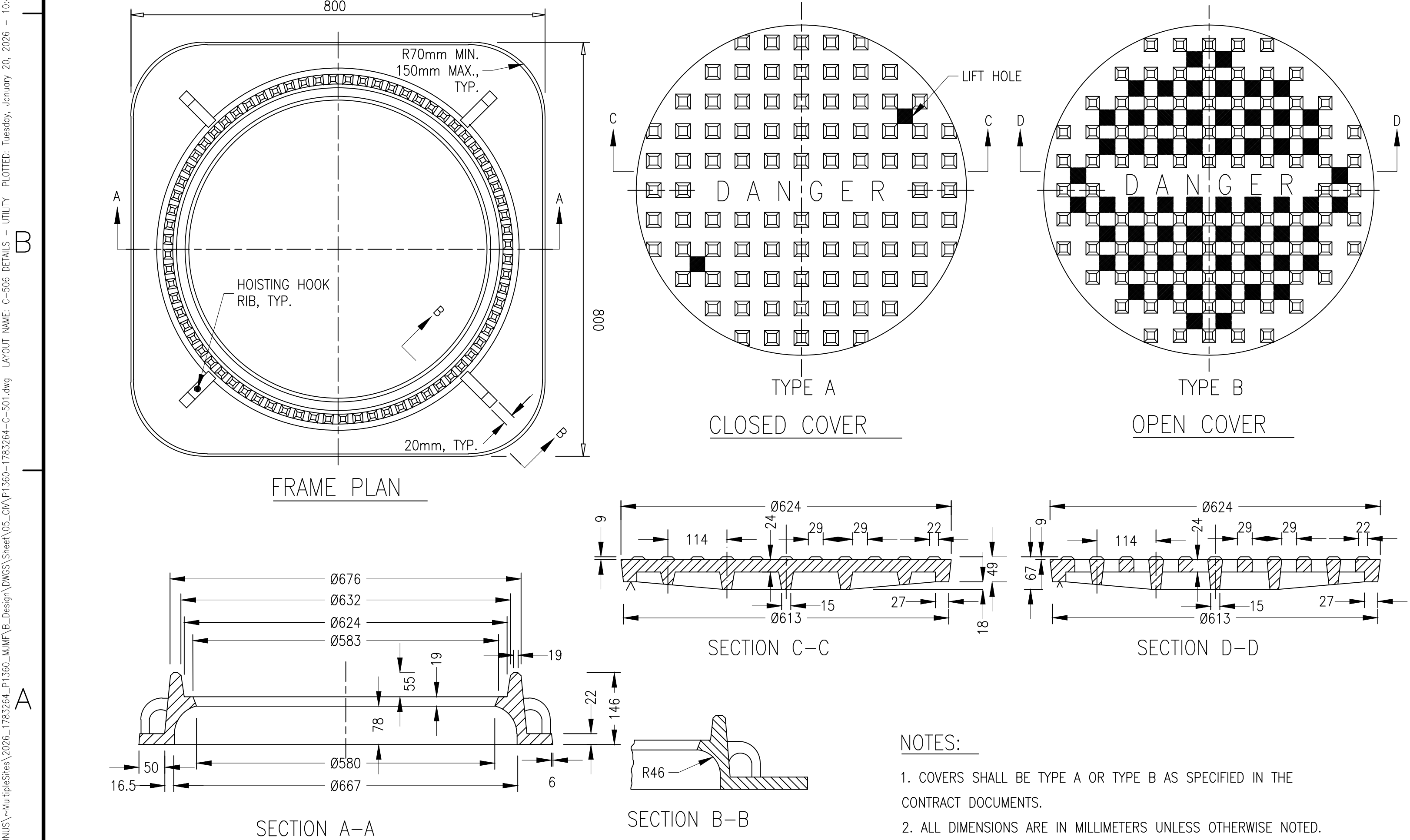
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

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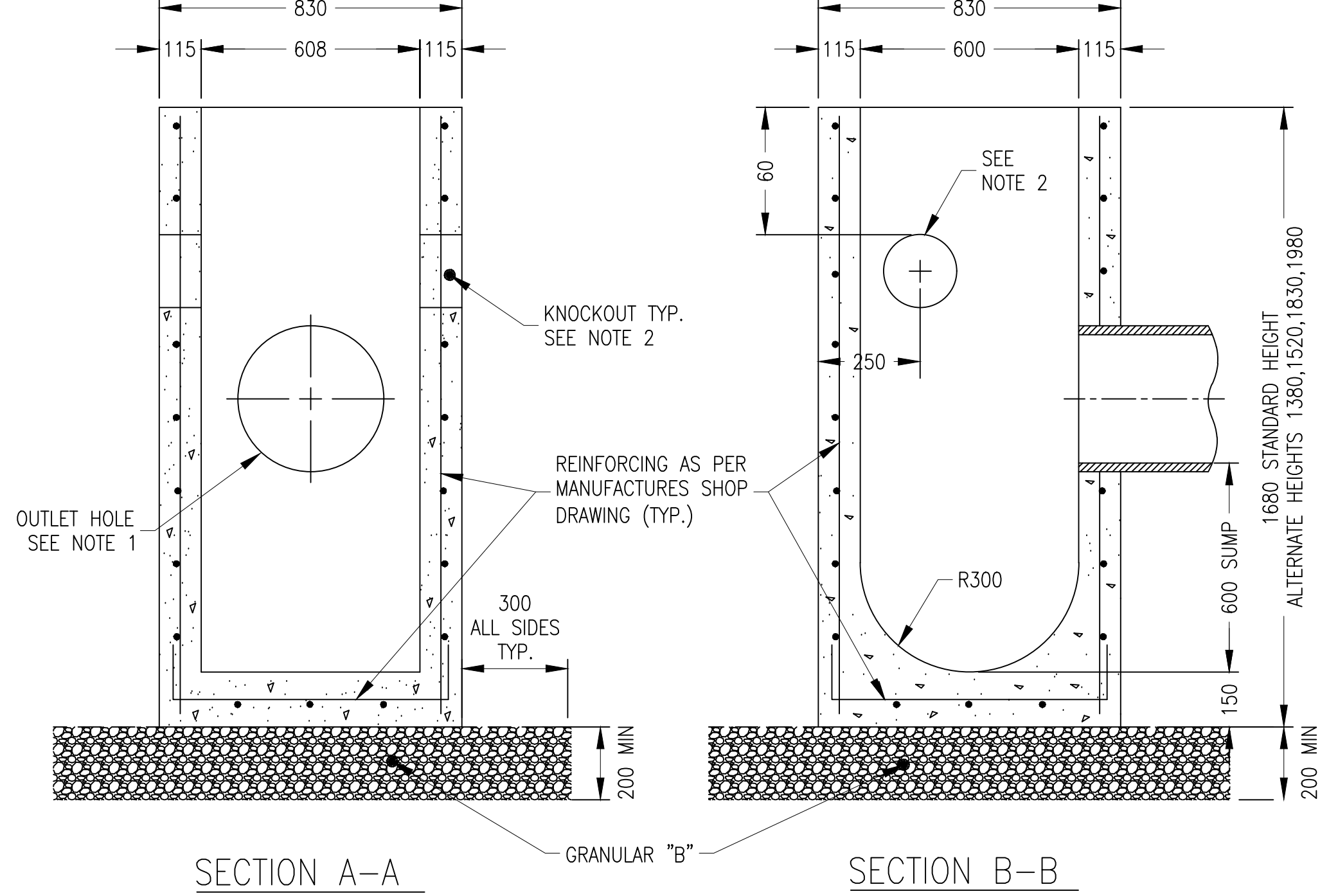
C1 CAST IRON, SQUARE FRAME WITH GRATE FOR CATCH BASINS
SCALE: NOT TO SCALE

DRAWING NUMBER 04420
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS



A1 CAST IRON, SQUARE FRAME WITH CIRCULAR CLOSED OR OPEN COVER FOR MAINTENANCE HOLES
SCALE: NOT TO SCALE

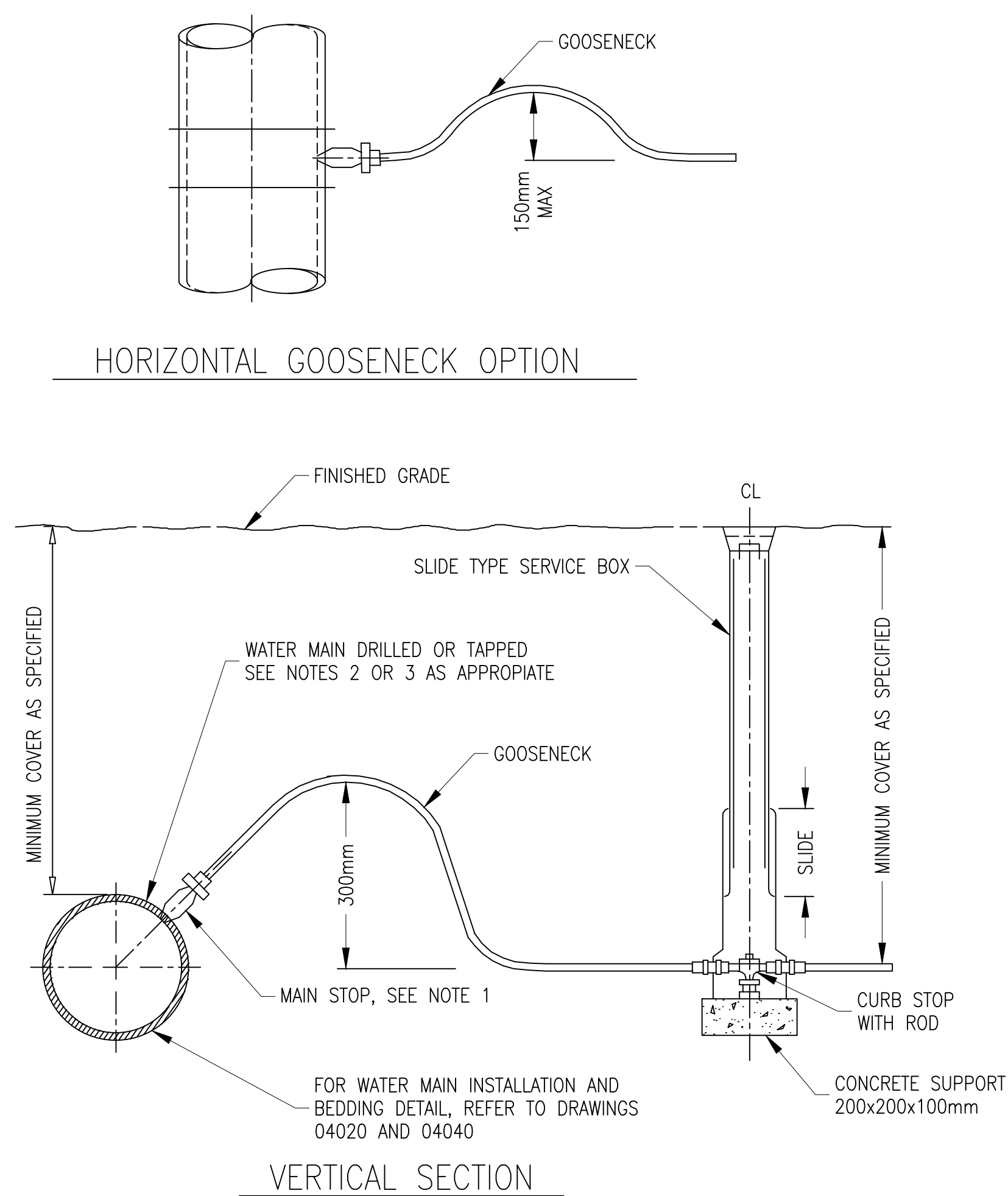
DRAWING NUMBER 04330
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS



A4 PRECAST CONCRETE CATCH BASIN
SCALE: NOT TO SCALE

DRAWING NUMBER 04350
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

APPROVED		DATE		APPROVED	
Win B Forbes		2026.01.28		DATE	
FOR COMMANDER NAFAF		16:55:54-05:00		DESCRIPTION	
ACTIVITY		D. EVANS VIA EMAIL		SEAL	
SATISFACTORY TO DATE		01/15/2026		DATE	
DESIGNED		MAD		DRAWN	
CHECKED		DTK		CHECKED	
BRANCH MANAGER		JMW		BRANCH MANAGER	
DESIGNED		WB		DESIGNED	
FIRE PROTECTION ENGINEER		DSN		FIRE PROTECTION ENGINEER	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND	
PLANNING, DESIGN AND CONSTRUCTION		VARIABLE LOCATIONS		VARIABLE LOCATIONS	
JOINT MARITIME FACILITY		DETAILS - UTILITY		DETAILS - UTILITY	
SCALE: AS NOTED		PROJECT NO.: 1838720		CONSTR. CONTR. NO.	
NAFAF DRAWING NO. 14157096		SHEET 34 OF 170		C-506	

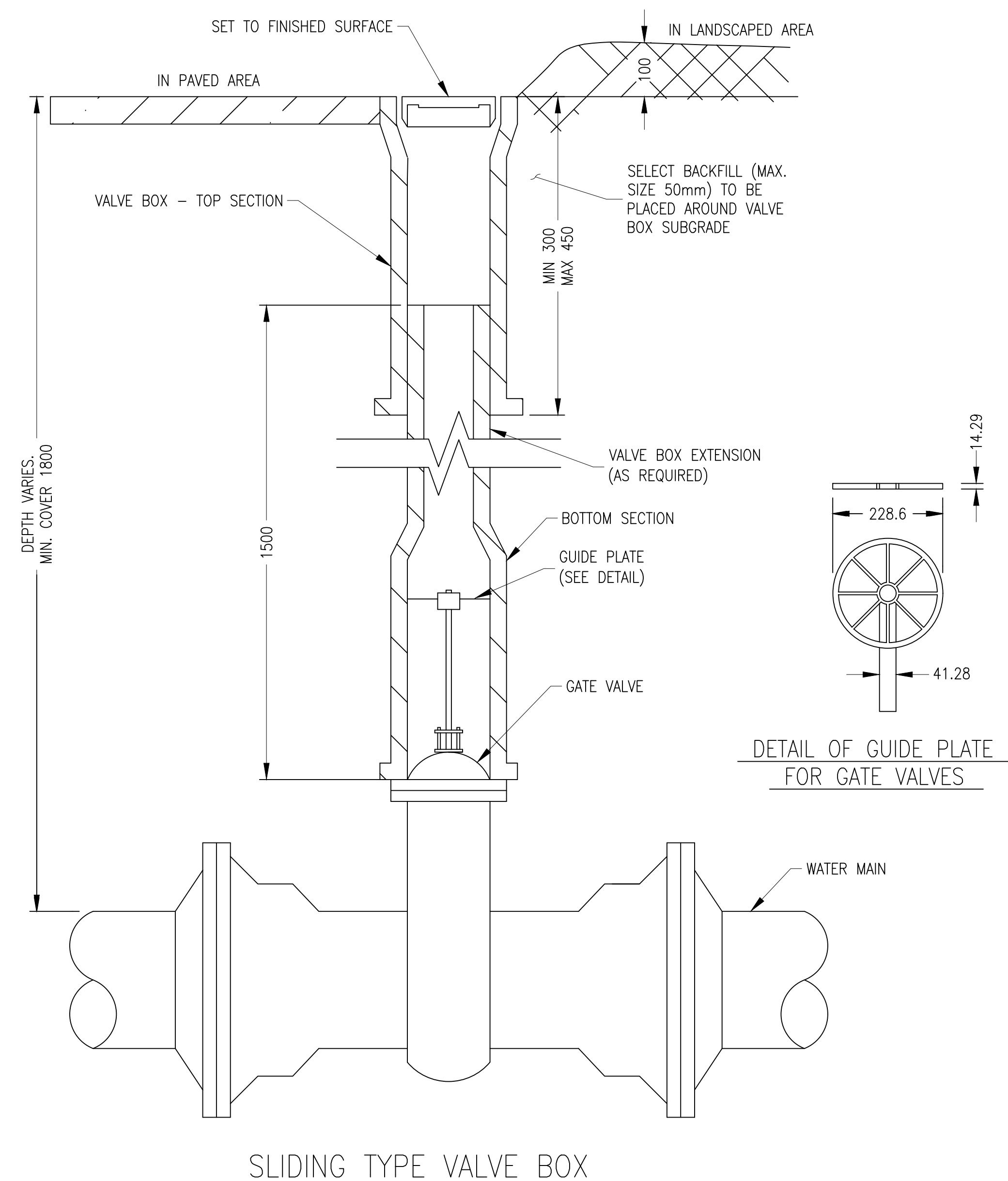


- NOTES:**
1. FOR PLASTIC SERVICE PIPE, INSTALL MAIN STOP AT 45° ABOVE HORIZONTAL WITH A MINIMUM 1.2m LONG GOOSENECK.
 2. DIRECT TAP DUCTILE IRON PIPE WITH APPROVED TOOL WITH STANDARD AWWA INLET THREAD.
 3. SERVICE CONNECTIONS TO PLASTIC WATER MAINS SHALL BE MADE USING SERVICE SADDLES OR FACTORY MADE TEES.
 4. COUPLINGS SHALL NOT BE PERMITTED UNLESS THE SERVICE LENGTH EXCEEDS 20m BETWEEN THE MAIN STOP AND CURB STOP.
 5. ALL WATER SERVICES SHALL BE INSTALLED 90° TO THE LONGITUDINAL AXIS OF THE WATER MAIN.
 6. BACKFILL MATERIAL WITHIN 500mm OF SERVICE BOX SHALL BE NATIVE OR IMPORTED, AS SPECIFIED.
 7. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.

A1 WATER SERVICE CONNECTION
SCALE: NOT TO SCALE

SCALE: NOT TO SCALE

DRAWING NUMBER 04520
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

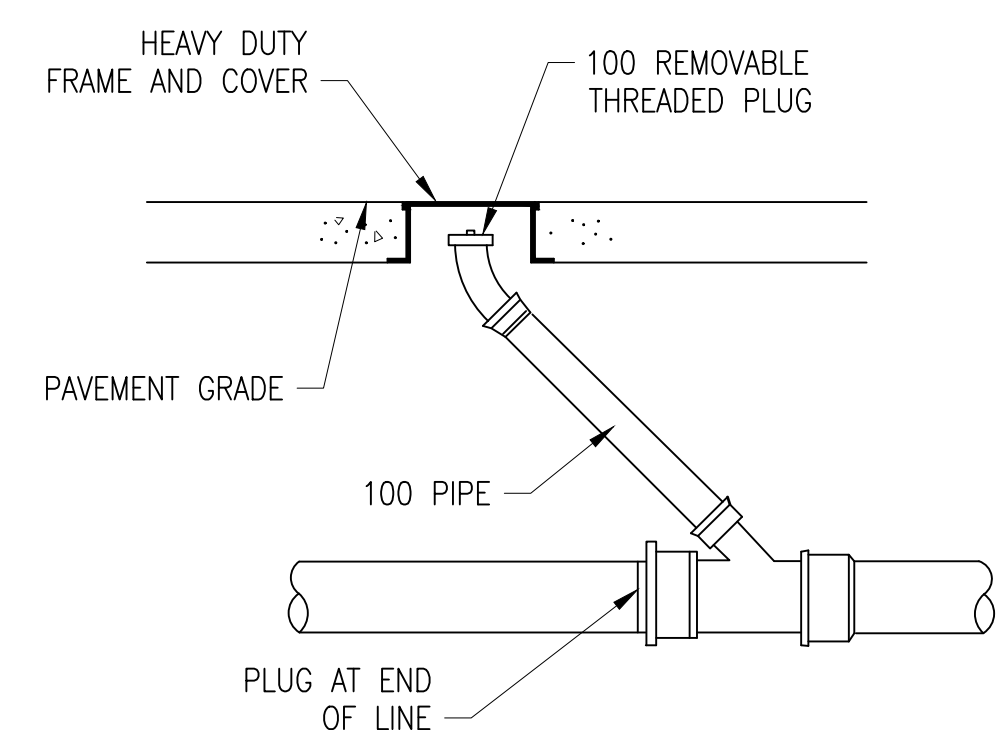


- NOTES:
1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.

A3 SLIDING TYPE VALVE BOX
SCALE: NOT TO SCALE

SCALE: NOT TO SCALE

DRAWING NUMBER 04540
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS



A5 CLEANOUT
SCALE: NOT TO SCALE

SCALE: NOT TO SCALE

[illegible]

FILE NAME: V:\DDO\N5-MultipleSta\2026_1783264-P1360_MMFB_B_Design\DWG\Sheet\05_CIV\F1360-1783264-C-501.dwg LAYOUT NAME: C-508 DETAILS - UTILITY PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: maryann.durant

D

C

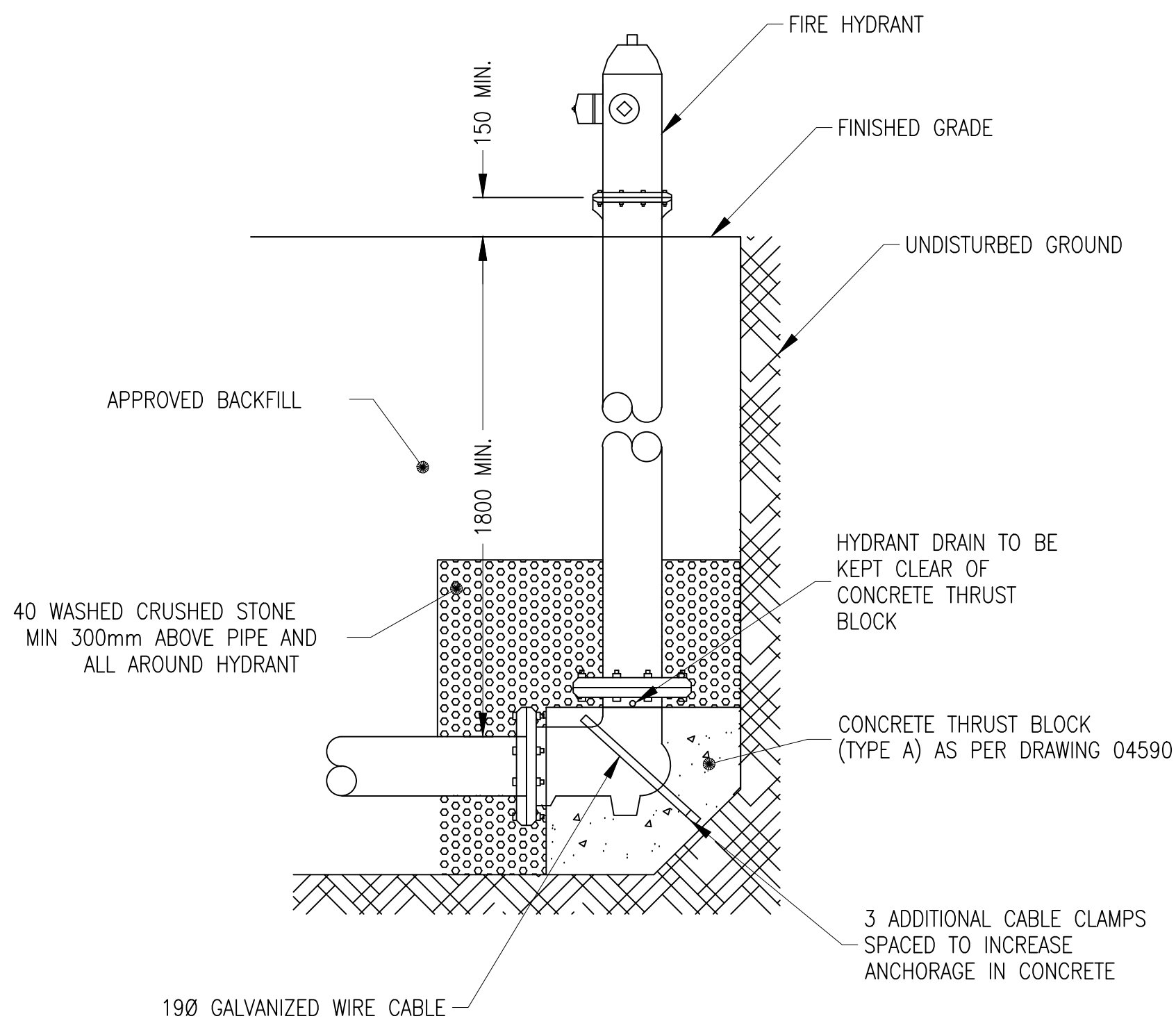
B

A

A1 TYPICAL HYDRANT DETAIL

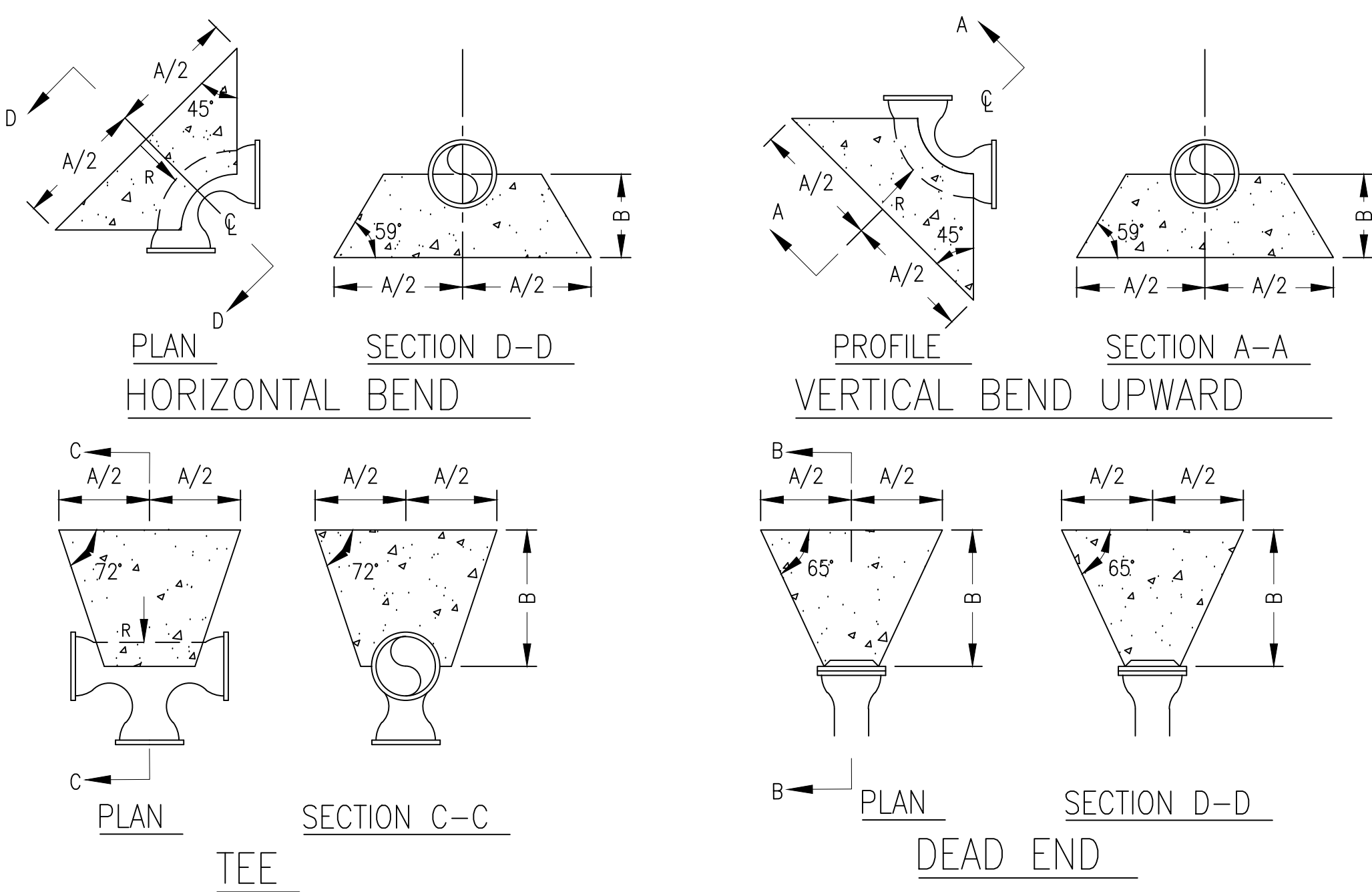
- NOTE:
- MECHANICAL JOINT RESTRAINTS CAN BE USED IN LIEU OF THRUST BLOCKS AT THE DISCRETION OF THE DESIGNER.

DRAWING NUMBER 04550
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS



A3 WATER MAIN THRUST BLOCKS TYPE 'A'

SCALE: NOT TO SCALE



ABBREVIATIONS:

R - REACTION IN 1000kg
b - MINIMUM BEARING AREA AT SOIL TO CONCRETE FACE IN m²
AREA
CONC - VOLUME OF CONCRETE IN m³
A & B - DIMENSION OF CONCRETE IN mm UNLESS OTHERWISE SHOWN

- NOTES:
- CONCRETE SHALL BE 25MPa 28 DAY STRENGTH.
 - BLOCKS SHALL BE POURED DIRECTLY AGAINST UNDISTURBED SOIL AS INDICATED. (NO UNDISTURBED SOIL SHOWN)
 - DESIGN DATA - STATIC PRESSURE OF 1000KPa.
- MINIMUM SOIL BEARING CAPACITY 120 KPa.
 - ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.

DRAWING NUMBER 04590
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

C3 WATER MAIN THRUST BLOCKS TYPE 'B'

SCALE: NOT TO SCALE

DRAWING NUMBER 04600
REFERENCE: NEWFOUNDLAND, LABRADOR
MUNICIPAL MASTER SPECIFICATIONS

NOMINAL DIAMETER (mm)	EFFECTIVE AREA (m ²)	BENDS --- HORIZONTAL & VERTICAL --- UP																				TEE & DEAD END					
		90°					45°					22 1/2°					11 1/4°										
		R	b AREA	A	B	CONC	R	b AREA	A	B	CONC	R	b AREA	A	B	CONC	R	b AREA	A	B	CONC	R	b AREA	A	B	CONC	
100	0.012	1.74	0.14	375	300	0.04	0.94	0.08	300	300	0.04	0.37	0.03	300	300	0.04	0.24	0.02	300	300	0.04	1.23	0.10	375	450	0.04	
150	0.024	3.61	0.30	525	300	0.04	1.95	0.16	450	300	0.04	1.00	0.08	300	300	0.04	0.50	0.04	300	300	0.04	2.55	0.21	450	450	0.04	
200	0.042	6.21	0.51	750	450	0.19	3.36	0.27	525	450	0.04	1.71	0.14	375	450	0.04	0.86	0.07	300	450	0.04	4.39	0.36	600	450	0.08	
250	0.063	9.21	0.75	900	450	0.19	5.03	0.41	675	450	0.08	2.61	0.21	450	450	0.04	1.29	0.11	375	450	0.04	6.58	0.54	750	450	0.19	
300	0.088	13.24	1.09	1100	450	0.38	7.12	0.58	750	450	0.19	3.65	0.30	525	450	0.04	1.84	0.15	375	450	0.04	9.34	0.76	900	525	0.19	

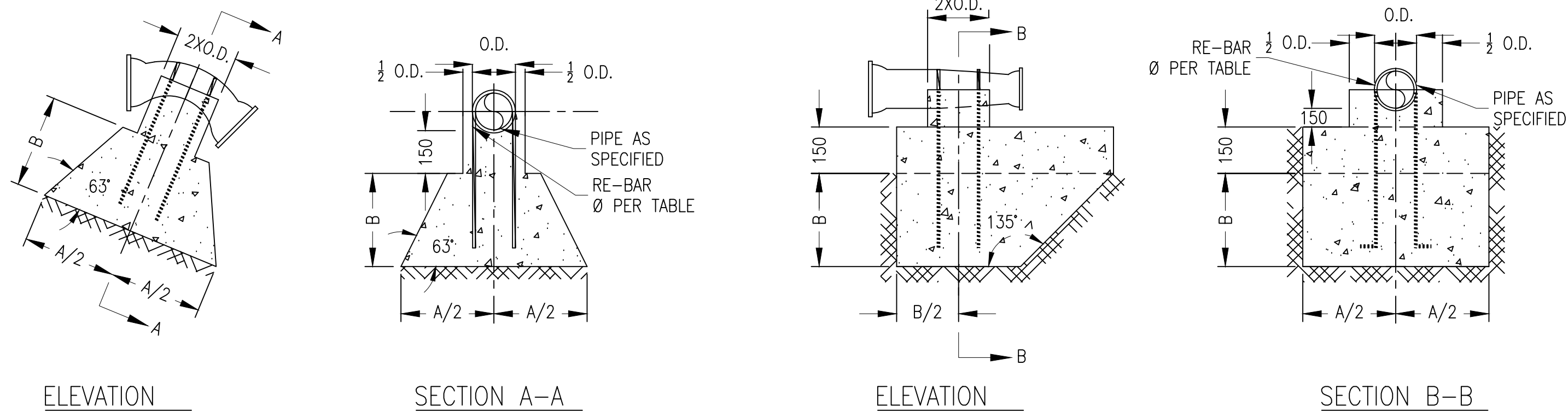
NOTES:

- CONCRETE SHALL BE 25 MPa 28 DAY STRENGTH.
- 200 MICRON THICK POLYETHYLENE TO BE PLACED AROUND FITTING AND BETWEEN THRUST BLOCK.
- RE-BARS, REINFORCING STEEL STRUCTURAL GRADE 125 MPa MINIMUM WORKING STRESS WHEN EXPOSED TO SOIL BARS SHALL BE COATED WITH ASPHALT PAINT TO PREVENT CORROSION.
- BLOCKS SHALL BE POURED DIRECTLY AGAINST UNDISTURBED SOIL AS INDICATED.
- DESIGN DATA - STATIC PRESSURE 1000KPa.
- ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED.

ABBREVIATIONS:

R - REACTION IN 1000 Kg.
CONC. VOLUME OF CONCRETE IN m³.
A & B DIMENSION OF CONCRETE IN mm UNLESS OTHERWISE NOTED.

NOMINAL DIAMETER (mm)	EFFECTIVE AREA (m ²)	VERTICAL BENDS DOWN																		NOMINAL DIAMETER (mm)	EFFECTIVE AREA (m ²)	REDUCERS						
		90°					45°					22 1/2°					11 1/4°					R	RE-BAR	A	B	CONC		
		R	RE-BAR	A	B	CONC	R	RE-BAR	A	B	CONC	R	RE-BAR	A	B	CONC	R	RE-BAR	A								B	CONC
100	0.012	1.74	15M	1150	525	0.75	0.94	15M	900	450	0.38	0.37	15M	750	450	0.19	0.24	15M	750	450	0.19	100 X 150	0.012	1.32	15M	900	450	0.57
150	0.025	3.61	20M	1525	750	1.72	1.95	20M	1225	600	0.98	1.00	15M	900	450	0.38	0.50	15M	900	450	0.38	150 X 200	0.017	1.84	15M	900	600	0.57
200	0.042	6.21	25M	1825	900	2.87	3.36	20M	1450	675	1.53	1.71	15M	1075	525	0.76	0.86	15M	900	450	0.38	150 X 250	0.038	4.03	15M	1200	750	1.15
250	0.063	9.21	25M	2050	975	4.20	5.03	25M	1675	750	2.29	2.61	20M	1375	675	1.15	1.29	15M	1075	525	0.57	150 X 300	0.064	6.80	20M	1200	900	1.34
300	0.088	13.24	25M	2275	1150	5.92	7.12	25M	1900	900	3.25	3.65	20M	1525	750	1.72	1.84	20M	1150	525	0.76	200 X 300	0.047	4.97	15M	1200	750	1.15



APPROVED: Digitally signed by Win B Forbes Date: 2026.01.28 16:55:54-05'00'

ACTIVITY

D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES: MAD DRW: DTK CHK: JWN

PM/DM: MJD/ATP

BRANCH MANAGER: JMW

DES PROJ DIR: WBF

FIRE PROTECTION ENGINEER: DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC

PLANNING, DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

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SCALE: NOT TO SCALE



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DESIGNED BY	DATE	BY	CHKD BY	DATE
MAD	DRW	DTK	CHK	JWN

JMW

PROTECTION ENGINEER DSN

51

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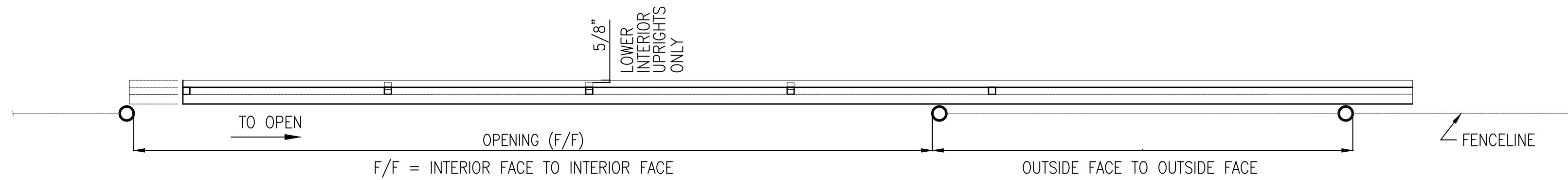
STR. CONTR. NO.

FAC DRAWING NO.
11157000

37 OF 170

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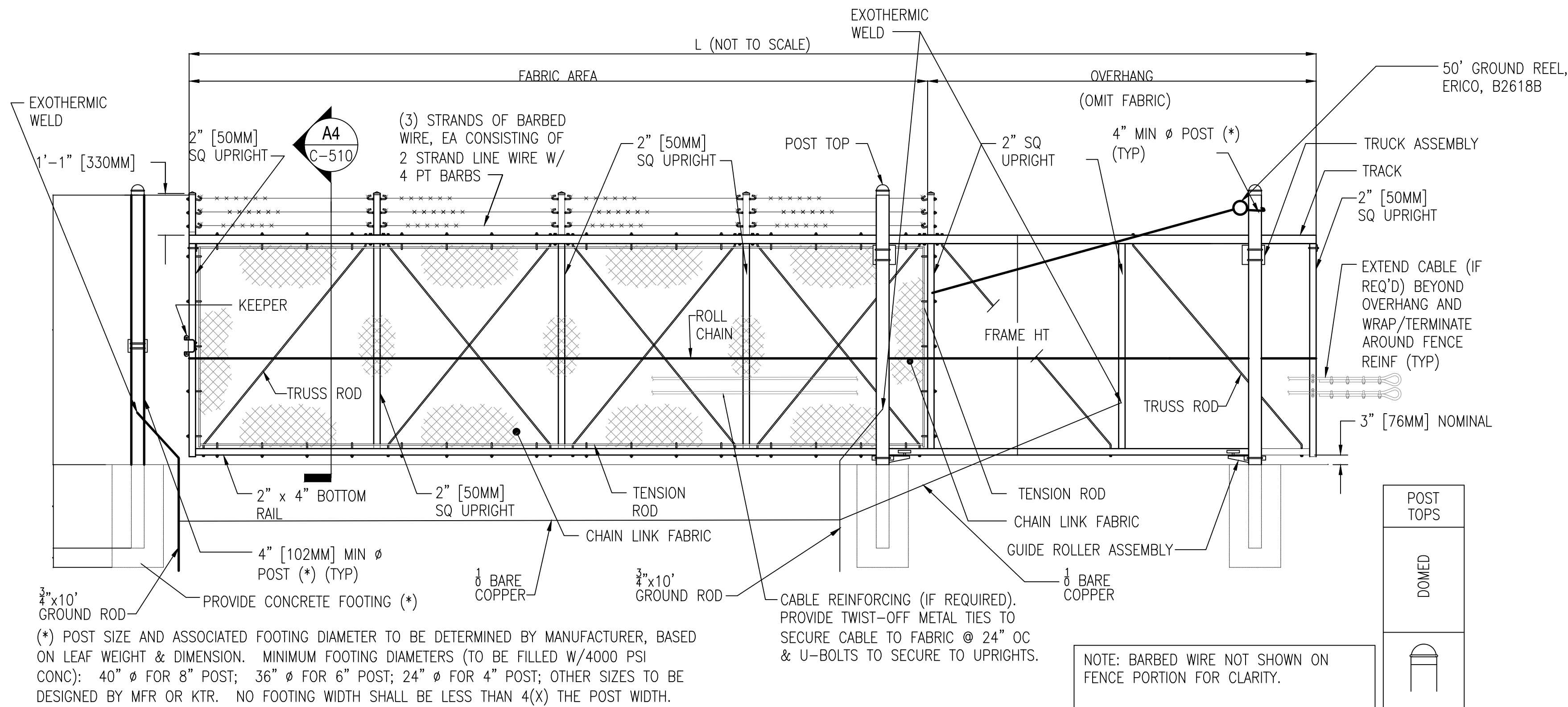
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C1 GATE PLAN
NOT TO SCALE

SAMPLING OF SIZES:

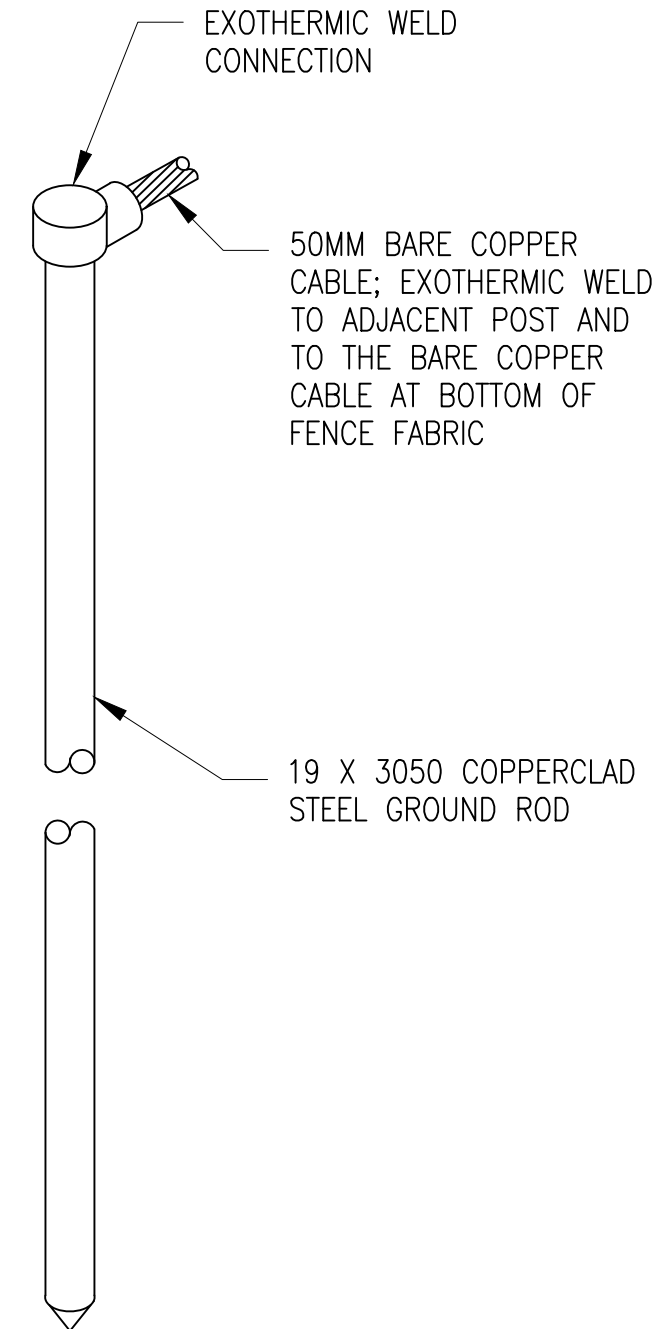
OPENING (F/F)	NOMINAL HEIGHT (H)	LENGTH (L)	GATE POSTS (G)	BAYS / OVERHANG
37'-0"	8'-0"	53'-3"	16'-0"	5 (FABRIC) / 2 (W/O)
30'-0"	8'-0"	42'-3"	12'-0"	4 (FABRIC) / 2 (W/O)
24'-0"	8'-0"	36'-3"	12'-0"	3 (FABRIC) / 2 (W/O)



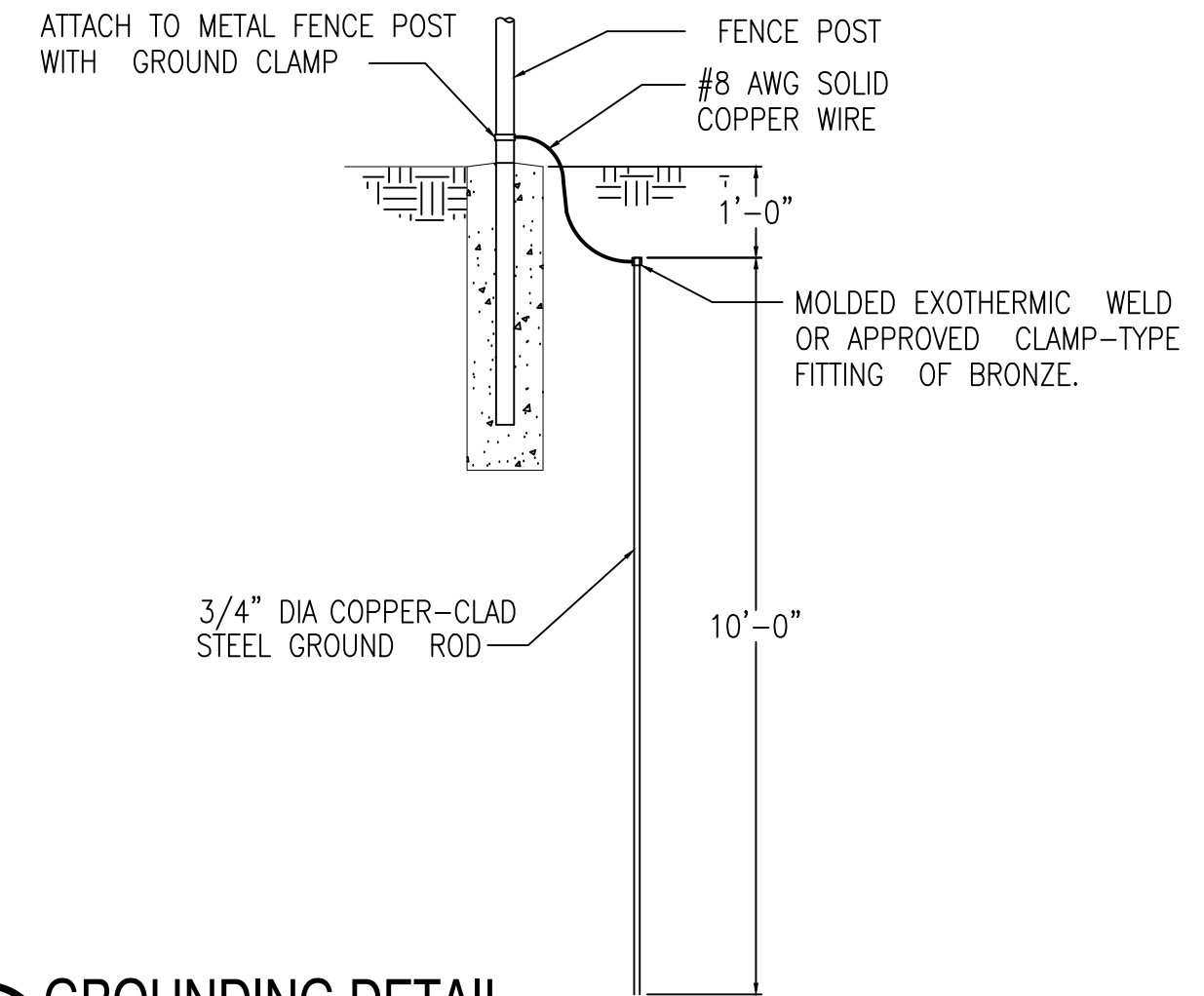
A1 TYPICAL SINGLE CANTILEVERED SLIDING GATE ELEVATION- (4) BAYS SHOWN
SCALE: NOT TO SCALE

WITH ROUND POSTS & WITH BARBED WIRE

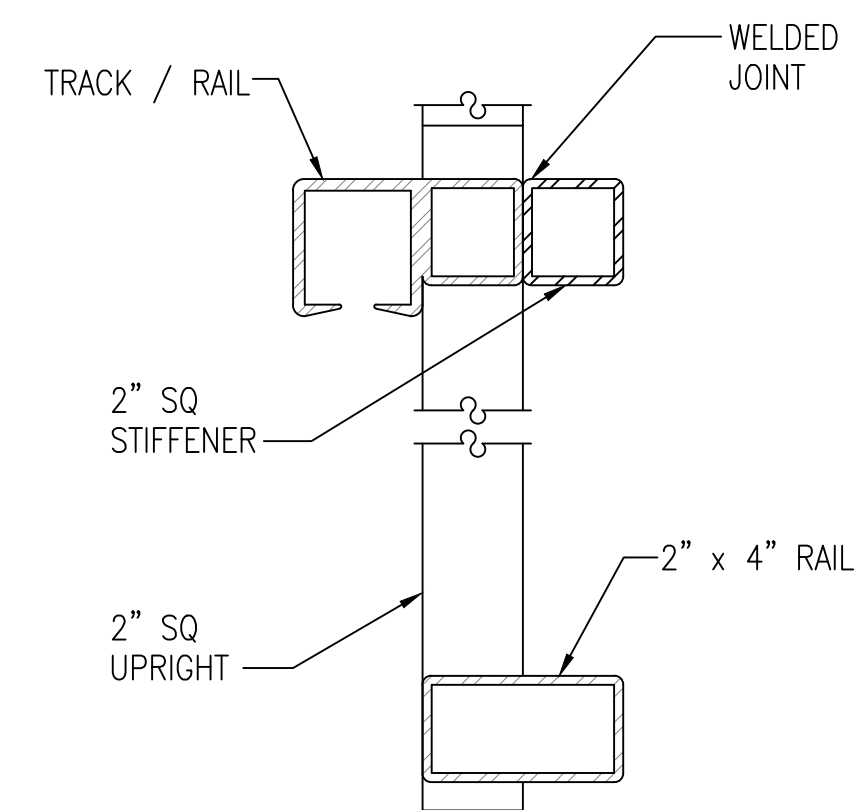
NOTE: BARBED WIRE NOT SHOWN ON FENCE PORTION FOR CLARITY.



C4 GROUND ROD CONNECTION DETAIL
SCALE: NOT TO SCALE



B4 GROUNDING DETAIL
SCALE: NOT TO SCALE



A4 SECTION (TYP)
NOT TO SCALE



APPROVED: **D. EVANS VIA EMAIL**
DATE: **01/15/2026**
DESIGNED: **MAD** DRAWN: **DTK** CHECKED: **JWN**
BRANCH MANAGER: **JMW**
DESIGN PRODUCTION: **WBF**
FIRE PROTECTION ENGINEER: **DSN**

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
DETAILS - FENCING

SCALE: **AS NOTED**
PROJECT NO.: **1838720**
CONSTRUCTION CONTRACT NO.:
NAVFAC DRAWING NO.: **14157100**
SHEET **38** OF **170**
C-510

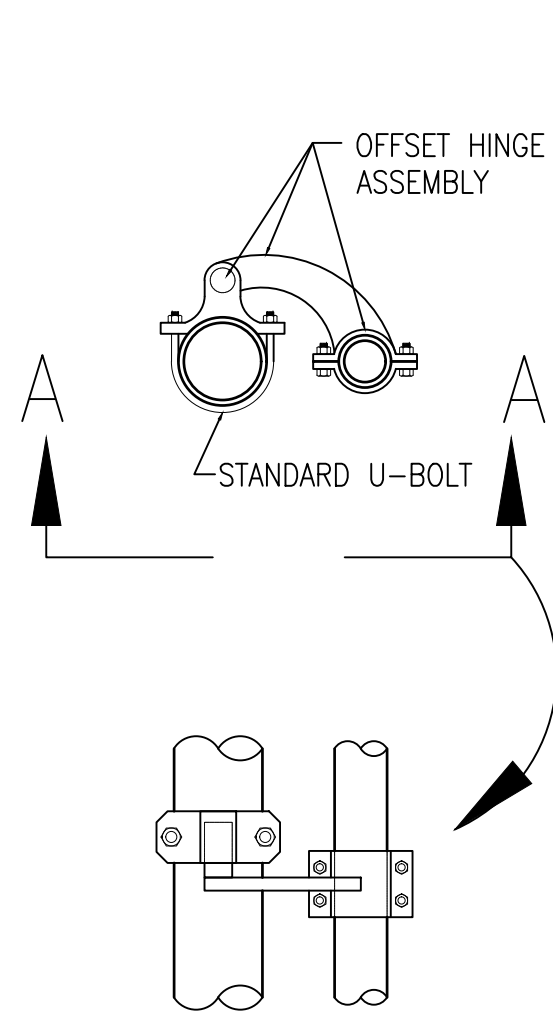
FILE NAME: V:\DDONUS\Multiples\A 2026_1783264_P1360_MMF\B_Design\DWG\Sheet\05_CIV\F1360-1783264-C-501.dwg LAYOUT NAME: C-511 DETAILS - FENCING PLOTTED: Tuesday, January 20, 2026 - 10:46am USER: maryam.durant

CHAIN LINK FENCING NOTES

- FABRIC:** THE STANDARD FENCE FABRIC SHALL BE VINYL, ZINC OR ALUMINUM-COATED STEEL WIRE CHAIN LINK WITH MESH OPENINGS NOT LARGER THAN TWO INCHES PER SIDE AND A TWISTED AND BARBED SELVAGE AT TOP AND BOTTOM IN ACCORDANCE WITH THE SPECIFICATIONS. UTILIZE 6-GAUGE FOR BASE PERIMETER OR HEIGHTENED SECURITY ZONES AND 9-GAUGE FOR BASE INTERIOR OR WHEN JOINING AN EXISTING FENCE WHICH IS ALREADY 9-GAUGE.
- FABRIC TIES:** ONLY 12-GAUGE STEEL TIES SHALL BE USED. COATING OR PLATING WILL BE ELECTROLYTICALLY COMPATIBLE WITH THE FENCE FABRIC TO INHIBIT CORROSION.
- REINFORCEMENT:** TENSION WIRES SHALL BE INSTALLED AND INTERWOVEN (OR AFFIXED WITH FABRIC TIES) ALONG THE TOP & BOTTOM OF THE FENCE FOR STABILIZATION OF THE FENCE FABRIC.
- FENCE HEIGHT:** CHAIN LINK FABRIC SHALL BE 7' HIGH WITH AN ADDITIONAL 1' IN HEIGHT COMPOSED OF 3 STRANDS OF BARBED WIRE AS REQUIRED. THE TOTAL FENCE HEIGHT SHALL BE 8'.
- GROUND CLEARANCE:** BOTTOM OF THE FENCE FABRIC SHALL BE WITHIN TWO INCHES OF FIRM SOIL.
- TOP GUARDS:** A TOP GUARD IS AN OVERHANG OF BARBED WIRED ALONG THE TOP OF A FENCE, FACING OUTWARD (AWAY FROM PROTECTED SITE) AND UPWARD AT APPROX. 45° ANGLE. TOP GUARD SUPPORTING ARMS WILL BE PERMANENTLY AFFIXED TO THE TOP OF FENCE POSTS TO INCREASE THE OVERALL HEIGHT OF THE FENCE AT LEAST 1 FOOT. THREE STRANDS OF 12-GAUGE BARBED WIRE, EQUALLY SPACED, SHALL BE INSTALLED ON THE SUPPORTING ARMS.
- FENCE POSTS:** SHALL BE ASTM F1043 OR F1083 ROUND PIPE OR SQUARE TUBE AND SHALL BE GALVANIZED IN ACCORDANCE WITH THE SPECIFICATIONS. FENCE POST SPACING AND SIZE (DIAMETER) SHALL BE DETERMINED IN ACCORDANCE WITH CHAIN LINK FENCE MANUFACTURERS' INSTITUTE (WLG 2445). SPACING SHALL NOT EXCEED 10'-0" OC. SIZE (DIAMETER) SHALL NOT BE LESS THAN THAT SPECIFIED.

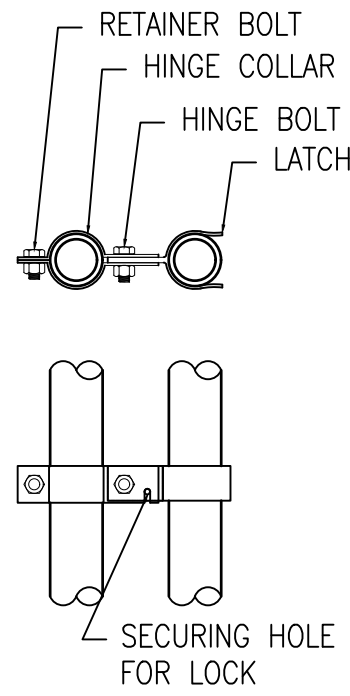
C2 OFFSET HINGE DETAIL

SCALE: NOT TO SCALE



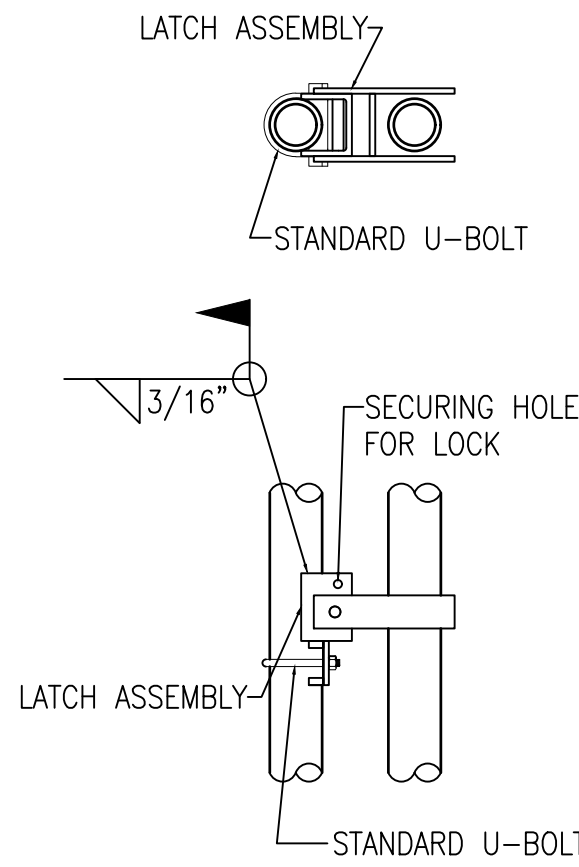
D3 GATE LATCH DETAIL

SCALE: NOT TO SCALE



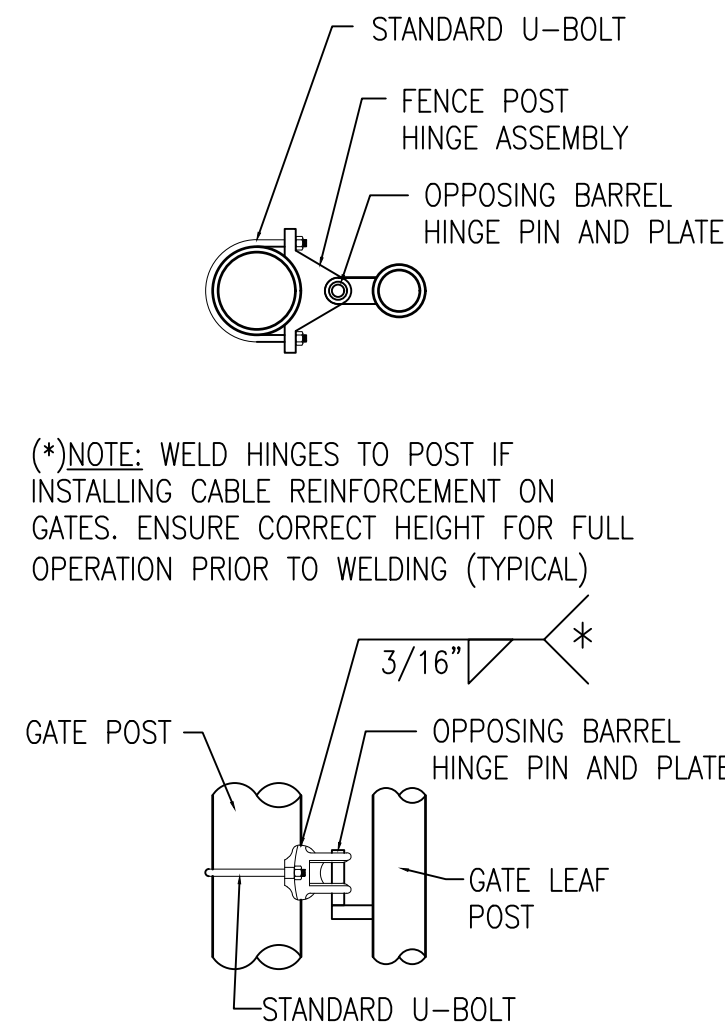
D4 FULCRUM LATCH DETAIL

SCALE: NOT TO SCALE



C3 GATE HINGE DETAIL

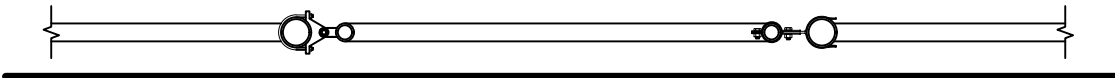
SCALE: NOT TO SCALE



(*)NOTE: WELD HINGES TO POST IF INSTALLING CABLE REINFORCEMENT ON GATES. ENSURE CORRECT HEIGHT FOR FULL OPERATION PRIOR TO WELDING (TYPICAL)

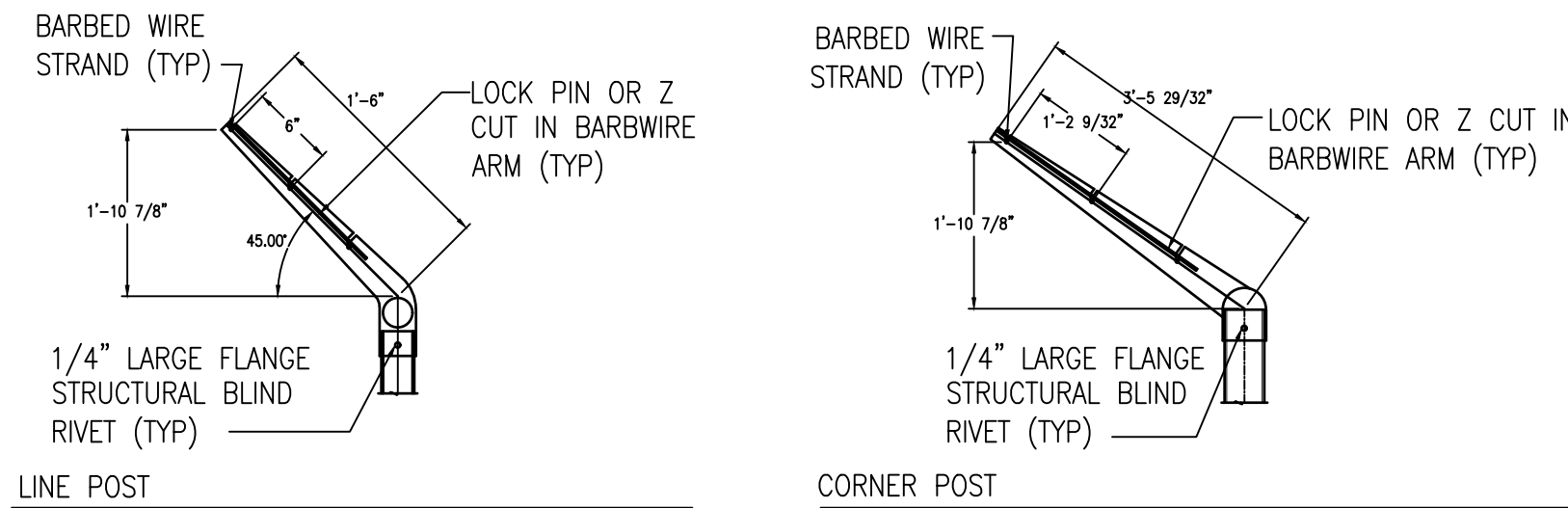
C4 PLAN SINGLE SWING GATE

SCALE: NOT TO SCALE



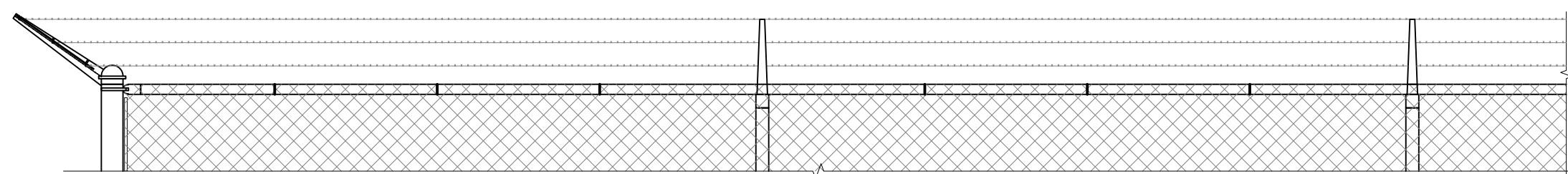
B3 SINGLE EXTENSION ARM DETAILS

SCALE: NOT TO SCALE



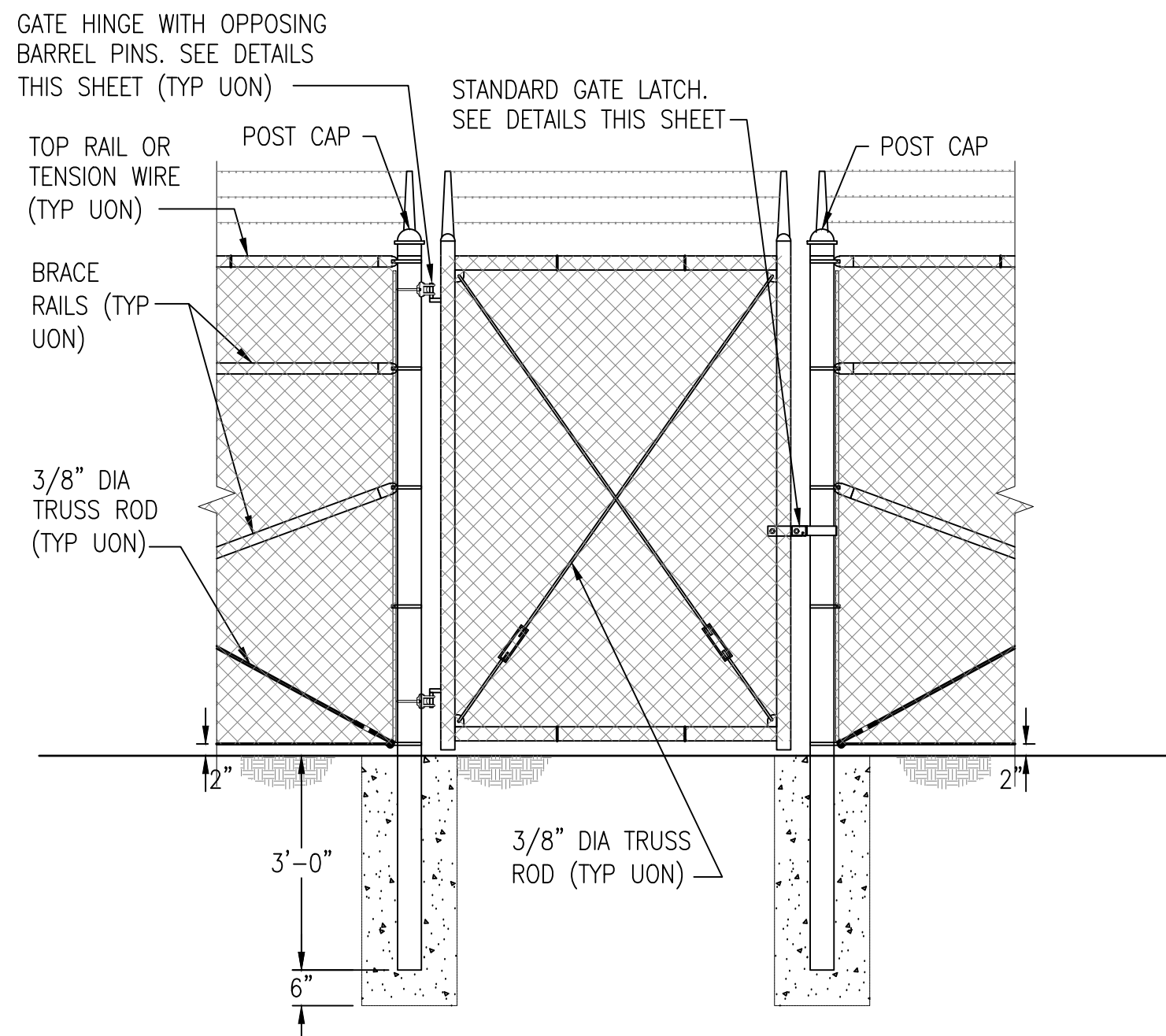
A3 TYPICAL 3 STRAND BARBED WIRE AND SINGLE EXTENSION ARM CONFIGURATION

SCALE: NOT TO SCALE



A1 TYPICAL FENCE SINGLE SWING GATE ELEVATION

SCALE: NOT TO SCALE



SINGLE OR DOUBLE LEAF GATES		
NOM HEIGHT (H)	UPRIGHT HT (U)	FRAME HT (F)
NOM HT INCLUDING BARBED WIRE	ACTUAL DIM	ACTUAL DIM
8'-0" [2438MM]	7'-10" [2388MM]	6'-8 1/2" [2045MM]

SINGLE LEAF GATES		
OPENING	GATE POSTS	HINGE SPACE (S)
FACE TO FACE	SQUARE & ROUND SIZES	POST TO UPRIGHT
3'-0" [914MM] THROUGH	2.5" [63.5MM] SQ x 3/16" TH OR 2.875" [73MM] OD	FOR SQUARE & ROUND GATE POSTS: 2 1/4" [57MM]
6'-0" [1829MM] THROUGH	4" [102MM] SQ x 3/16" TH OR 4" [102MM] OD	FOR SQUARE & ROUND GATE POSTS: 2 1/4" [57MM]
12'-0" [3657MM] THROUGH	6" [153MM] SQ x 3/16" TH OR 6.625" [168MM] OD	FOR SQUARE & ROUND GATE POSTS: 2 1/4" [57MM]
19'-0" [5790MM] THROUGH	8" [203MM] SQ x 1/4" TH OR 8.625" [219MM] OD	FOR SQUARE & ROUND GATE POSTS: 2 1/4" [57MM]

GATE POSTS & FOUNDATIONS: GATE POST SIZE AND ASSOCIATED FOOTING DIAMETER TO BE DETERMINED BY MANUFACTURER, BASED ON LEAF WEIGHT & DIMENSION, BUT NOT LESS THAN DIAMETER SHOWN ON THESE DRAWINGS. MINIMUM FOOTING DIAMETERS (TO BE FILLED W/4000 PSI CONC): 40" Ø FOR 8" POST; 36" Ø FOR 6" POST; 24" Ø FOR 4" POST; OTHER SIZES TO BE DESIGNED BY MFR OR KTR. NO FOOTING WIDTH SHALL BE LESS THAN 4(X) THE POST WIDTH.

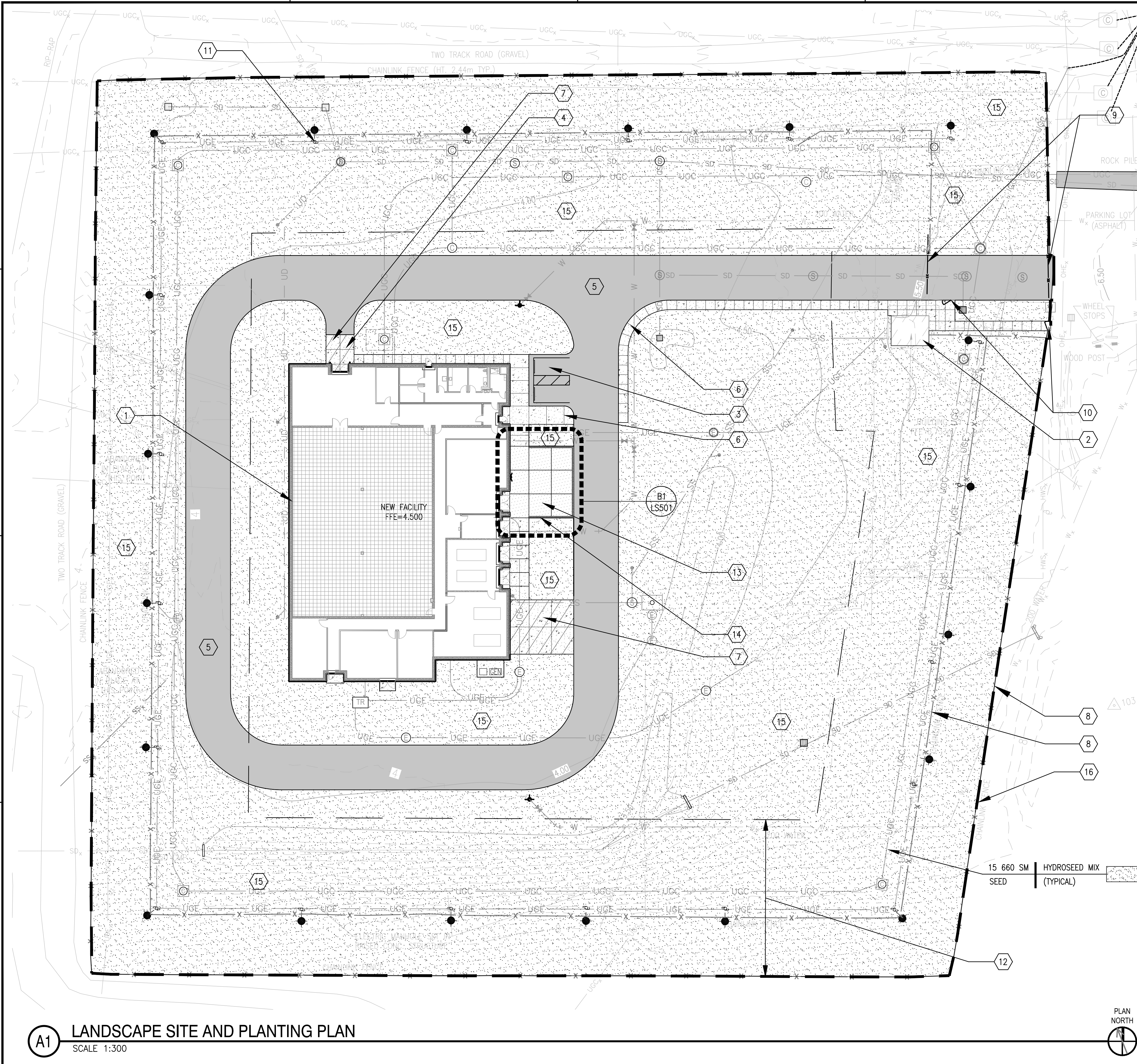
NOTE: IF GATE HINGES ARE NOT OPPOSING (AS SHOWN ABOVE) OR LEAF IS NOT LOCKED MECHANICALLY TO THE HINGES, WELD AN ANGLE, PLATE, OR BLOCK ABOVE HINGE TO RESTRICT LEAF FROM BEING REMOVED OR LIFTED OFF. RESTRICTION SHALL NOT HINDER OPERATION OF GATE.

APPROVED	DATE	APPROVED	DATE
Win B. Forbes	2026.01.28	Win B. Forbes	2026.01.28
FOR COMMANDER NAFAC			
ACTIVITY			
D. EVANS VIA EMAIL			
SUBSTANTIALITY TO DATE 01/15/2026			
DES	MAD	DRW	DTK
PM/DM	MJD/ATP	CHK	JWN
BRANCH MANAGER			
DES PROJ DIR			
FIRE PROTECTION ENGINEER			
DSN			
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC			
PLANNING, DESIGN AND CONSTRUCTION			
VARIOUS LOCATIONS			
VARIOUS LOCATIONS			
JOINT MARITIME FACILITY			
DETAILS - FENCING			
SCALE: AS NOTED			
PROJECT NO.: 1838720			
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO.			
14157101			
SHEET 39 OF 170			
C-511			

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FILE NAME: Y:\CONUS\Multiplesheets\2026_1783264_P1360_JAMFB_B_Design\DWG\Sheet\UG_LA\P1360-1783264-LS101.dwg LAYOUT NAME: LS101 PLOTTED: Friday, January 16, 2026 - 10:46am USER: nicole.welsh@nava.com

A1 LANDSCAPE SITE AND PLANTING PLAN
SCALE 1:300



GENERAL SHEET NOTES

- COORDINATE ADDITIONAL SITE WORK WITH GEOTECHNICAL, CIVIL, STRUCTURAL, ARCHITECTURAL, INTERIOR DESIGN, MECHANICAL, AND ELECTRICAL DRAWINGS.

WORK KEYNOTES

- BUILDING, REFER TO ARCHITECTURAL DRAWINGS
- EXISTING GUARD BUILDING TO REMAIN
- VEHICULAR PARKING, REFER TO CIVIL DRAWINGS
- LOADING AREA, REFER TO CIVIL DRAWINGS
- ASPHALT PAVEMENT, REFER TO CIVIL DRAWINGS
- CONCRETE PAVEMENT, REFER TO CIVIL DRAWINGS
- HEAVY DUTY CONCRETE PAVEMENT, REFER TO CIVIL DRAWINGS
- EXISTING CHAIN LINK FENCE TO REMAIN
- SLIDING SECURITY GATE, REFER TO CIVIL DRAWINGS
- PEDESTRIAN SECURITY GATE, REFER TO CIVIL DRAWINGS
- SITE LIGHTING, REFER TO ELECTRICAL DRAWINGS
- SECURITY FENCE CLEAR ZONE, REFER TO CIVIL DRAWINGS
- UTILITY YARD, REFER TO CIVIL AND MECHANICAL DRAWINGS
- PRECAST CONCRETE SCREEN WALL, SEE LS501
- HYDROSEED GRASS MIX (TYPICAL SYMBOL), SEE LS501
- LIMITS OF LANDSCAPE WORK

GRAPHIC SCALE



DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
LANDSCAPE SITE PLAN
VARIOUS LOCATIONS

APPROVED: Digitally signed by signature
Wm B Forbes
Date: 2026.01.28
FOR COMMANDER NAVAFLS: 16:55:55-05'00'

ACTIVITY
D. EVANS VIA EMAIL

SATISFACTORY TO: DATE: 01/15/2026

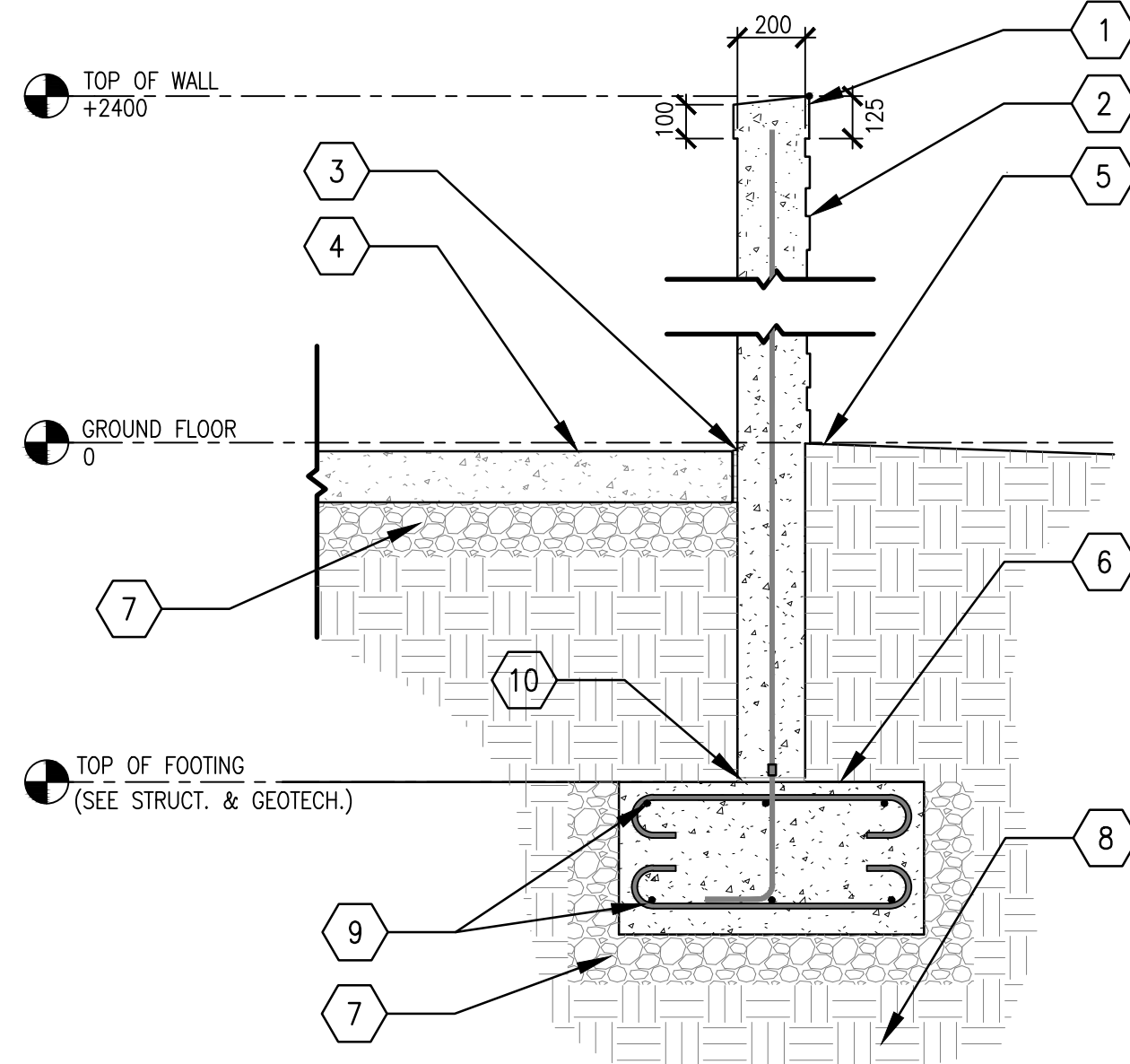
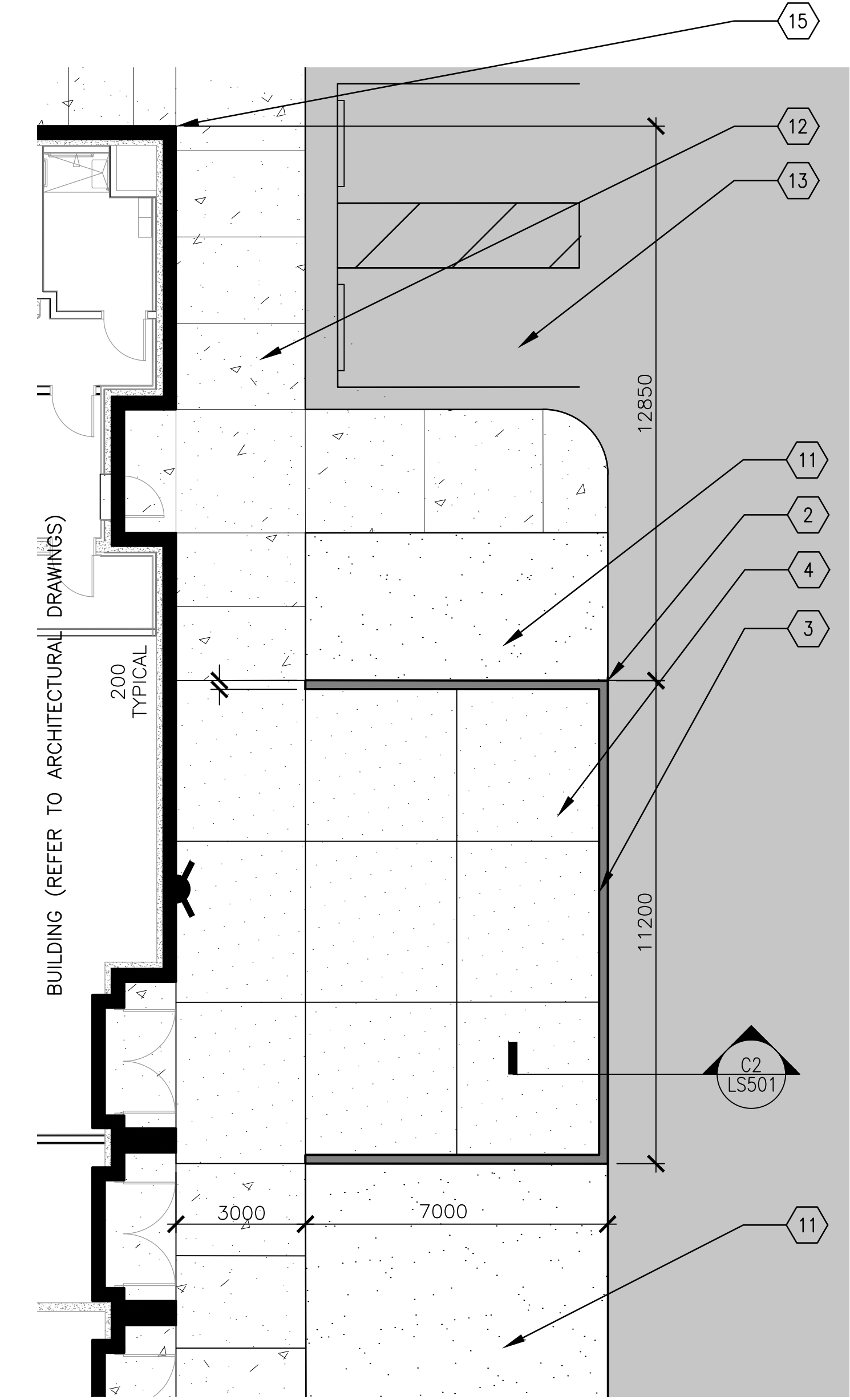
DES: NKW DRW: NKW CHK: MPO
PM/DM: MJD/ATP
BRANCH MANAGER: KMC
DES PROD DIR: WBF
FIRE PROTECTION ENGINEER: DSN

SCALE: AS NOTED
EPROJEST NO.: 1838720
CONSTR. CONTR. NO.

NAVFAC DRAWING NO.: 14157103
SHEET 41 OF 170
LS101

NAVFAC METRIC DRAWING REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\MultipleSheets\2026_1783264_P1362_JMMP\B_Design\DWG\Sheet\06_LA\P1362-1783264-LS501.dwg LAYOUT NAME: LS501 PLOTTED: Friday, January 16, 2026 - 10:48am USER: nicole.weston@nava



PLANTING NOTES

- ALL PLANTING MATERIAL MUST BE INSTALLED AS SHOWN ON THE PLANS AND IN ACCORDANCE WITH THE SPECIFICATIONS, DESIGN CRITERIA, AND ALL OTHER APPLICABLE STANDARDS.
- PLANT COUNTS ARE FOR THE CONVENIENCE OF CONTRACTOR ONLY. PROVIDE ALL NECESSARY PLANT MATERIAL TO FULFILL THE DESIGN INTENT OF THESE DRAWINGS. GET APPROVAL OF FINAL LAYOUT OF PLANT MATERIALS IN THE FIELD BY THE CONTRACTING OFFICER PRIOR TO PLANTING.
- INFORM CONTRACTING OFFICER IMMEDIATELY OF ANY CHANGED CONDITIONS OCCURRING ON-SITE WHICH ARE NOT REFLECTED ON THE PLANS.
- ALL PLANTING AREAS EXCEEDING A SLOPE OF 3:1 MUST RECEIVE EROSION CONTROL FABRIC PER SPECIFICATIONS.
- COORDINATE EXISTING AND PROPOSED UTILITIES WITHIN THE PROJECT LIMITS THAT MAY BE AFFECTED BY INSTALLATION. IMMEDIATELY CONTACT THE CONTRACTING OFFICER IF A CONFLICT IS EVIDENT.
- LEAVE THE SITE IN A CLEAN CONDITION REMOVING ALL USED MATERIALS, TRASH, AND TOOLS ON A DAILY BASIS.
- UPON COMPLETION OF ALL WORK, CONTACT THE CONTRACTING OFFICER TO ARRANGE FOR A WALK-THROUGH TO VERIFY ALL ASPECTS OF WORK ARE COMPLETED. WORK MUST BE FULLY COMPLETED ACCORDING TO ALL PLANS AND SPECIFICATIONS AND MUST BE ACCEPTED BY THE CONTRACTING OFFICER IN WRITING PRIOR TO BEGINNING THE 365-DAY ESTABLISHMENT PERIOD.
- PRIOR TO THE END OF THE ESTABLISHMENT PERIOD, CONTACT THE CONTRACTING OFFICER TO ARRANGE FOR A FINAL WALK THROUGH. ALL MAINTAINED AREAS MUST BE ACCEPTED IN WRITING PRIOR TO THE END OF THE MAINTENANCE PERIOD.

GENERAL SHEET NOTES

- COORDINATE ADDITIONAL SITE WORK WITH CIVIL, STRUCTURAL, ARCHITECTURAL, INTERIOR DESIGN, MECHANICAL, AND ELECTRICAL DRAWINGS.
- SEE CIVIL SITE PLAN FOR SITE LAYOUT DIMENSIONS.

WORK KEYNOTES

- PRECAST CONCRETE WALL CAP, SMOOTH FINISH, WITH A 6mm REVEAL AND SINGLE SLOPE TO BACK. CAP TO BE INTEGRALLY CAST WITH WALL. COLOR TO MATCH BUILDING. REFER TO ARCHITECTURAL DRAWINGS.
- 200mm WIDE PRECAST CONCRETE WALL WITH HORIZONTAL SHIPLAP WOODEN TEXTURE AND COLOR TO MATCH BUILDING, REFER TO ARCHITECTURAL DRAWINGS. PROVIDE 100mm X 200mm HORIZONTAL OPENING THROUGH WALL AT LOW POINT OF SLAB TO ALLOW FOR DRAINAGE.
- EXPANSION JOINT, REFER TO CIVIL DRAWINGS
- MECHANICAL YARD, REFER TO MECHANICAL AND STRUCTURAL DRAWINGS
- FINISH GRADE, REFER TO CIVIL DRAWINGS
- CONCRETE FOOTING, REFER TO STRUCTURAL DRAWINGS
- AGGREGATE BASE, REFER TO STRUCTURAL DRAWINGS
- COMPACTED SUBGRADE 95% TO 300mm DEEP, REFER TO STRUCTURAL DRAWINGS
- REINFORCEMENT, REFER TO STRUCTURAL DRAWINGS
- CONNECTION BY MANUFACTURER
- TURF, TYP., SEE SHEET LS101 AND THIS SHEET
- CONCRETE PAVEMENT, REFER TO CIVIL DRAWINGS
- PARKING AREA, REFER TO CIVIL DRAWINGS
- ASPHALT DRIVE, REFER TO CIVIL DRAWINGS
- POINT OF BEGINNING (BUILDING CORNER), REFER TO CIVIL DRAWINGS

APPROVED	Digitally signed by signature Date: 2026.01.28 FOR COMMANDER NAVAFLC 16:55:55-05'00'
ACTIVITY	D. EVANS VIA EMAIL
SATISFACTORY TO	DATE 01/15/2026
DES NKW	DRW NKW
PM/DM	MJD/ATP
BRANCH MANAGER	KMC
DES PROD DIR	WBF
FIRE PROTECTION ENGINEER	DSN
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PLANNING, DESIGN AND CONSTRUCTION	NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC NORFOLK, VA VARIOUS LOCATIONS VARIOUS LOCATIONS JOINT MARITIME FACILITY LANDSCAPE DETAILS
SCALE:	AS NOTED
APPROVED NO.:	1838720
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14157104
SHEET	42 OF 170
LS501	

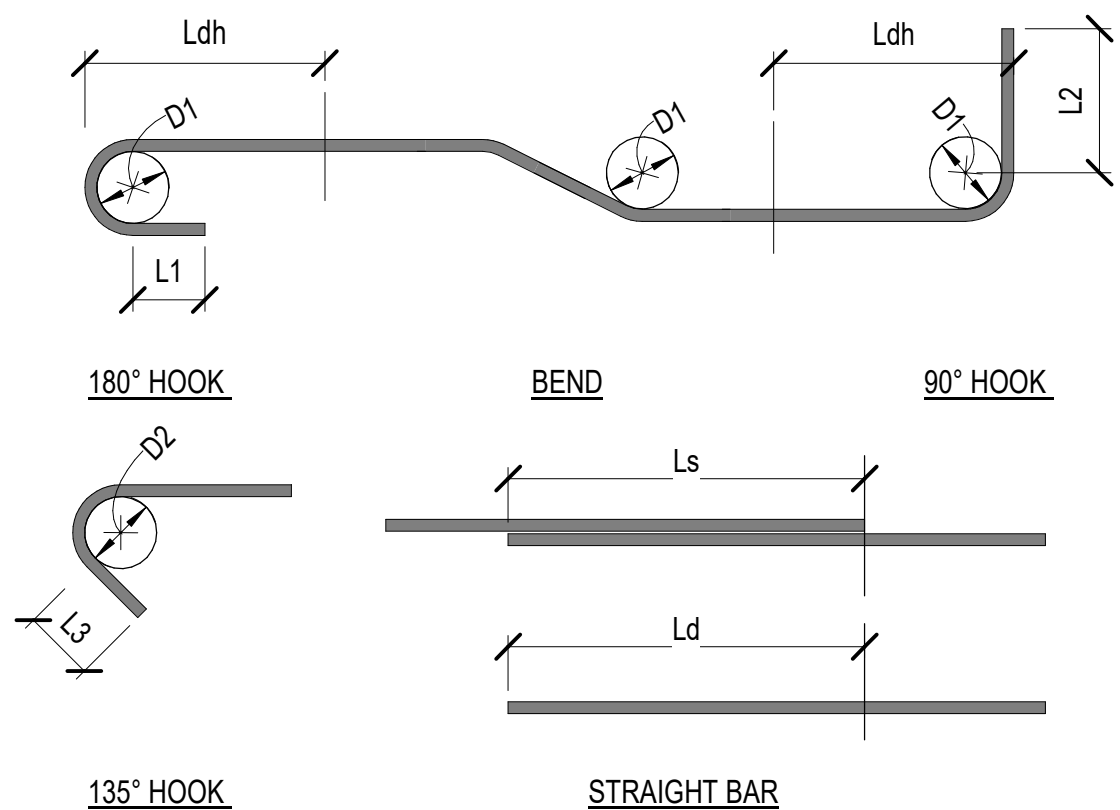
TYPICAL REINFORCEMENT BEND AND SPLICE DETAILS

W/ WHS	WITH WELD HEADED STUD
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NOTES:

- VALUES ARE BASED ON FY = 400 MPa REINFORCING BARS AND NORMAL WEIGHT CONCRETE.
- CASE 1 IS APPLICABLE WHEN ONE OF THE FOLLOWING CONDITIONS ARE SATISFIED:
 - CLEAR COVER ≥ 1.0 db, CENTER TO CENTER SPACING $\geq 2db$, AND STIRRUPS OR TIES THROUGHOUT Ld NOT LESS THAN THE CODE MINIMUM.
 - CLEAR COVER ≥ 1.0 db AND CENTER TO CENTER SPACING ≥ 3.0 Db
- CASE 2 IS APPLICABLE WHEN CASE 1 CONDITIONS ARE NOT MET.
- TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 300mm OF FRESH CONCRETE PLACED BELOW.
- FOR 35M BZARS AND SMALLER Ldh MAY BE MULTIPLIED BY 0.63 IF CENTER TO CENTER SPACING IS $\geq 6db$
- FOR 35M BARS AND SMALLER Ldh MAY BE MULTIPLIED BY 0.8 IF SIDE COVER NORMAL TO PLANE OF HOOK $\geq 6db$.
- THE TOTAL REDUCED DEVELOPMENT Ldh FOR NOTES 5 AND 6 MUST NOT BE LESS THAN THE GREATER OF 8db AND 150mm.

BAR SIZE	STANDARD 90 AND 180 DEGREE HOOKS			STIRRUPS, TIES, AND HOOPS			
	D1	L1	L2	D2	L1	L2	L3
10M	70	70	140	50	70	80	80
15M	100	80	200	65	70	100	100
20M	120	90	240	120	80	240	120
25M	150	120	300	150	100	300	150
30M	240	140	360	180	120	360	220
35M	285	160	430	220	150	430	270
45M	440	200	530	---	---	---	---
55M	565	250	680	---	---	---	---



IMMATURE
TIC
FOLK,
ON

LEGENDS AND ABBREVIATIONS

3-002

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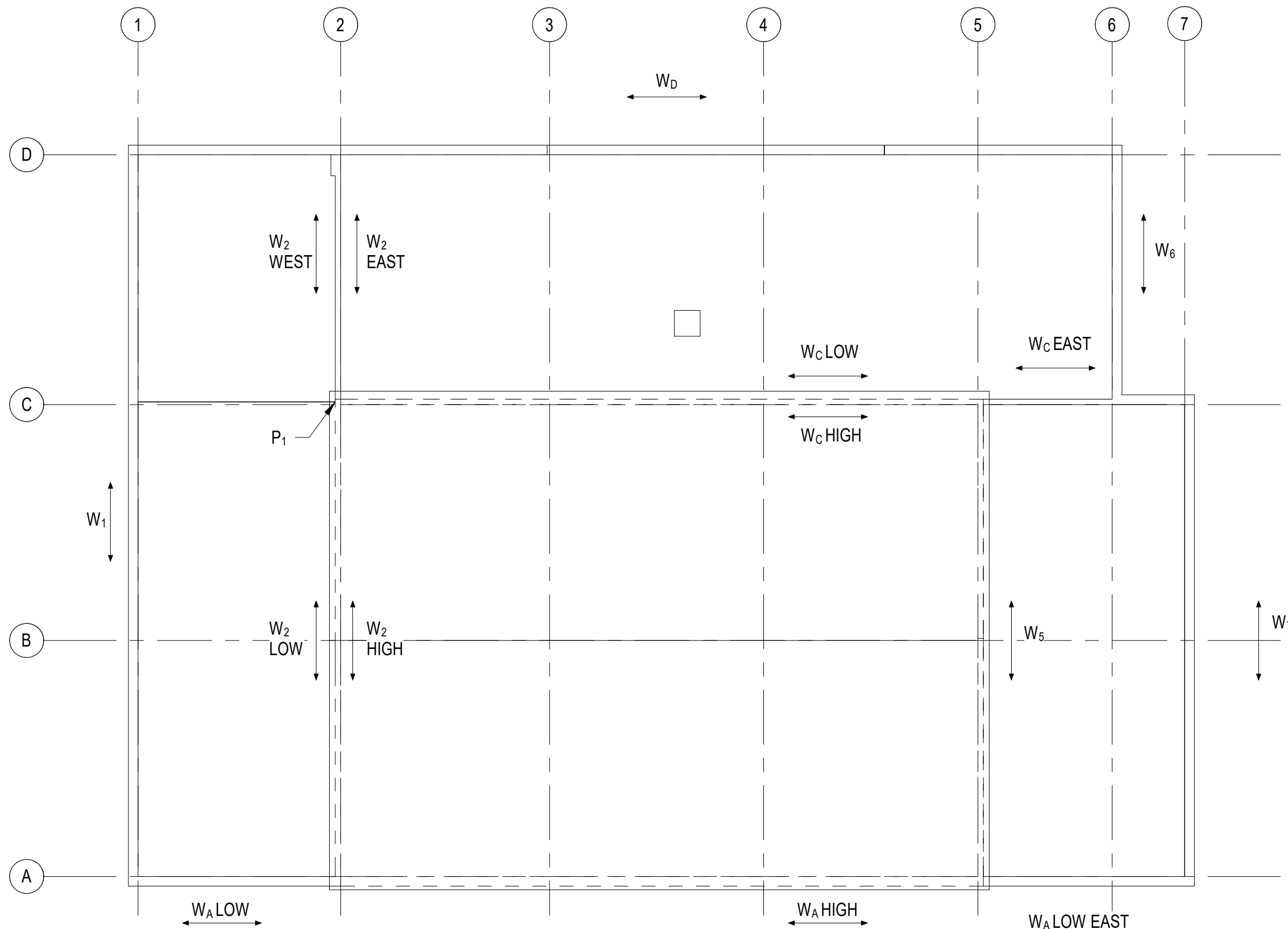
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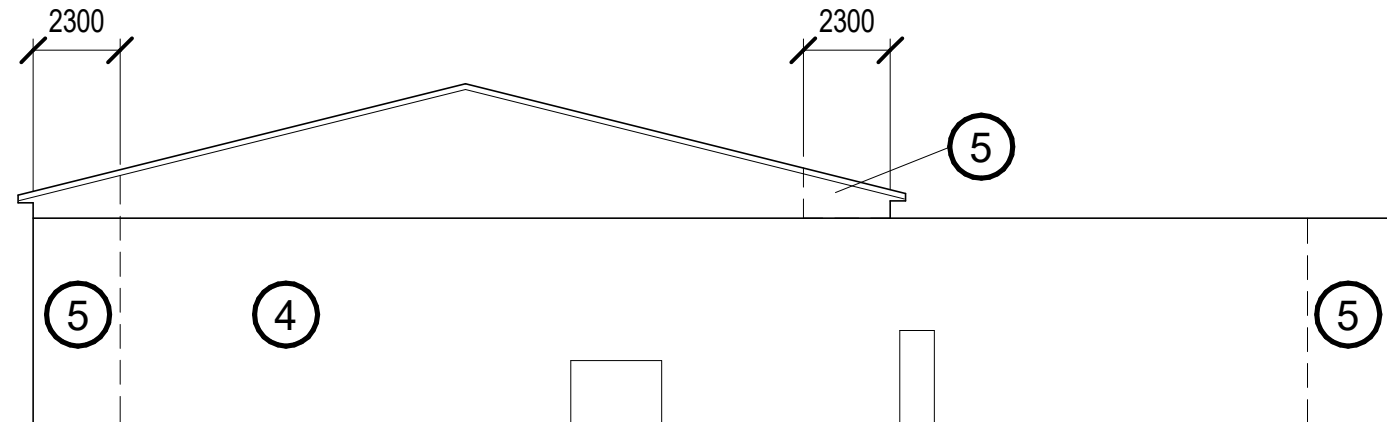
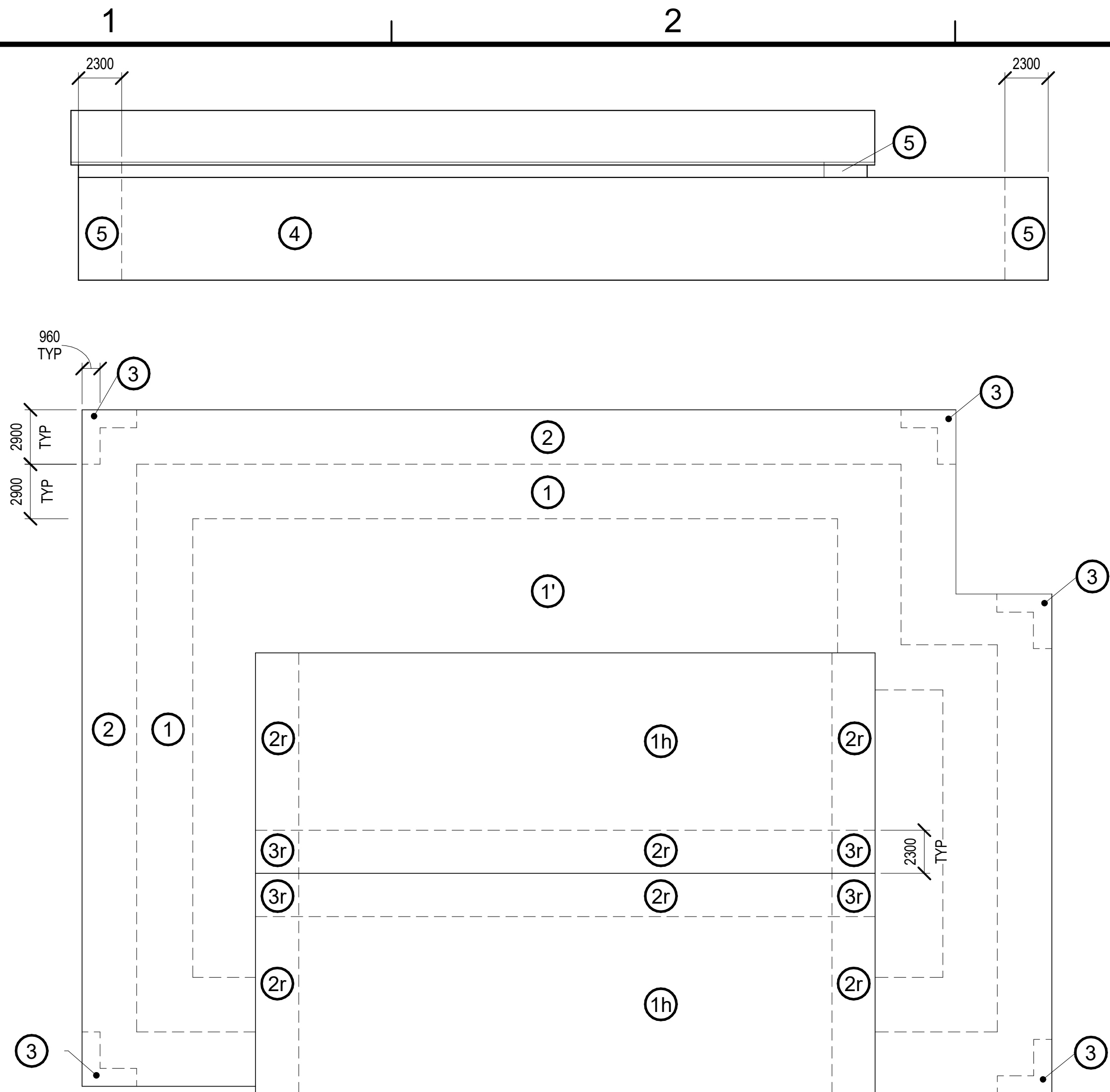
B1 COMPONENTS AND CLADDING WIND PRESSURE

NOT TO SCALE



A1 SHEAR WALL LOADING

NOT TO SCALE



WALL AND ROOF COMPONENTS AND CLADDING WIND PRESSURES						
WIND ZONE	EFFECTIVE WIND AREA					
	0.9 SM	1.9 SM	4.6 SM	9.3 SM	46.5 SM	
1	+1.11 -4.33	+1.03 -4.03	+0.94 -3.67	+0.88 -3.38	+0.88 -2.72	
1'	+1.11 -2.49	+1.03 -2.49	+0.94 -2.49	+0.88 -2.49	+0.88 -1.68	
1h	+1.81 -5.49	+1.62 -5.49	+1.39 -3.34	+1.21 -1.71	+1.21 -1.71	
2	+2.49 -5.71	+2.37 -5.33	+2.23 -4.86	+2.12 -4.49	+1.86 -3.64	
2r	+1.81 -8.01	+1.62 -6.93	+1.39 -5.49	+1.21 -4.41	+1.21 -2.97	
3	+2.49 -5.71	+2.37 -5.33	+2.23 -4.86	+2.12 -4.49	+1.86 -3.64	
3r	+1.81 -9.52	+1.62 -8.16	+1.39 -6.35	+1.21 -4.99	+1.21 -4.99	
4	+2.97 -3.22	+2.84 -3.08	+2.66 -2.92	+2.53 -2.78	+2.22 -2.47	
5	+2.97 -3.98	+2.84 -3.70	+2.66 -3.36	+2.53 -3.09	+2.22 -2.47	

NOTE:

1. WIND PRESSURES SHOWN IN UNITS OF kPa.
2. PRESSURES ARE GROSS VALUES AND DO NOT INCLUDE DEAD LOADS.

LOADING NOTES

- REFER TO GENERAL NOTES ON S-001 FOR WIND AND SEISMIC LOAD DESIGN CRITERIA.
- WIND AND SEISMIC LOADS ARE ULTIMATE VALUES PER NBC.
- DESIGN PRECAST CONCRETE SHEAR WALLS AND CONNECTIONS FOR THE NBC-2020 LOAD COMBINATION THAT PRODUCES THE MOST UNFAVORABLE EFFECT USING THE PROVIDED WIND AND SEISMIC IN-PLANE SHEAR LOADS, BEAM END REACTIONS, AND ROOF WIND LOADS TRIBUTARY TO THE WALL. THE WALL SELF-WEIGHT MUST ALSO BE INCLUDED IN THE DESIGN.
- SHEAR WALL LOADS CAN ACT IN EITHER DIRECTION AND BOTH DIRECTIONS MUST BE CONSIDERED IN THE DESIGN OF THE SHEAR WALLS.
- DESIGN PRECAST CONCRETE WALLS AND CONNECTIONS FOR THE NBC-2020 LOAD COMBINATION THAT PRODUCES THE MOST UNFAVORABLE EFFECT USING THE PROVIDED WALL WIND PRESSURE, BEAM END REACTIONS, SOIL ACTIVE PRESSURE, AND WIND ROOF PRESSURE TRIBUTARY TO THE WALL. WALL DESIGN MUST ALSO INCLUDE SELF-WEIGHT AND OUT OF PLANE SEISMIC FORCES CALCULATED PER NBC-2020.
- WALL DESIGN AND CONNECTIONS MUST ACCOUNT FOR A ECCENTRICITY OF THE BEAM END REACTIONS TAKEN AT THE FACE OF THE WALL.

BEAM SHEAR REACTION DESIGN LOADS (KN)

BEAM	DEAD	LIVE	SNOW (LINES 1&D)	SNOW (LINES 2, C, & A)
W410X39	5.6	5.0	23.5	29.6
W410X46	5.9	5.1	28.4	44.9
W530X66	10.0	8.0	32.5	49.4
W530X66 UPPER	10.5	8.9	-	30.6
W200x46	17.0	5.0		

NOTES:

- END REACTIONS ARE IN KN.
- END REACTIONS ARE UNFACTORED.
- FOR SNOW LOADING (#.#) DENOTES THE COLUMN LINEWHERE THE BEAM END REACTION OCCURS.

SHEAR WALL FORCES

WALL	WIND KN	SEISMIC KN
W1	47.2	26.3
W2 LOW	210.7	156.0
W2 HIGH	94.8	83.6
W2 WEST	63.5	43.5
W2 EAST	180	43.5
W5	227.6	170.2
W6	180	25.6
W7	47.1	26.3
WA LOW	71.2	31.7
WA HIGH	136.7	121.1
WA LOW EAST	20.2	22.4
WC LOW	214.5	188.6
WC EAST	37.4	37.1
WC HIGH	70.3	82.6
WD	96.1	42.5



APPROVED
Win B Forbes
2026.01.28
FOR COMMANDER

D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES MWW DRW MWW CHK LMM

PMCM MJD/ATP

BRANCH/MANAGER JTW

DES PROJ DR RLS

FIRE PROTECTION ENGINEER DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC

ATLANTIC PLANNING DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

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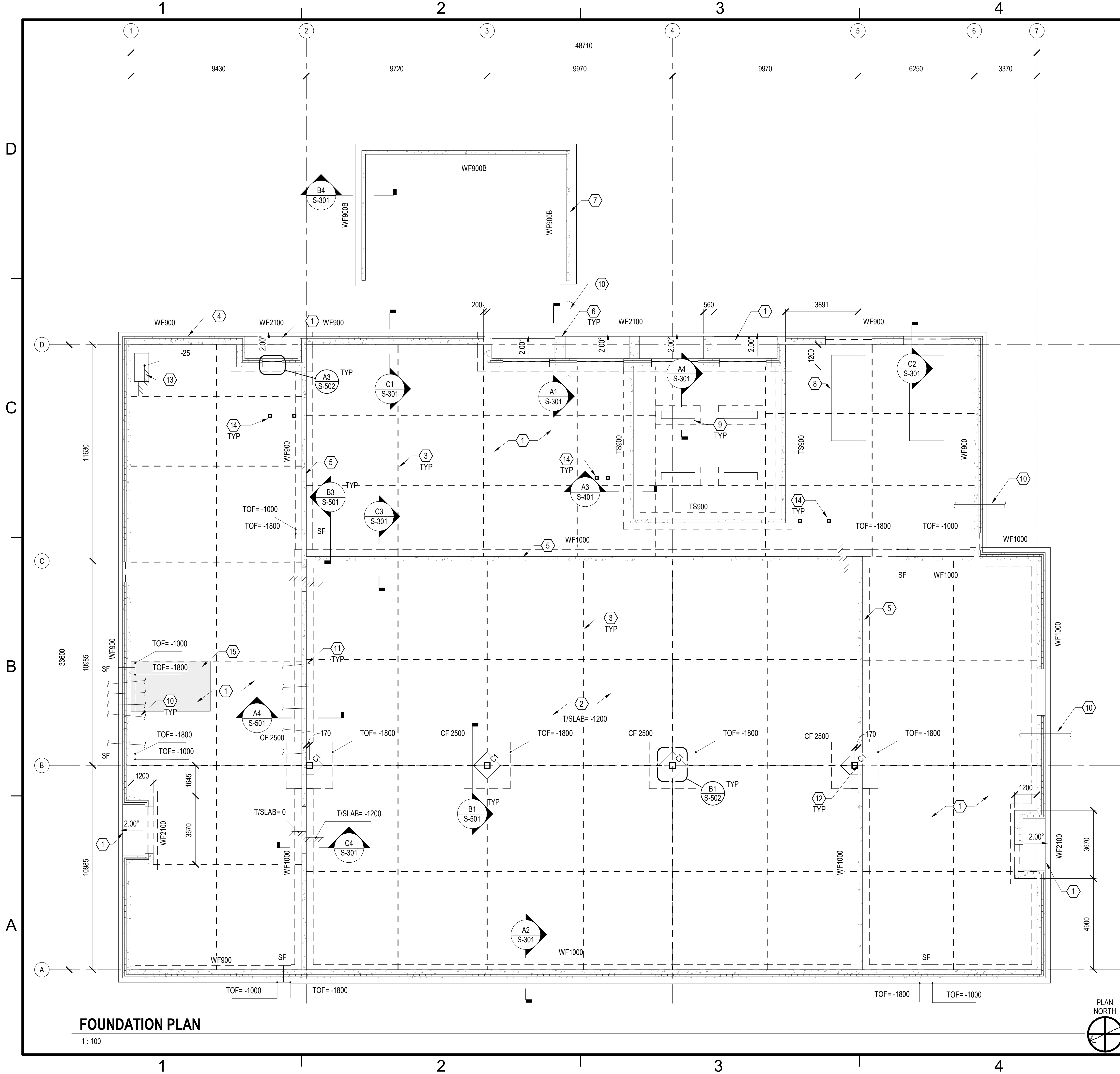
VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

VARIOUS LOCATIONS

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FOUNDATION PLAN

1: 100

FOUNDATION SHEET NOTES

- A. REFER TO S-001 FOR GENERAL NOTES AND S-501 AND S-502 FOR MISCELLANEOUS FOUNDATION DETAILS.
- B. ELEVATIONS SHOWN ON THE STRUCTURAL DRAWINGS ARE RELATIVE TO A REFERENCE ELEVATION OF 0.00 T/SLAB (TOP OF GROUND FLOOR SLAB).
- C. TOP OF FOOTING ELEVATION = -1000 UNO AS TOF. APPROXIMATE STEPPED FOOTING LOCATIONS ARE INDICATED AS "SF" ON PLAN. REFER TO TYPICAL STEPPED FOOTING DETAIL ON S-501.
- D. "CF " REFERENCES A CONCRETE COLUMN FOOTING. REFER TO SCHEDULE ON S-501.
- E. "WF " REFERENCES A CONCRETE WALL FOOTING. REFER TO SCHEDULE ON S-501.
- F. "C_" REFERENCES A STEEL COLUMN. REFER TO COLUMN SCHEDULE ON S-503.
- G. SEE ARCH FOR DIMENSIONS NOT SHOWN.

FOUNDATION KEYNOTES

- 1 150mm CONCRETE SLAB-ON-GRADE REINFORCED WITH ONE LAYER OF M10 @ 300mm OC EACH WAY REFER TO S-502 FOR DETAILS.
- 2 250mm CONCRETE SLAB-ON-GRADE REINFORCED WITH TWO LAYERS OF M10 @ 300 mm OC REFER TO S-502 FOR DETAILS. PROVIDE EQUIPOTENTIAL GROUND PLANE IN SLAB. COORDINATE WITH ELECTRICAL. SEE EG101.
- 3 PROVIDE CONSTRUCTION JOINT OR CONTROL JOINT IN AREAS INDICATED. CONSTRUCTION JOINT LOCATIONS ARE AT CONTRACTOR'S DISCRETION.
- 4 450mm THICK PRECAST CONCRETE INSULATED WALL. REFER TO S-504 FOR DETAILS.
- 5 250mm THICK PRECAST CONCRETE WALL. REFER TO S-504 FOR DETAILS.
- 6 SOLID PRECAST CONCRETE WALL/JAMB.
- 7 200mm THICK PRECAST CONCRETE FENCE WALL. REFER TO S-301 FOR DETAILS AND LA DWGS FOR LOCATION.
- 8 150mm EQUIPMENT PAD, SEE DETAIL ON S-502, COORDINATE SIZE WITH MANUF.
- 9 TANK SADDLE SUPPORT PIER, COORDINATE WITH TANK SADDLE LOCATION.
- 10 APPROXIMATE UTILITY LOCATION. COORDINAT EXACT LOCATION WITH CIVIL, PLUMBING, AND ELECTRICAL DRAWINGS. PROVIDE PIPE SLEEVES IN PRECAST WALLS AS REQUIRED.
- 11 APPROXIMATE LOCATION OF CONDUIT, COORDINATE EXACT LOCATION WITH ELECTRICAL.
- 12 PROVIDE SUFFICIENT OPENING IN WALL FOR BASEPLATE INSTALLATION.
- 13 DROPPED SLAB SEE ARCH. COORDINATE LOCATION WITH SHOWER.
- 14 LADDER SUPPORT/DOOR JAMB, SEE DETAIL ON S-504.
- 15 EQUIPOTENTIAL GROUND PLANE COORDINATE WITH ELECTRICAL, SEE EG101.

GRAPHIC SCALE(S)

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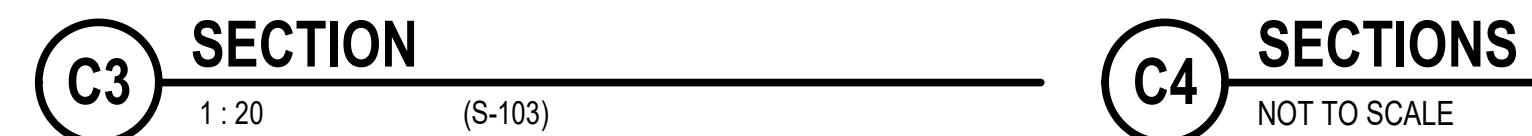
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SEAL			
FOUNDATION KEYNOTES			
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15 EQUIPOTENTIAL GROUND PLANE COORDINATE WITH ELECTRICAL, SEE EG101.			
GRAPHIC SCALE(S)			
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SCALE AS NOTED			
PROJECT NO. 14157108			
CONSTR CONTR NO.			
NAVFAC DRAWING NO. 14157108			
SHEET 46 OF 170			
S-101			
DRAWING REVISION: 03 MARCH 2024			

APPROVED

Win B. Forbes
2026.01.28
FOR COMMANDER

D. EVANS VIA EMAIL
01/15/2026
DES: MWW
DRW: MWW
CHK: LMM
PMCM: MJD/ATP
BRANCH/MANAGER: JTW
DES PROD OR: RLS
FIRE PROTECTION ENGINEER: DSN

NAVFACILITIES ENGINEERING SYSTEMS COMMAND
NAVFACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC
ATLANTIC PLANNING DESIGN AND CONSTRUCTION
NORFOLK, VA
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
VARIOUS LOCATIONS
FOUNDATION PLAN



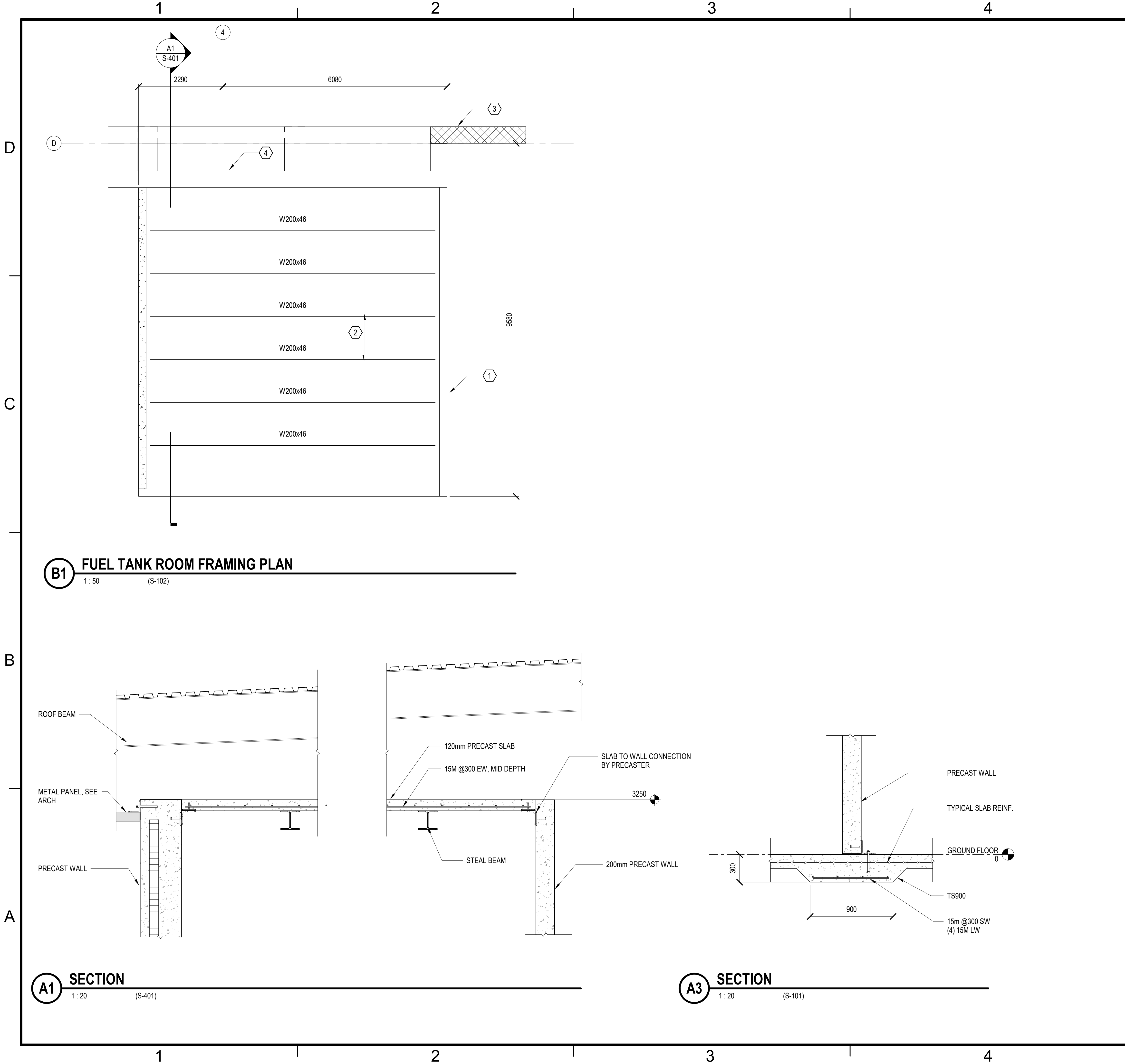
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SFACTORY TO		DATE 01/15/2026	
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CDM		MJD/ATP	
ANCH MANAGER		JTW	
S PROD DIR		RLS	
E PROTECTION ENGINEER		DSN	

DATE:	AS NOTED
PROJECT NO.:	1838720
INSTR. COUNT NO.	

50 OF 170

S-302

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FUEL TANK ROOM SHEET NOTES

- FRAMING EQUALLY SPACED BETWEEN WALLS.
- REFER TO PRECAST CONCRETE NOTES ON S-001 AND LOADING ON S-002.
- TYPICAL REINFORCEMENT SHOWN IS THE MINIMUM REQUIRED, PROVIDE ADDITIONAL REINFORCEMENT AS REQUIRED BY PRECAST MANUFACTURERS DESIGN AT NO ADDITIONAL COST.
- SIGNIFICANT DEVIATION FROM THE TYPICAL DETAILS MUST BE APPROVED BY CONTACTING OFFICER.
- DESIGN ROOF SLAB, AND CONNECTIONS FOR 1.0 kPa LIVE LOAD

KEYNOTES

- 200mm PRECAST INTERIOR WALL
- 120mm THICK PRECAST SLAB
- EXTERIOR WALL, SEE PLAN
- WALL/DOOR HEADER, SEE PLAN

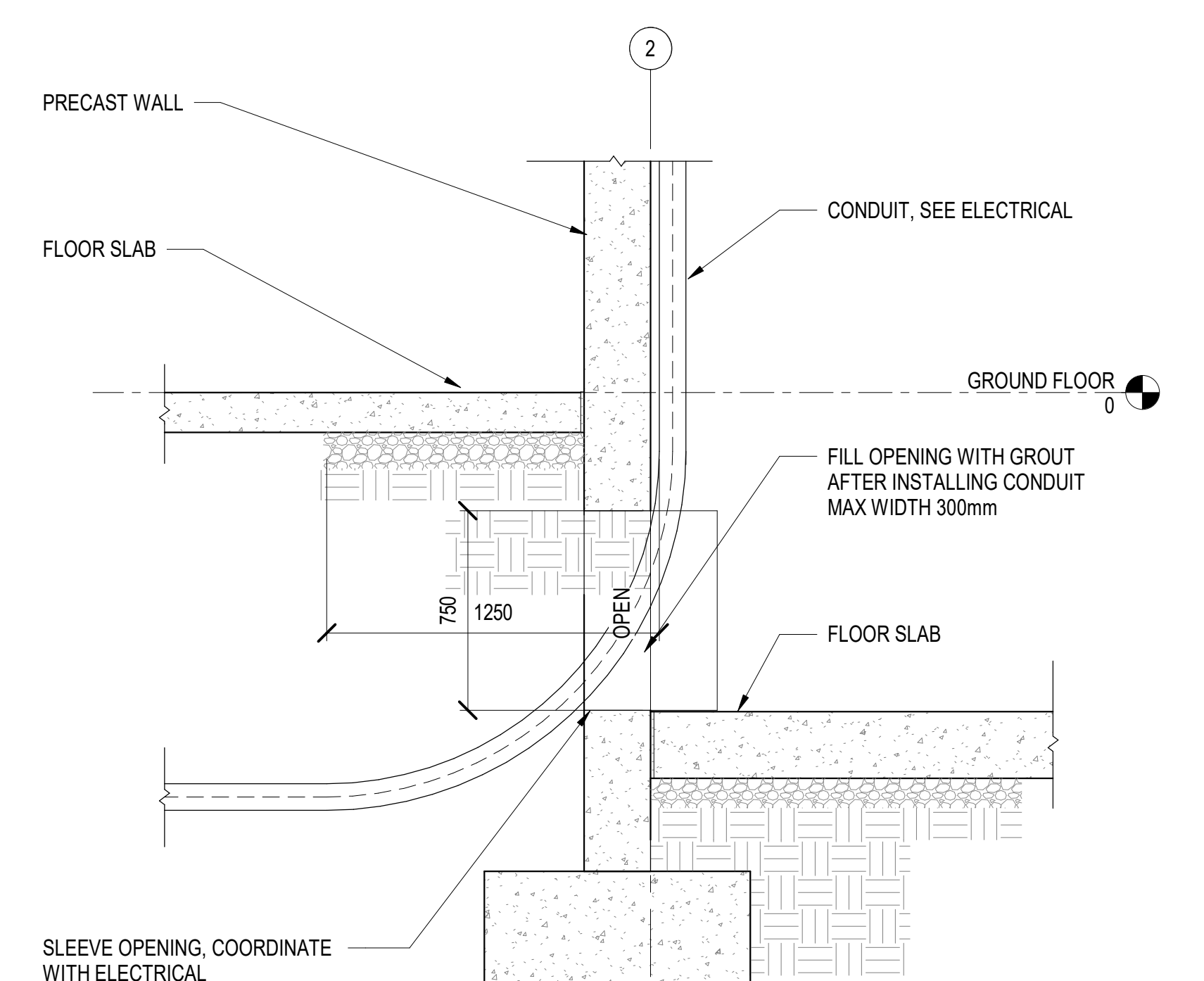
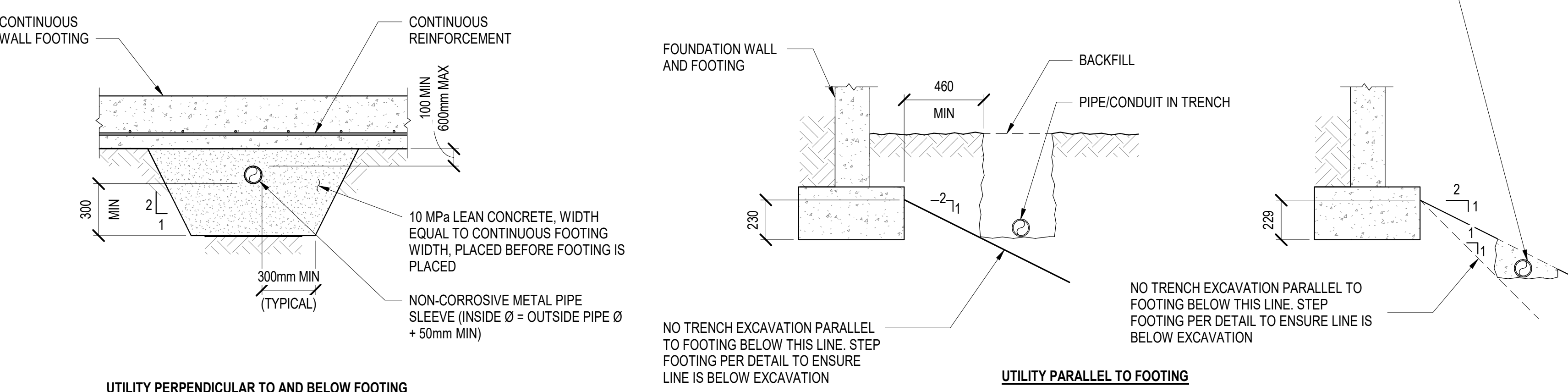
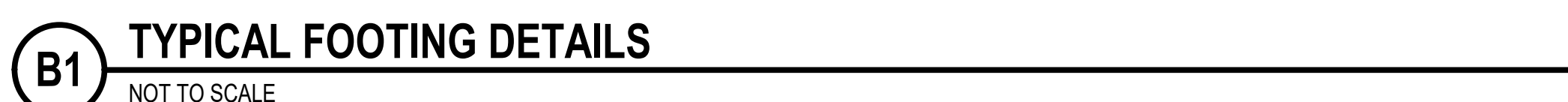
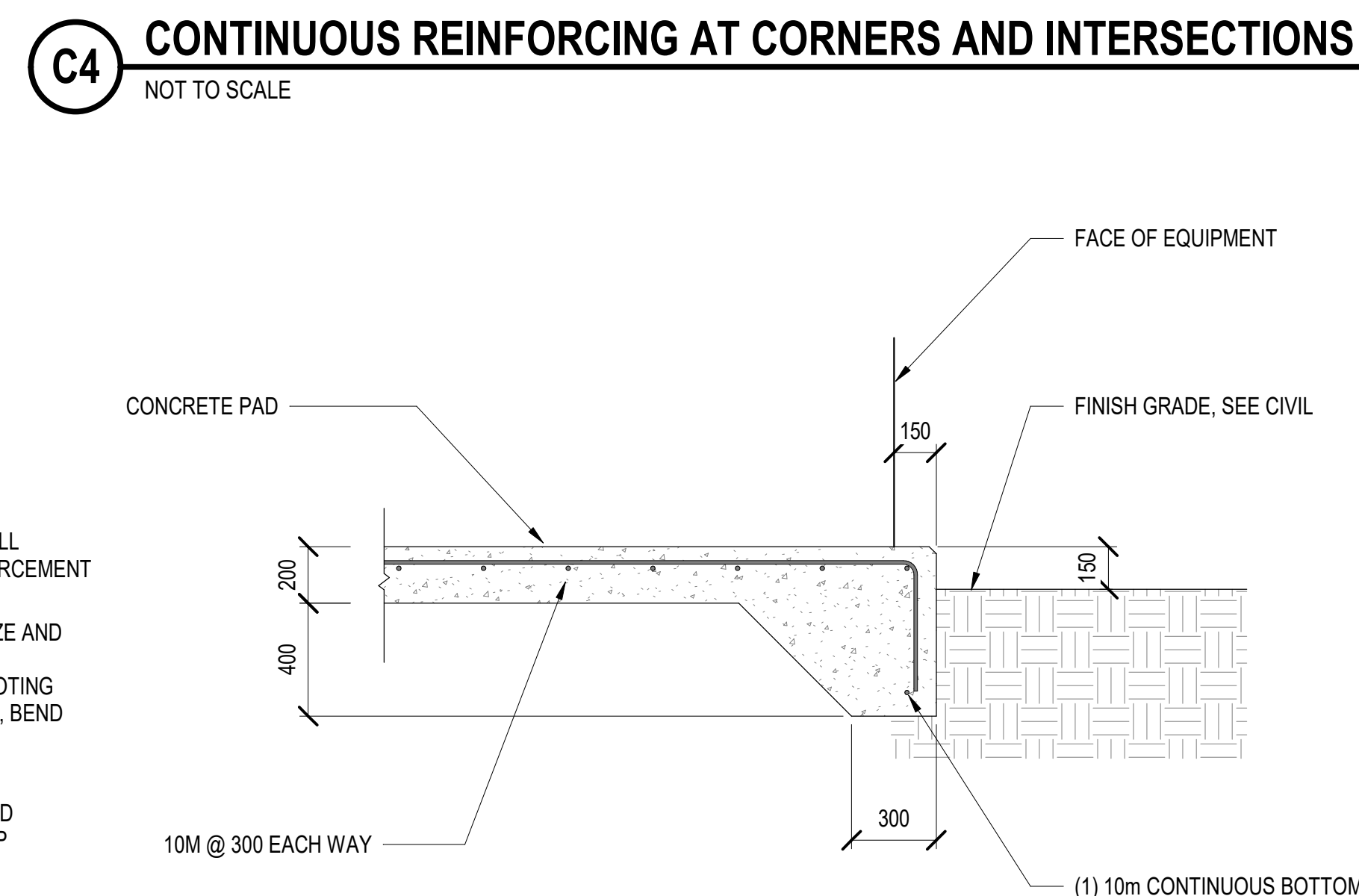
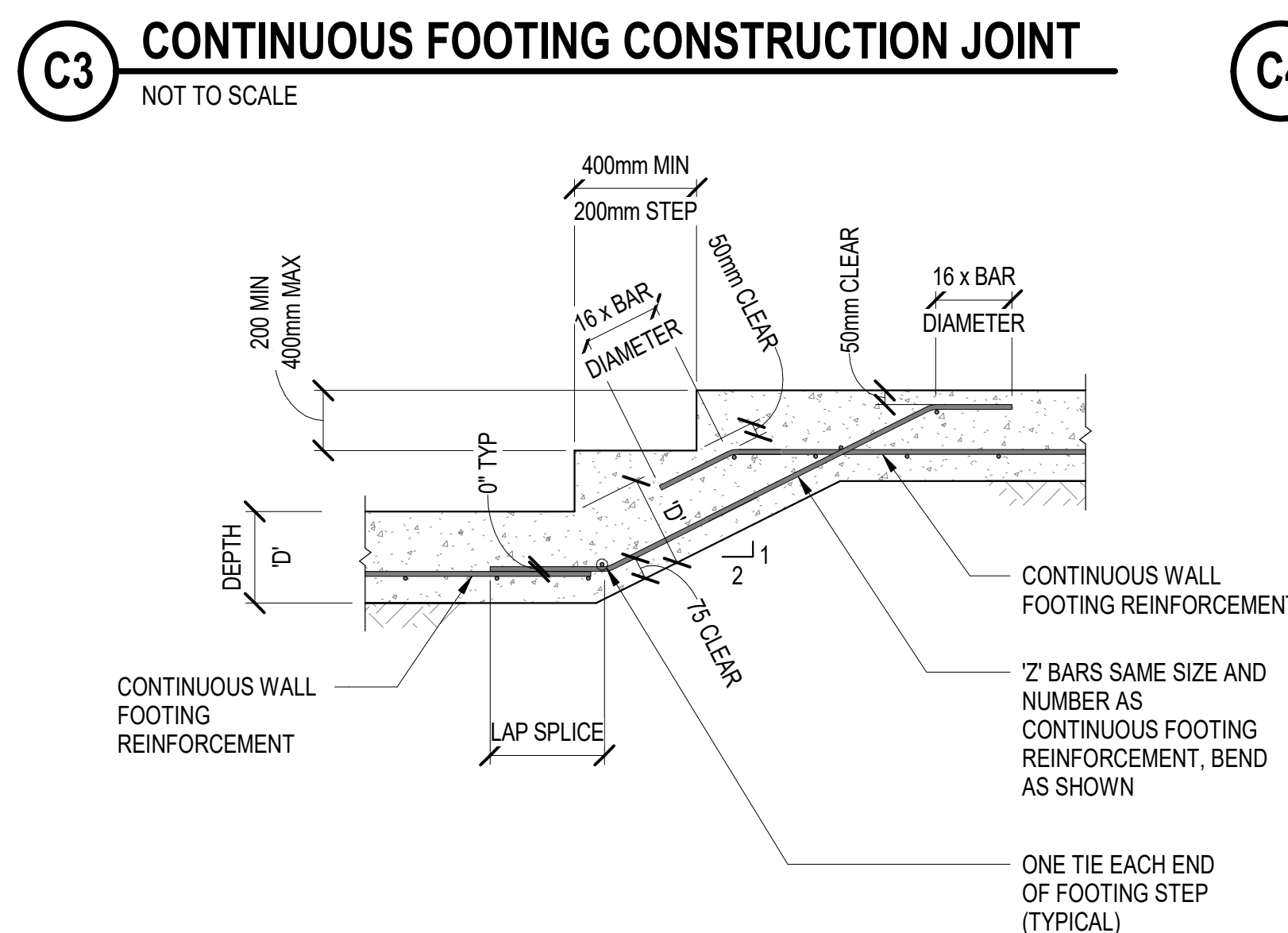
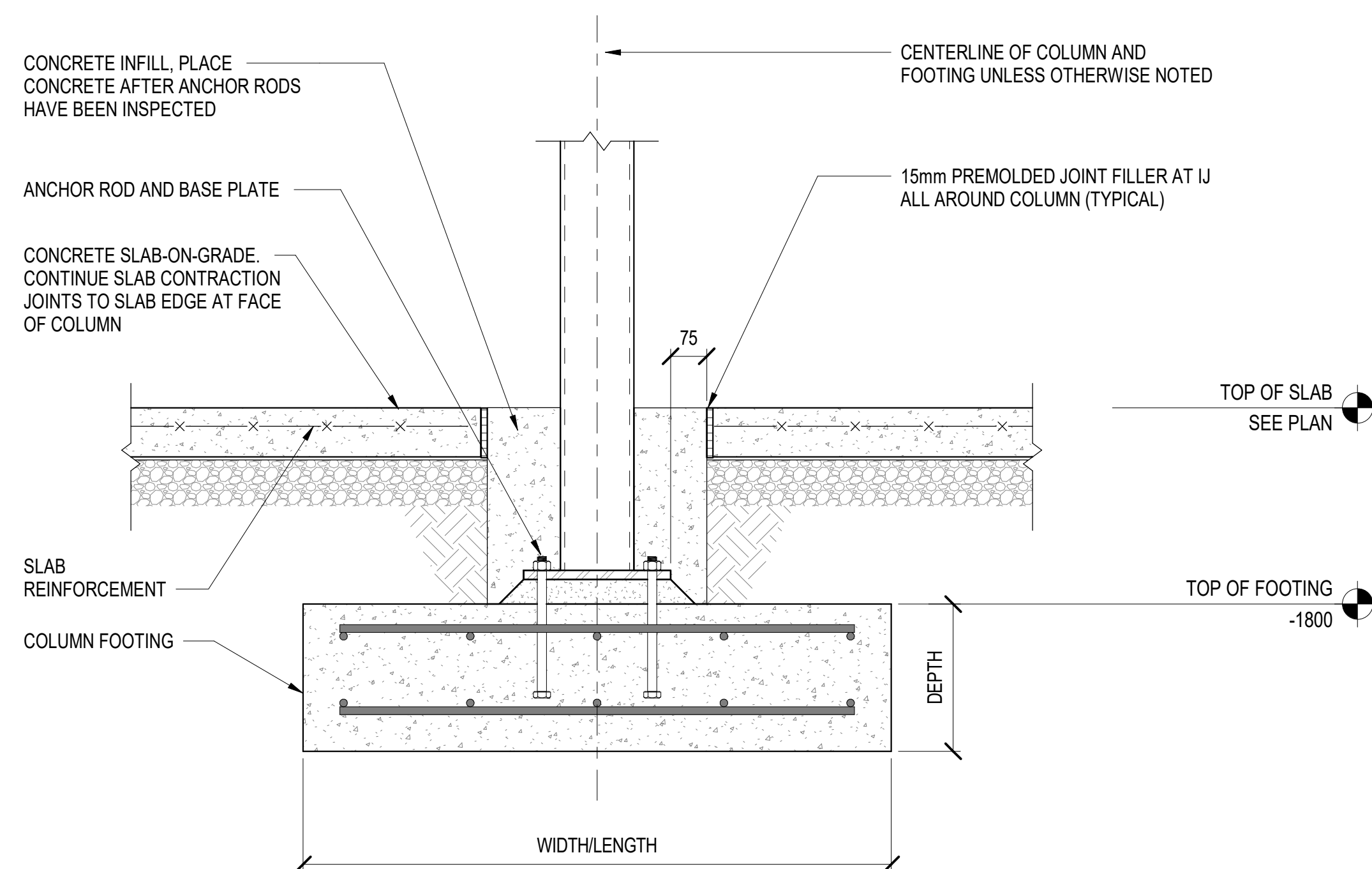
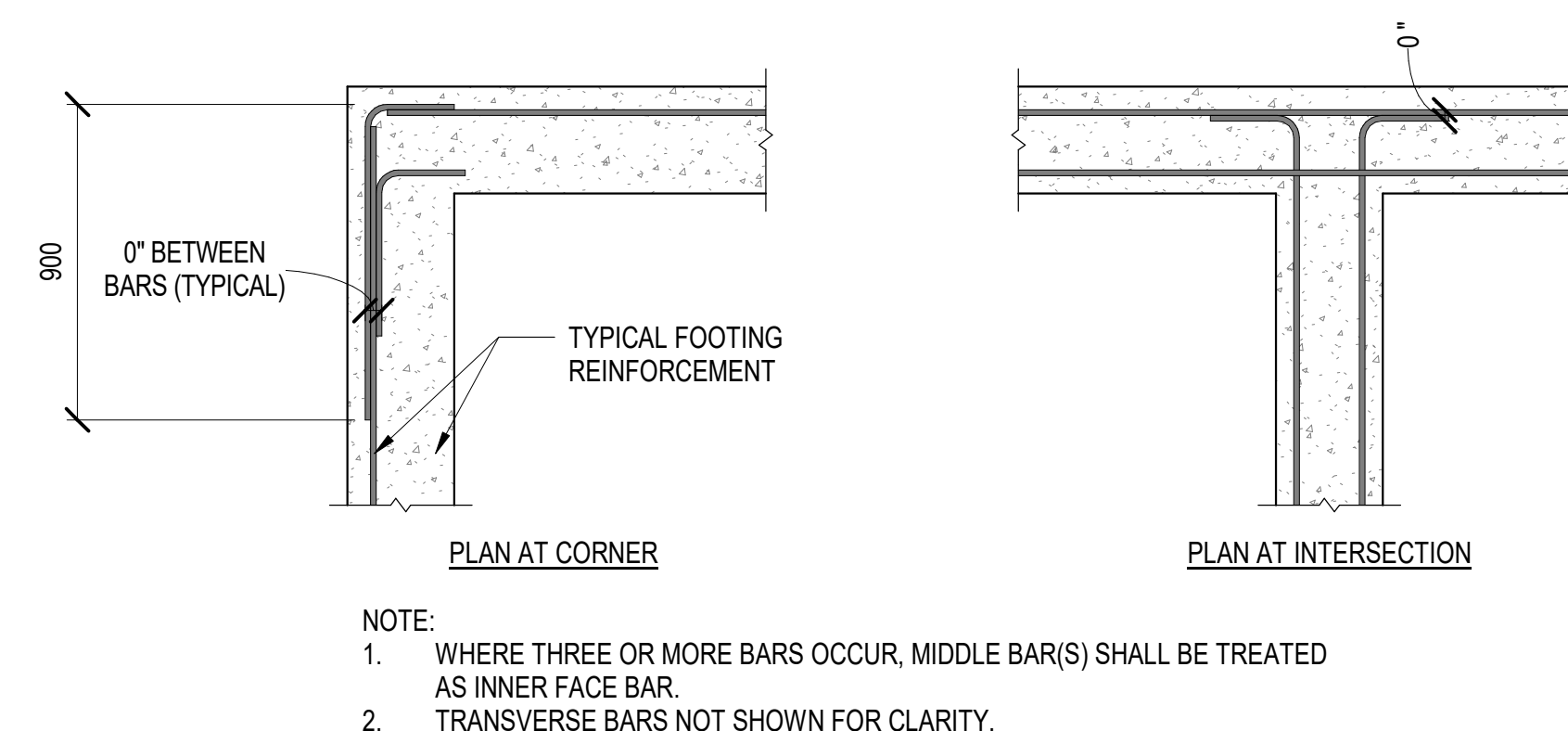
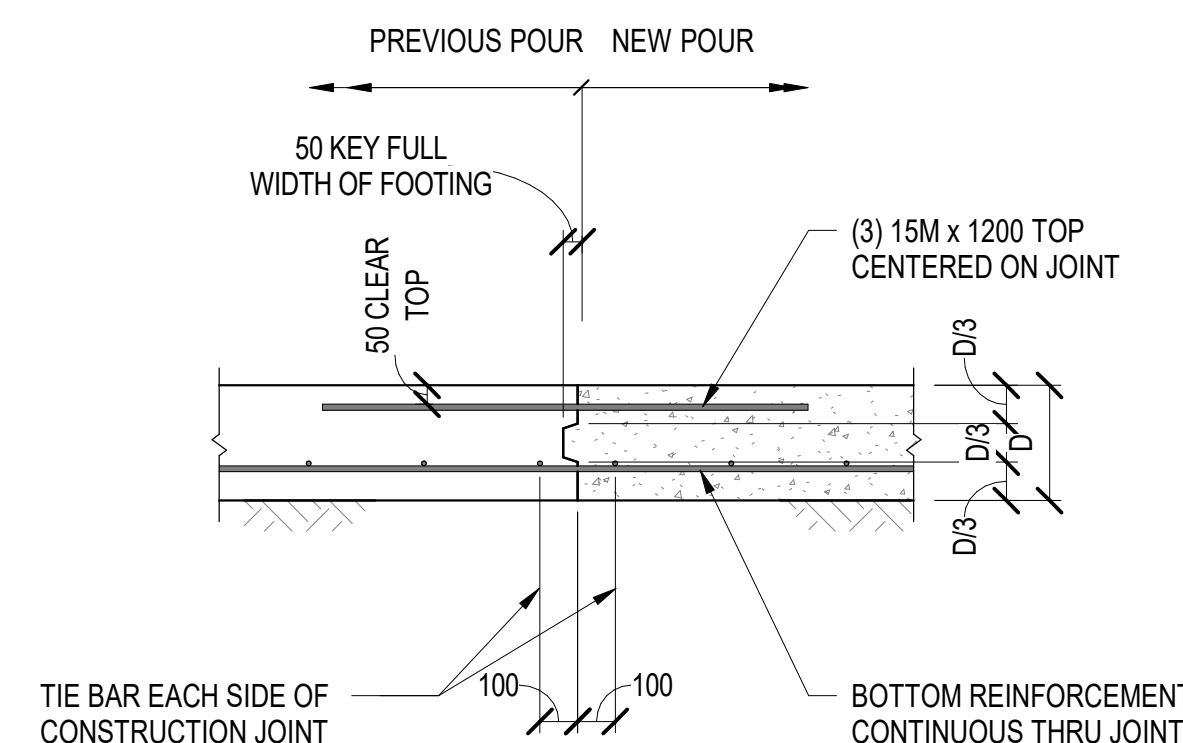
GRAPHIC SCALE(S)



APPROVED	DATE	APPROVED
SYN	DESCRIPTION	SYN
SEAL		
APPROVED		
Digitally signed by signature		
Win B forbes		
Date: 2026.01.28		
FOR COMMANDER		
ACTIVITY		
D. EVANS VIA EMAIL		
SATISFACTORY TO DATE 01/15/2026		
DESIGNER/DRW AUTH/CHK		
PMCM MJD/ATP		
BRANCH/MANAGER JT/W		
DES PROJ DIR RLS		
FIRE PROTECTION ENGINEER DSN		
DEPARTMENT OF THE NAVY		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC		
ATLANTIC PLANNING DESIGN AND CONSTRUCTION		
VARIOUS LOCATIONS		
VARIOUS LOCATIONS		
JOINT MARITIME FACILITY		
FUEL TANK ROOM FRAMING PLAN		
SCALE AS NOTED		
EPROJEST NO. 1838720		
CONSTR CONTR NO.		
NAVFAC DRAWING NO. 14157113		
SHEET 51 OF 170		
S-401		
DRAWFORM REVISION: 03 MARCH 2024		

WALL FOOTING SCHEDULE			
MARK	DIMENSIONS		REINFORCING
	WIDTH	DEPTH	
WF900	900	300	(3)15M CONTINUOUS BOTTOM WITH 15M TIES AT 300 ON CENTER
WF900B	900	450	(3)15M CONTINUOUS TOP & BOTTOM WITH 15M TIES AT 300 ON CENTER
WF1000	1000	450	(4)20M CONTINUOUS BOTTOM WITH 20M TIES AT 300 ON CENTER
WF2100	2100	450	(7)20M CONTINUOUS BOTTOM WITH 20M TIES AT 300 ON CENTER TOP AND BOTTOM

COLUMN FOOTING SCHEDULE				
MARK	DIMENSIONS			REINFORCING STEEL
	WIDTH	LENGTH	DEPTH	
CF 2500	2500	2500	800	(9)25M EACH DIRECTION TOP AND BOTTOM

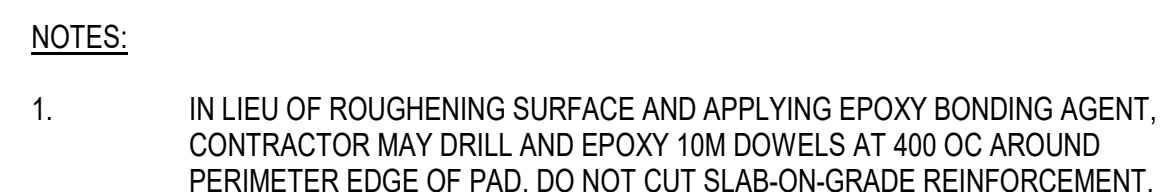
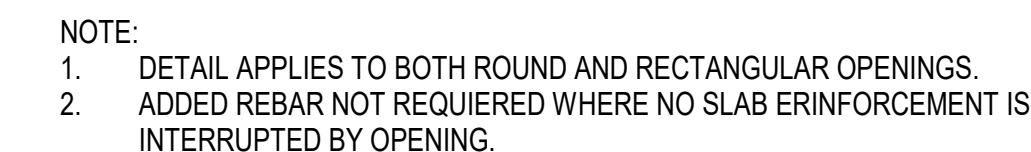




C



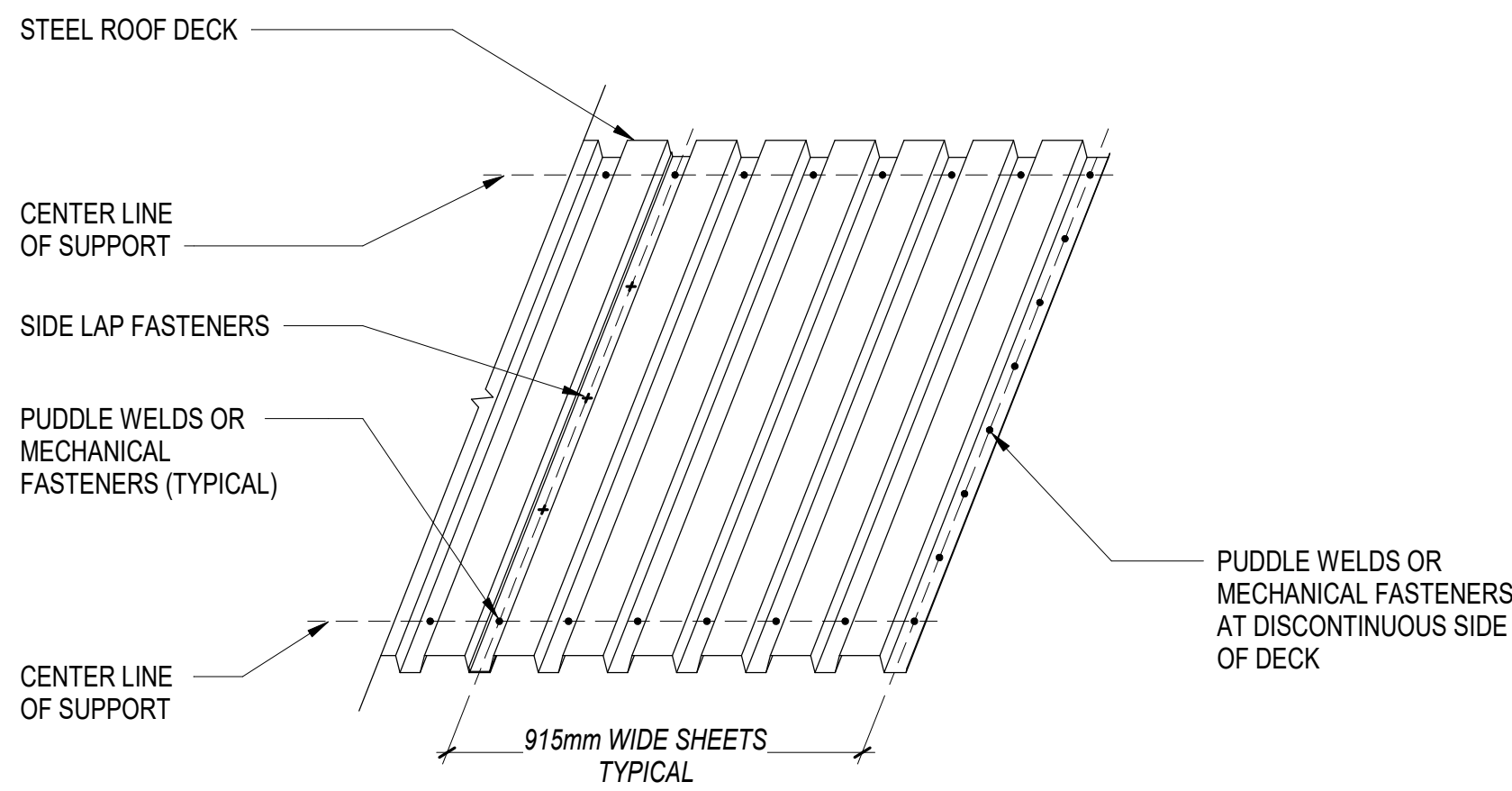
A



NOT TO SCALE

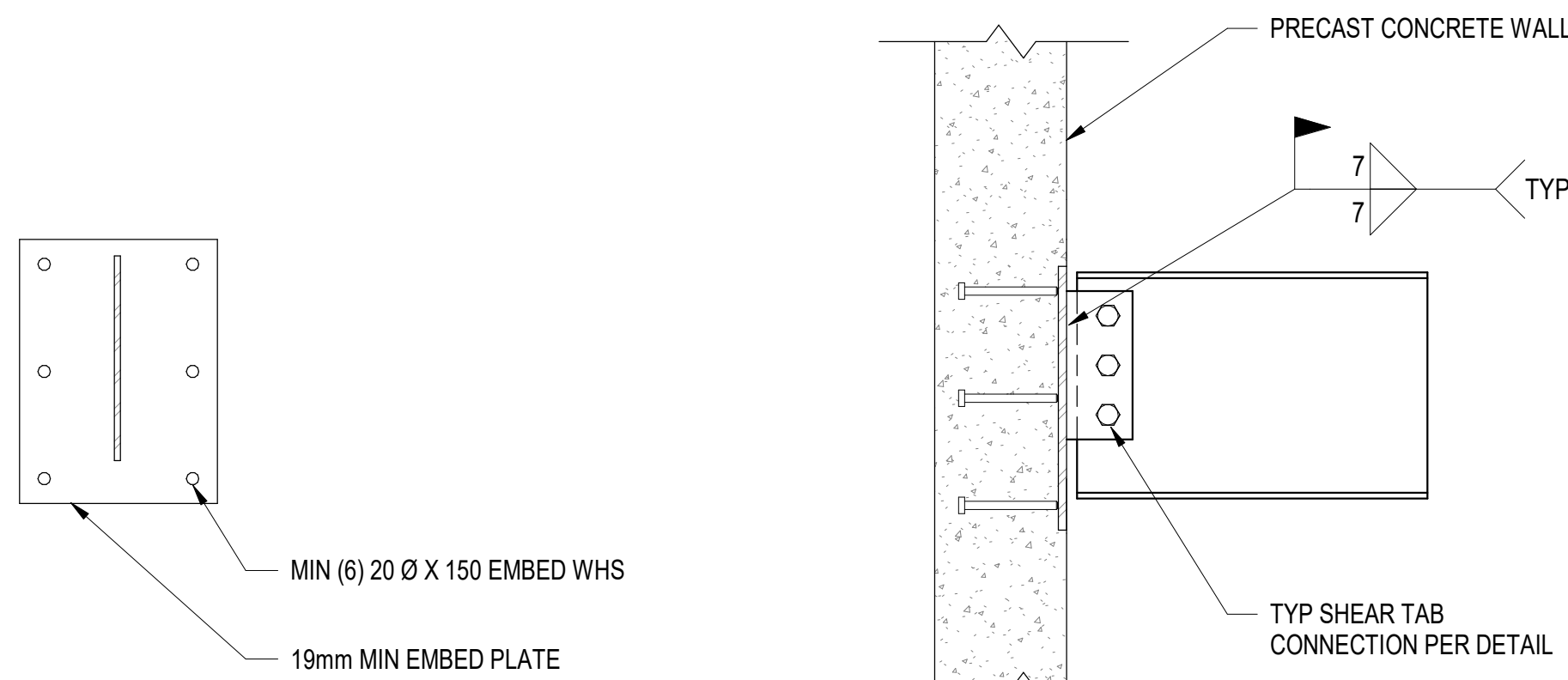


S-502



C1 ROOF DECK ATTACHMENT

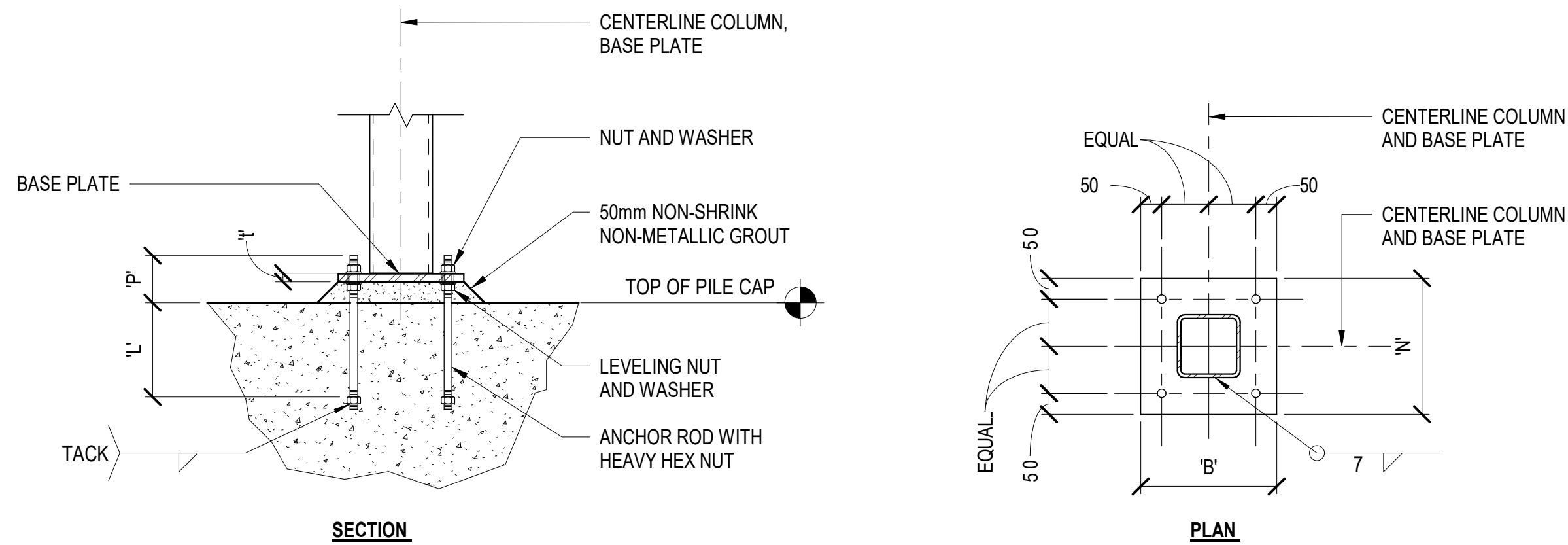
NOT TO SCALE



EMBED PLATE ELEVATION

EMBED PLATE SECTION

NOTE: EMBED PLATE AND WHS SHOWN ARE MINIMUM REQUIREMENT. PRECAST MANUFACTURER MUST DESIGN CONNECTION AND WALLS FOR MINIMUM BEAM END REACTIONS PROVIDED ON S-003.

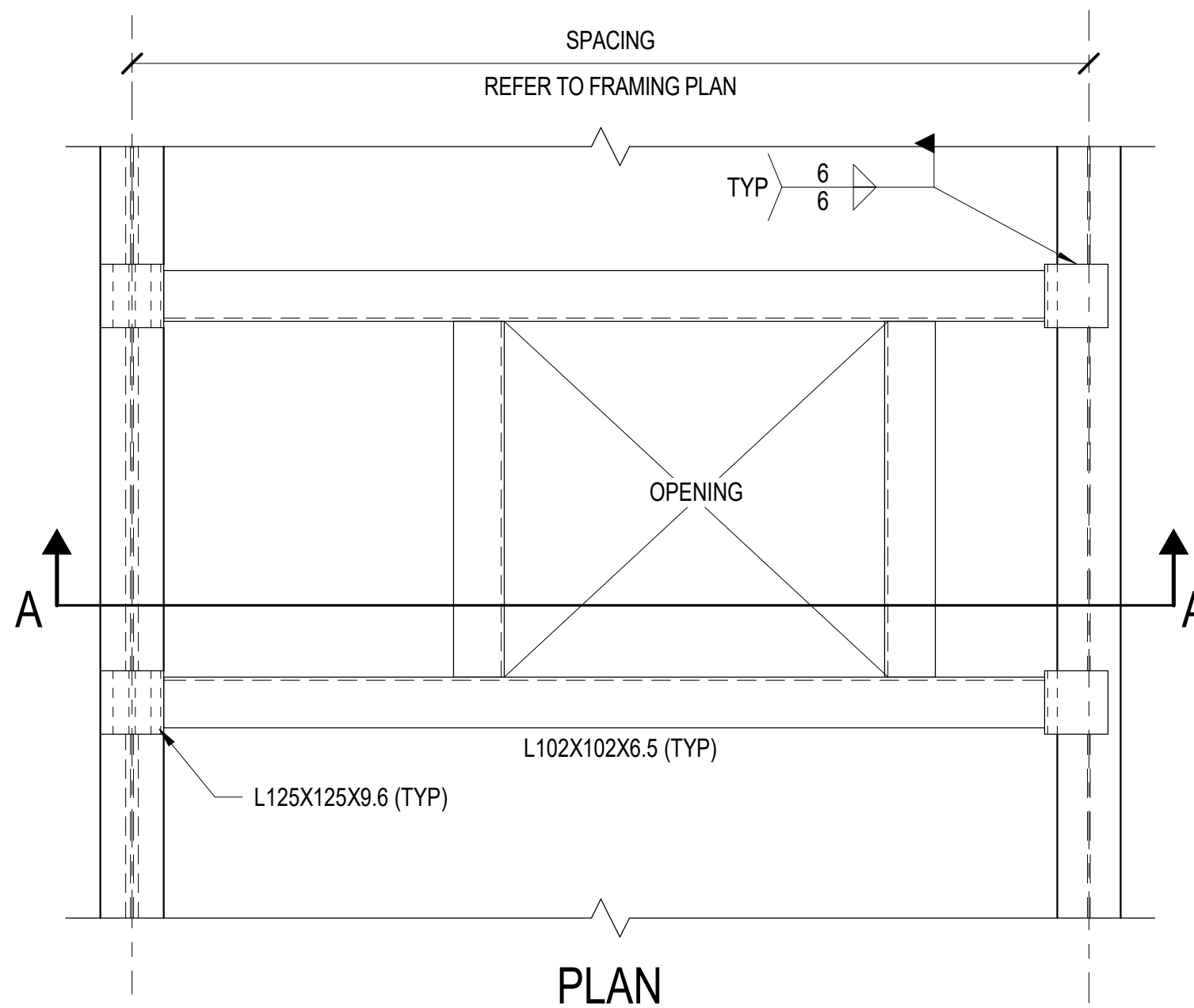


STEEL COLUMN SCHEDULE

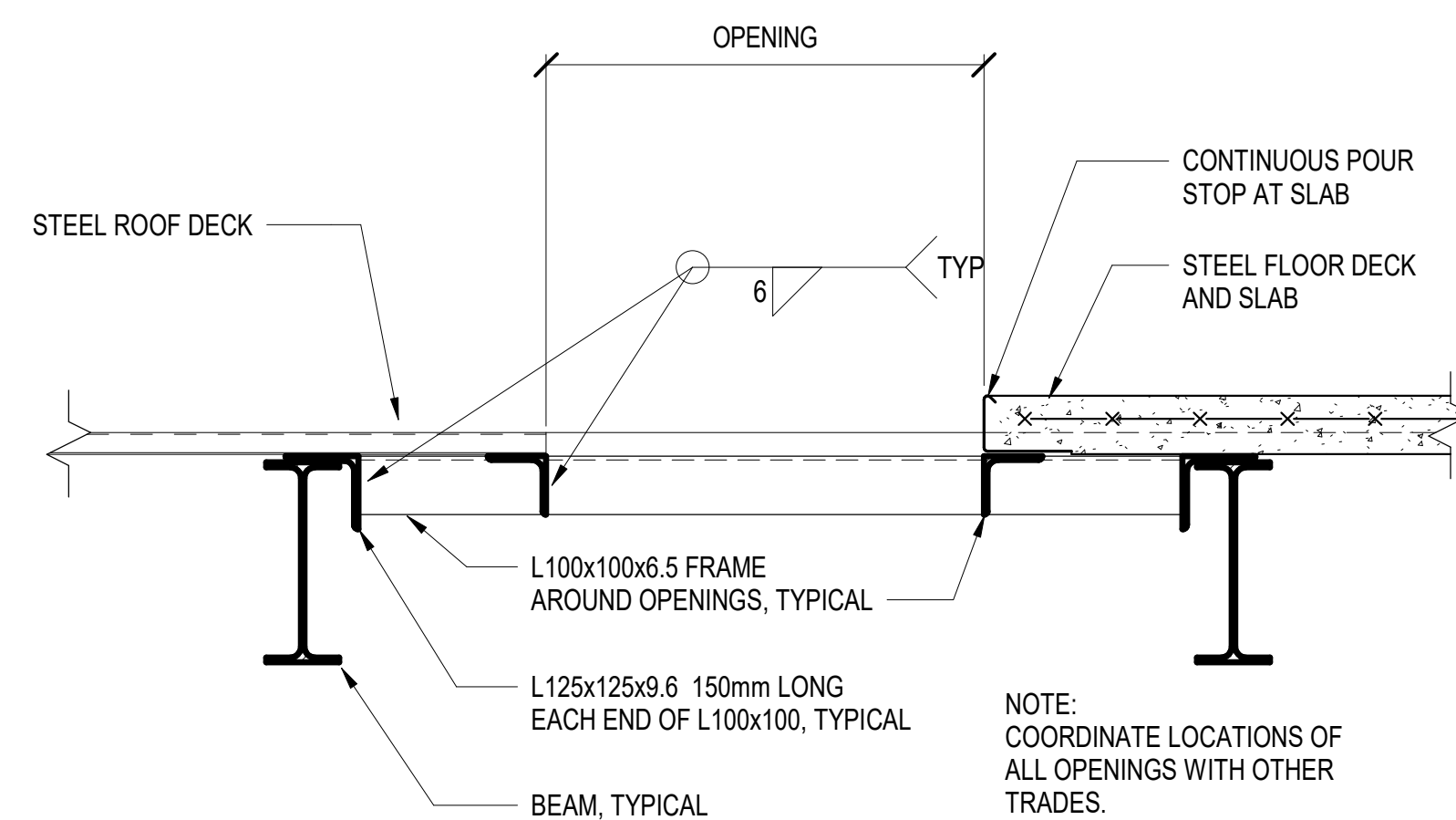
MARK	SIZE	ANCHOR RODS				BASE PLATE		
		QUANTITY	ROD DIAMETER	EMBEDMENT 'L'	PROJECTION 'P'	WIDTH 'B'	LENGTH 'N'	THICKNESS 't'
C1	HSS305X305X9.5	4	25	300	100	500	500	25

C3 TYPICAL STEEL COLUMN SCHEDULE AND BASE PLATE DETAIL

NOT TO SCALE



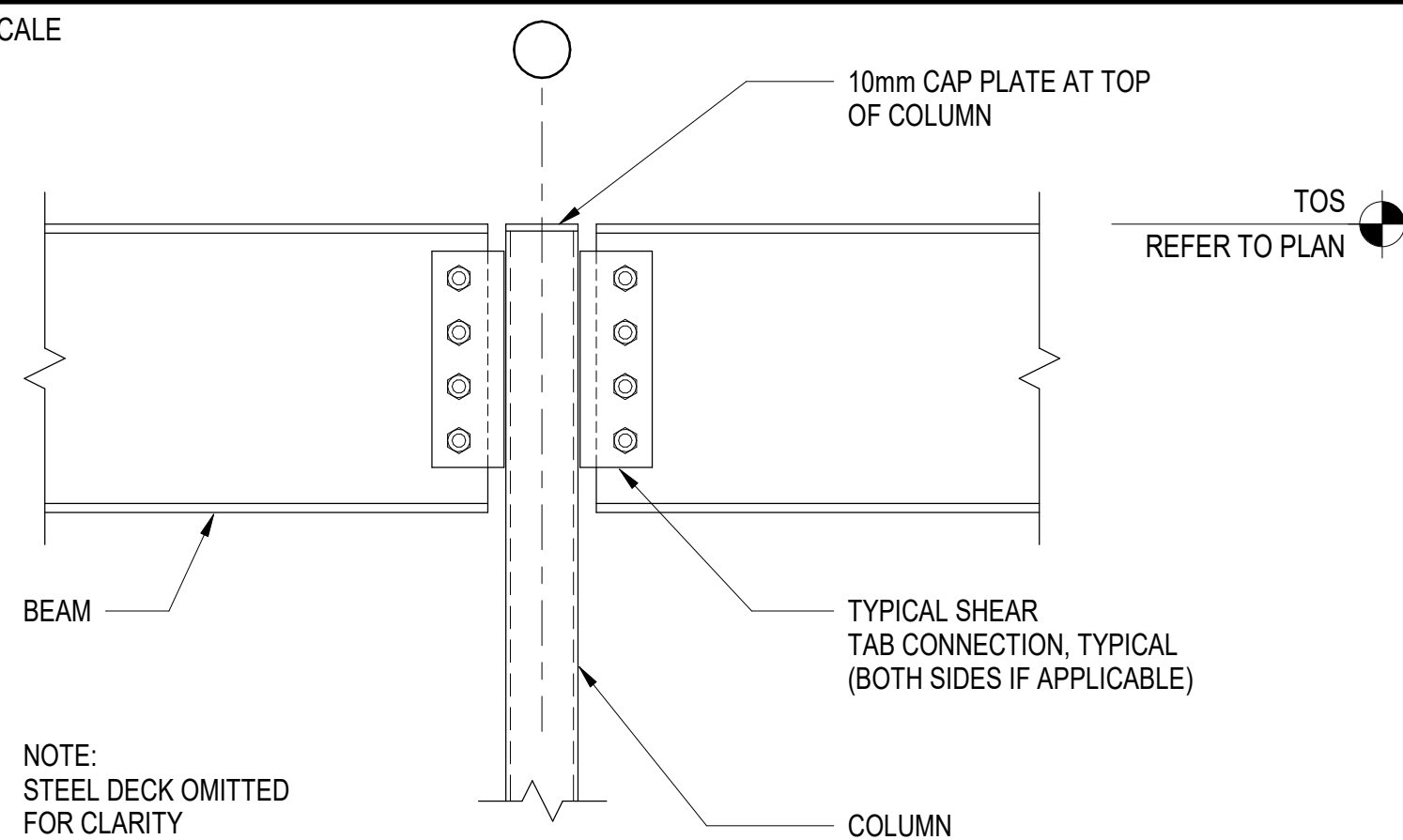
PLAN



SECTION A-A

B1 BEAM CONNECTION TO CONCRETE WALL

NOT TO SCALE

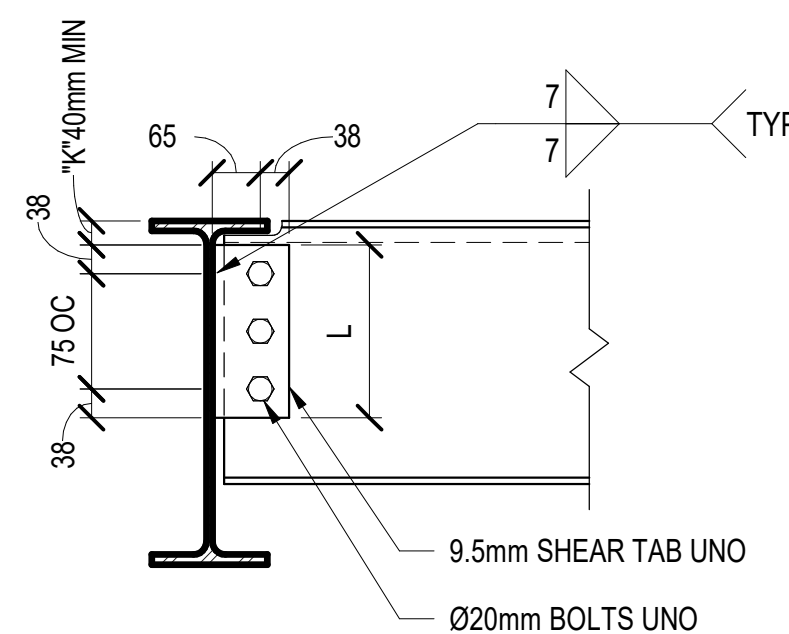


A1 TYPICAL BEAM TO COLUMN SHEAR CONNECTION

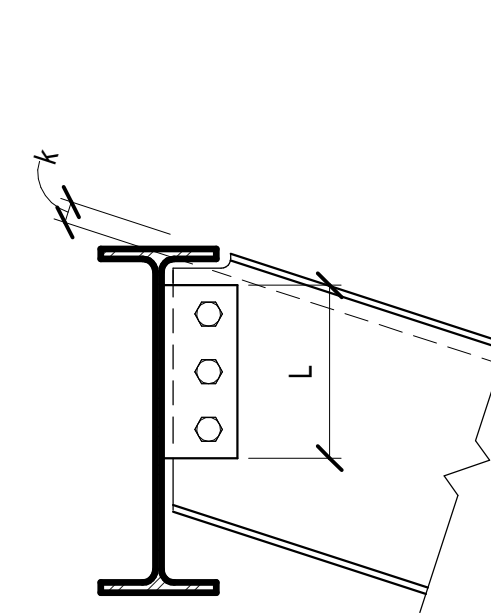
NOT TO SCALE

B2 TYPICAL DECK SUPPORT AT ROOF / FLOOR OPENINGS

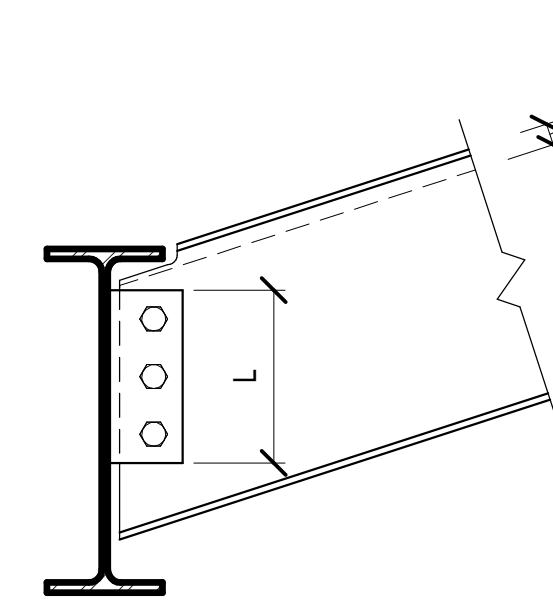
NOT TO SCALE



HORIZONTAL



SLOPE DOWN



SLOPE UP

MIN # OF BOLTS	
BEAM SIZE	NO OF ROWS
W200	2
W410	4
W530	6
W690	7

A3 TYP SHEAR TAB CONNECTION

NOT TO SCALE



APPROVED
Digitally signed by signature
Win B Forbes
Date: 2026.01.28
FOR COMMANDER
16:55:55-05'00'

ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026
DES MWW DRW MWW CHK LMM
PMCM MJD/ATP
BRANCHMANAGER JTW
DES PROD DR RLS
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC
ATLANTIC PLANNING DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
VARIOUS LOCATIONS
STEEL DETAILS
SCALE AS NOTED
EPROJEST NO: 1838720
CONSTR CONTR NO:
NAVIFAC DRAWING NO: 14157116
SHEET 54 OF 170
S-503
DRAWFORM REVISION: 03 MARCH 2024

1

2

3

4

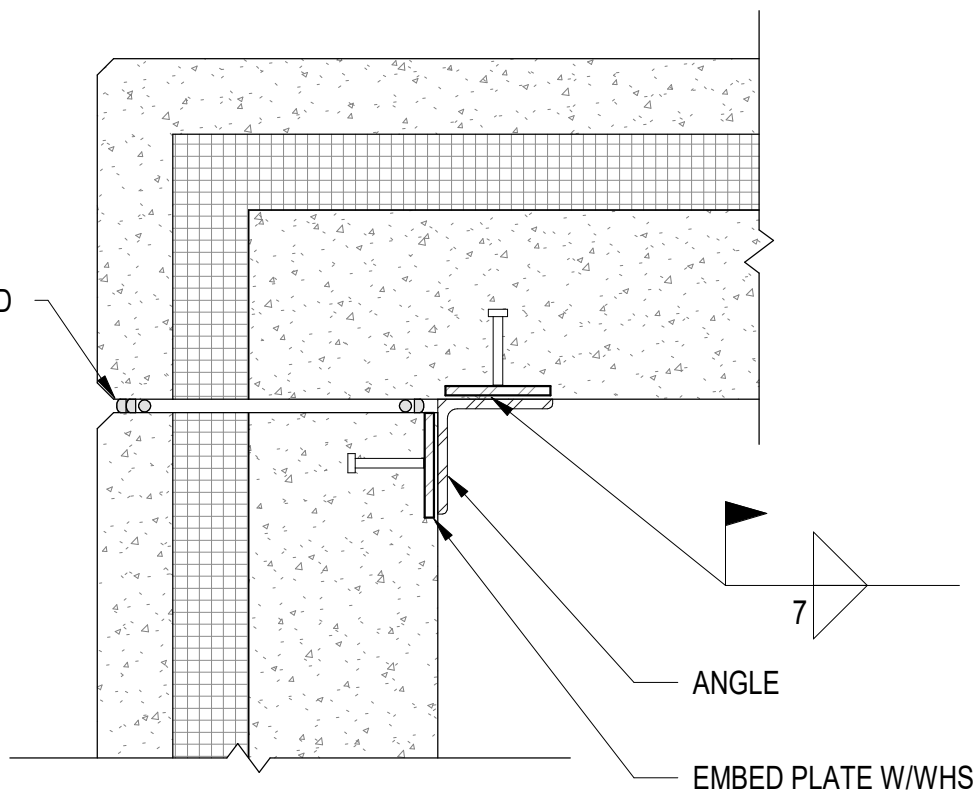
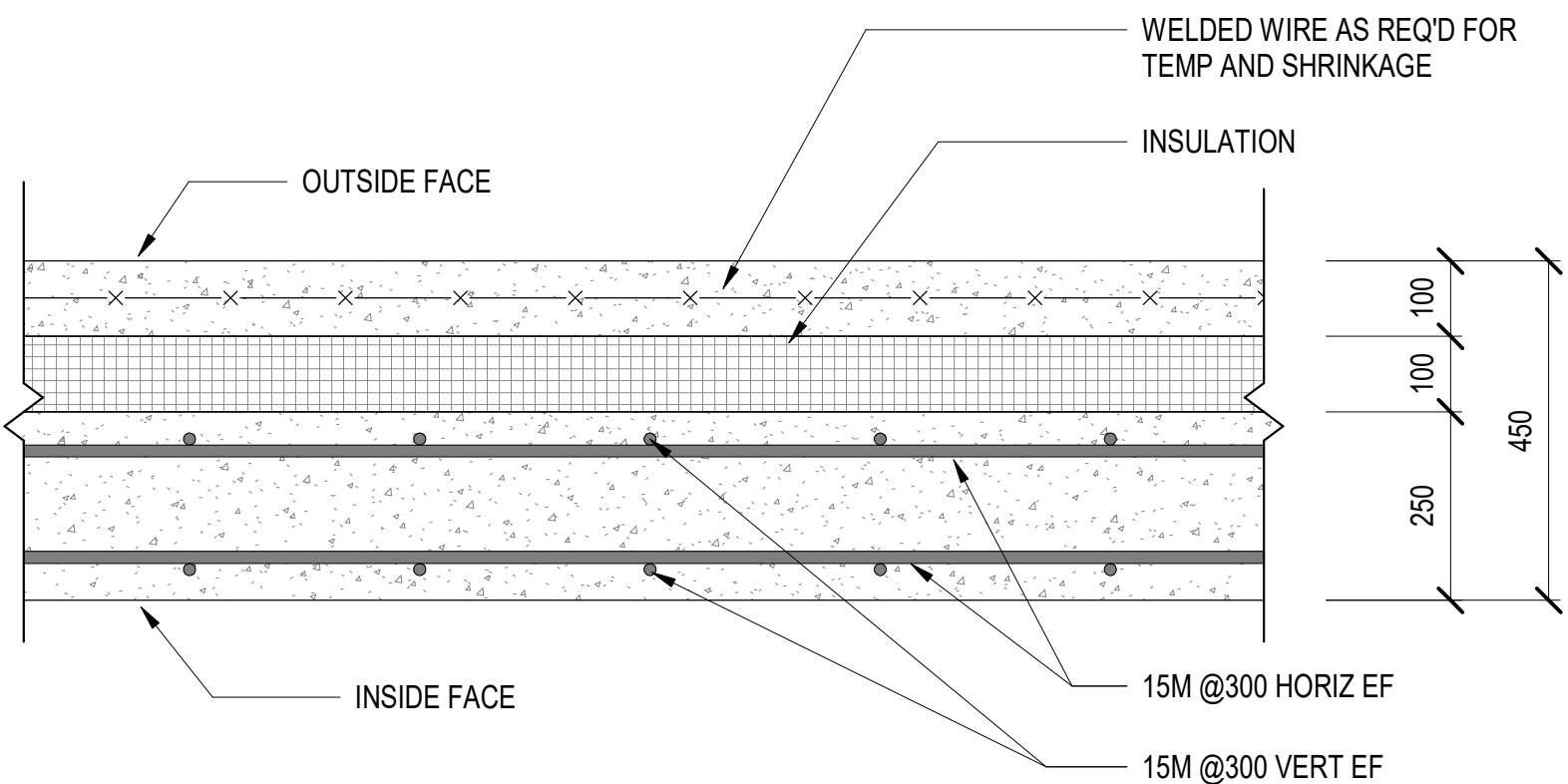
5

SHEET NOTES

- A. REFER TO PRECAST CONCRETE NOTES ON S-001 AND LOADING ON S-002.
- B. TYPICAL REINFORCEMENT AND CONNECTIONS SHOWN ARE THE MINIMUM REQUIRED, PROVIDE ADDITIONAL REINFORCEMENT AS REQUIRED BY PRECAST MANUFACTURERS DESIGN AT NO ADDITIONAL COST.
- C. SIGNIFICANT DEVIATION FROM THE TYPICAL DETAILS MUST BE APPROVED BY CONTACTING OFFICER.

D

D

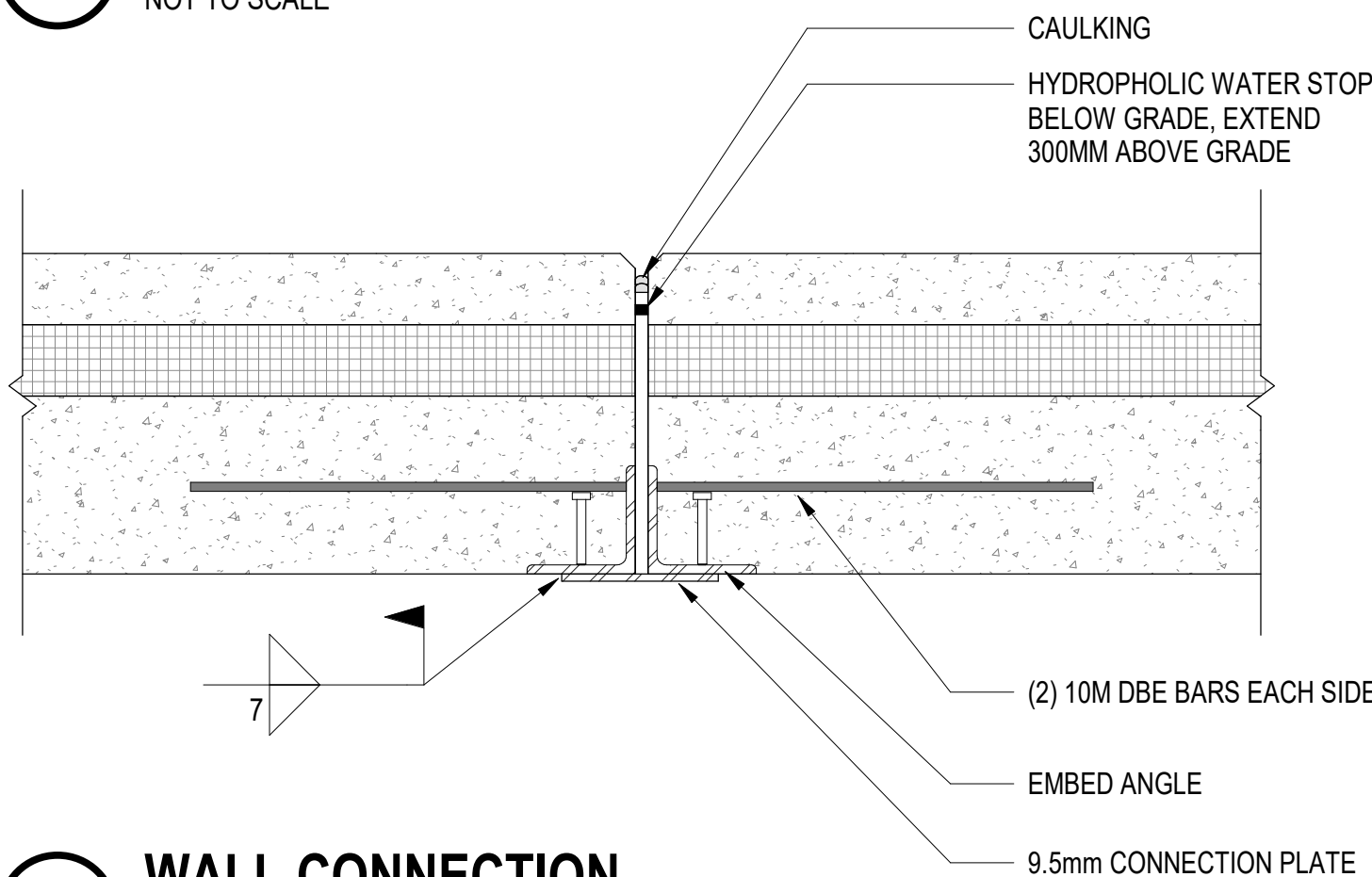
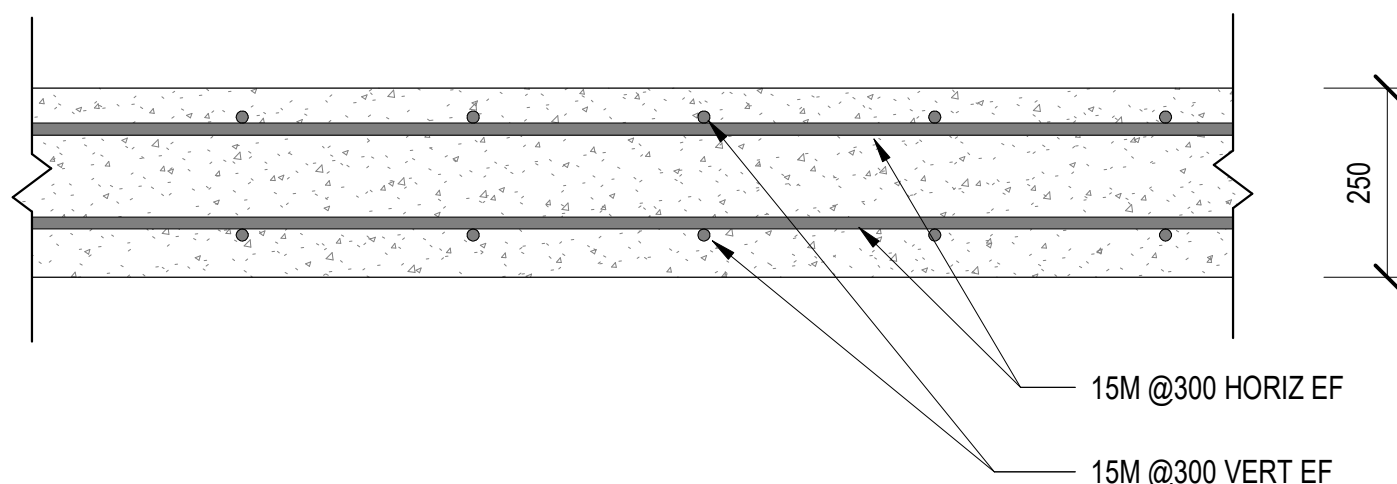


D1 TYPICAL 450mm WALL SECTION
NOT TO SCALE

D2 WALL CORNERS DETAILS
NOT TO SCALE

C

C

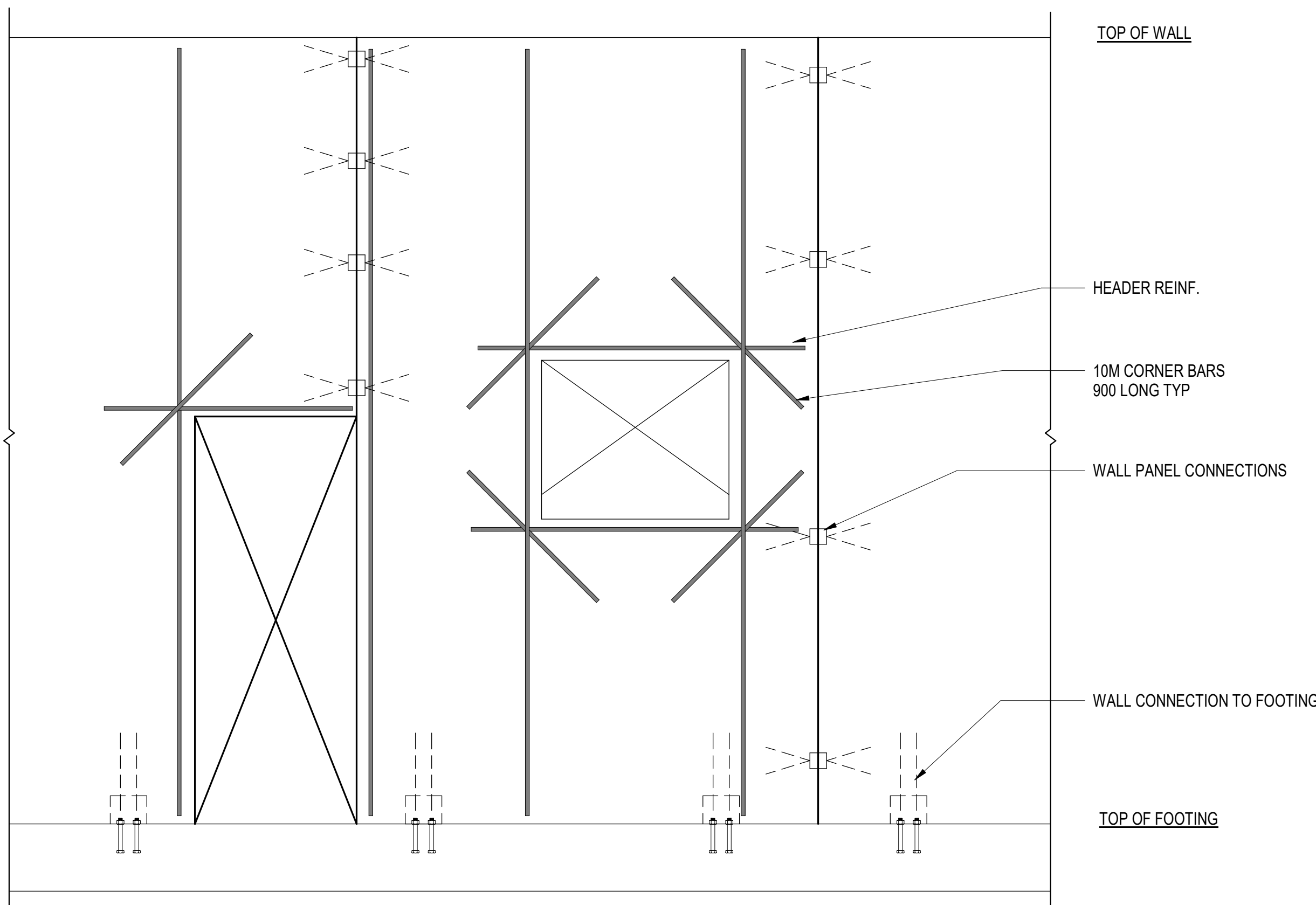


C1 TYPICAL 250mm WALL SECTION
NOT TO SCALE

C2 WALL CONNECTION
NOT TO SCALE

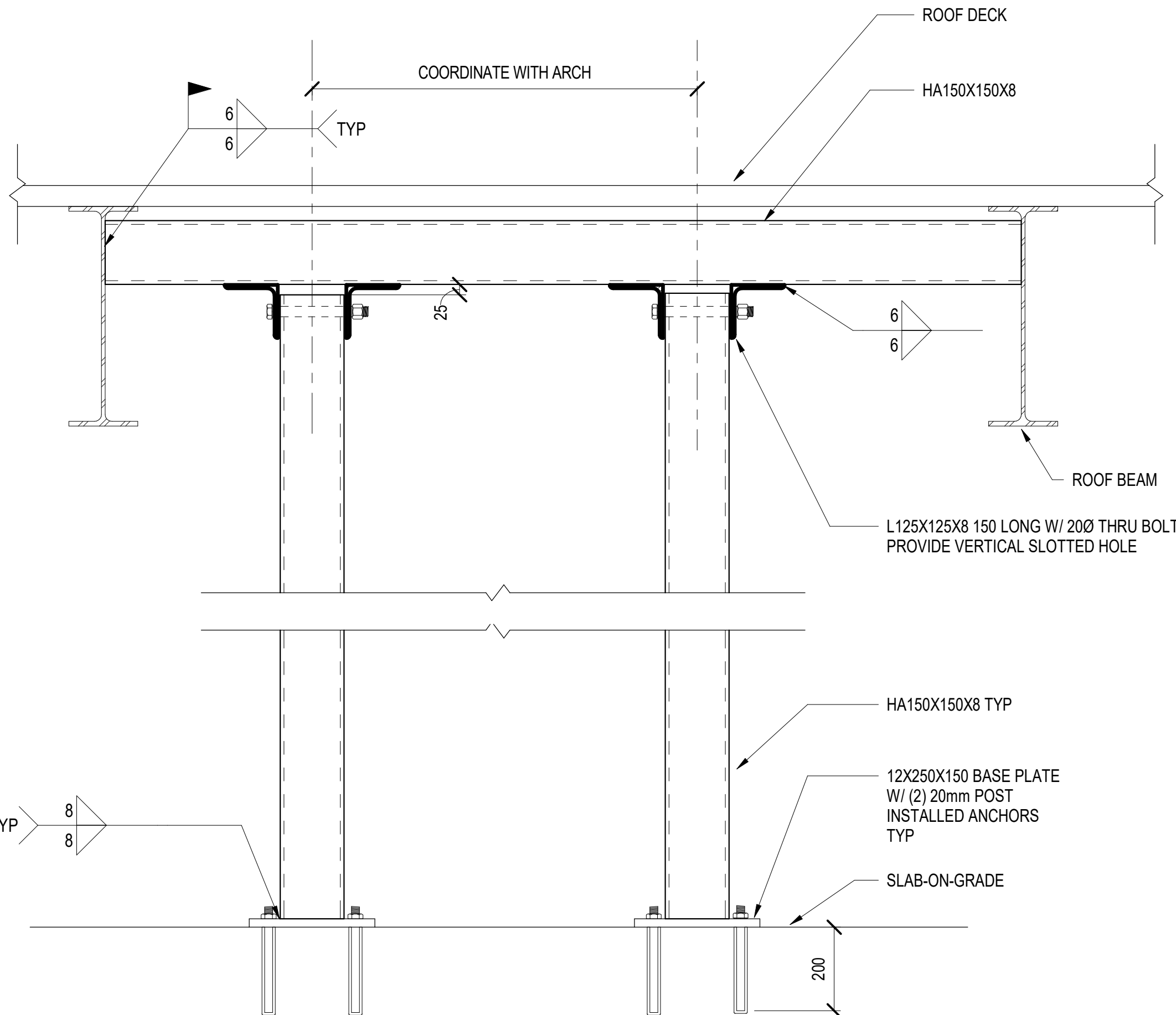
B

B



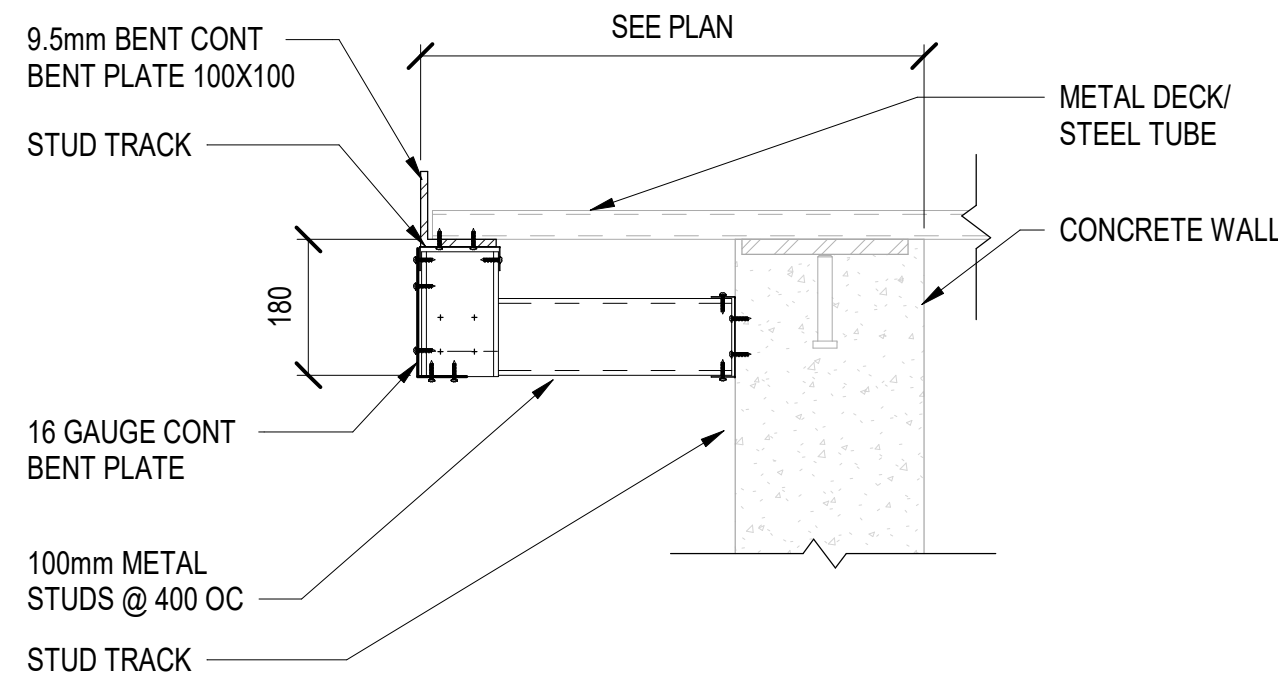
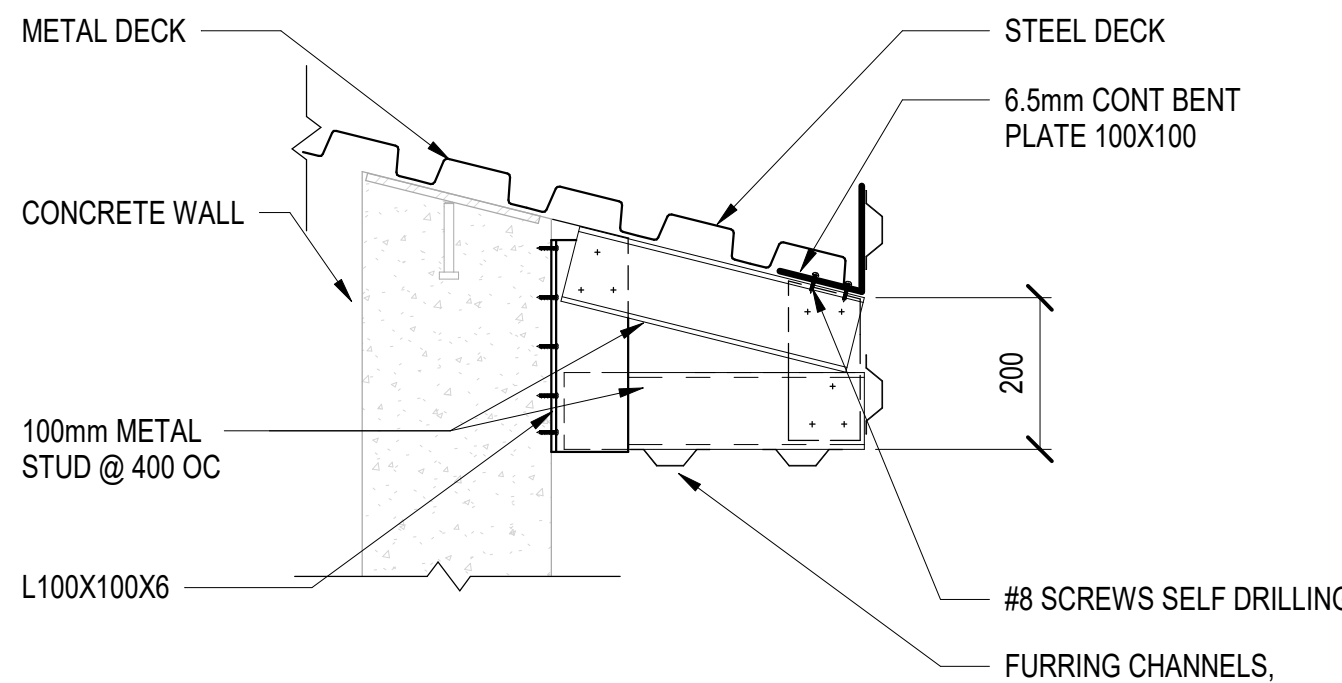
A1 TYPICAL WALL REINFORCEMENT AT OPENINGS AND EDGES
NOT TO SCALE

A3 ROOF OVERHANGS
NOT TO SCALE



- NOTE:
- COORDINATE LOCATION WITH LADDER AND DOORS, SEE ARCH.
 - ENSURE BEAM SUPPORTS AND BASE PLATES FIT WITHIN INTERIOR WALL

B4 LADDER SUPPORT/DOOR JAMBS
1:10



APPROVED
Digitally signed by signature
Win B. Forbes
Date: 2026.01.28
FOR COMMANDER
16:55:55-05'00"

ACTIVITY
D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES. MWW DRW. MWW CHK. LMM

PMCM MJD/ATP

BRANCH MANAGER JTW

DES. PROD. DR. RLS

FIRE PROTECTION ENGINEER DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC
ATLANTIC PLANNING, DESIGN AND CONSTRUCTION
NORFOLK, VA

VARIOUS LOCATIONS
JOINT MARITIME FACILITY
WALL DETAILS

SCALE: AS NOTED

PROJECT NO.: 1838720

CONSTR. CONTR. NO.

NAVIFAC DRAWING NO. 14157117

SHEET 55 OF 170

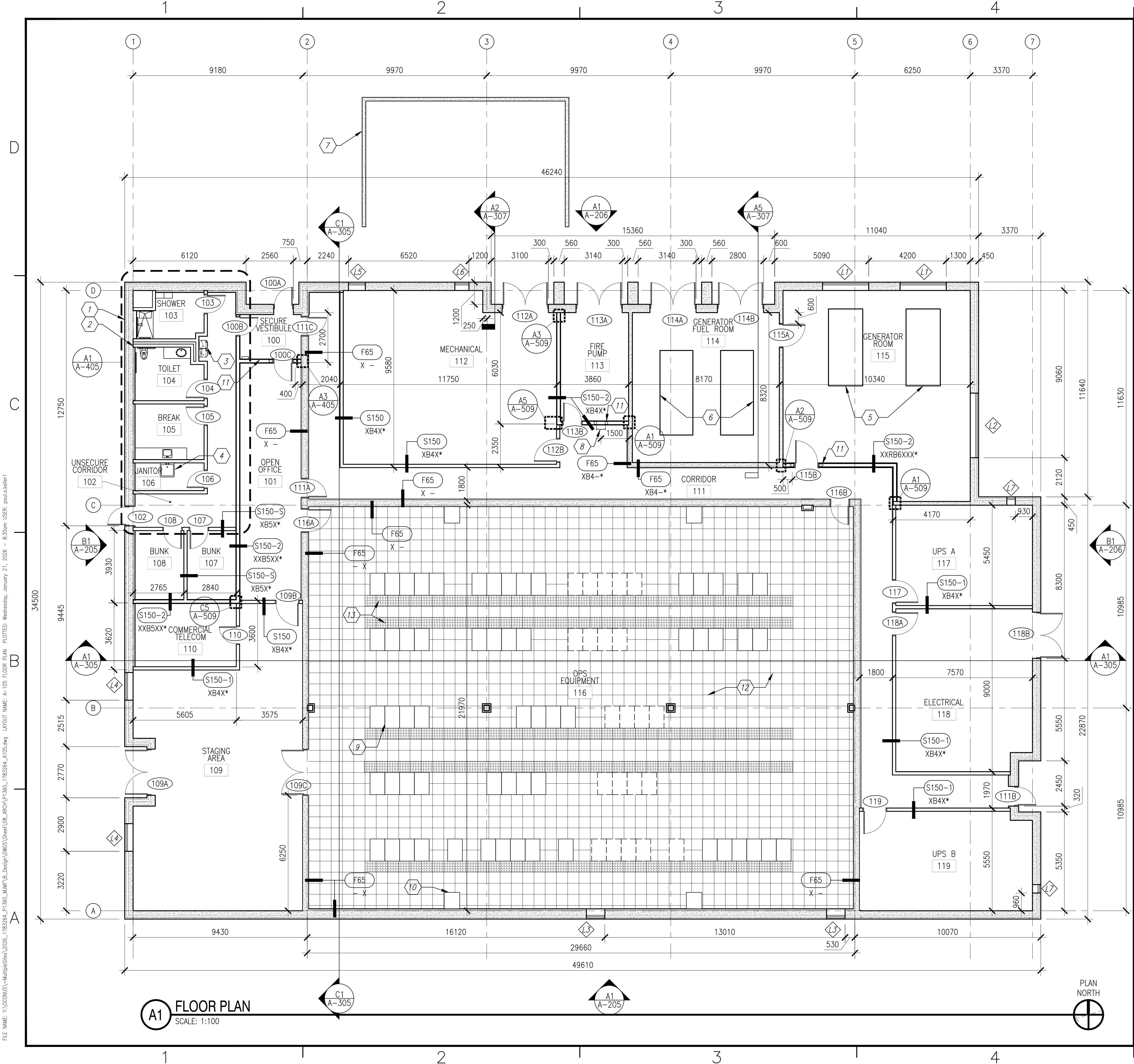
S-504

DRAWING REVISION: 03 MARCH 2024

A

[illegible]

FILE NAME: Y:\OCONUS\MultipleSites\2026_1783264_P1360_JMMP\B_Design\DWG\Sheet\08_ARCH\VP1360_1783264_A105.dwg LAYOUT NAME: A-105 FLOOR PLAN PLOTTED: Wednesday, January 21, 2026 - 8:30am USER: paul.a.keller1



A1 FLOOR PLAN
SCALE: 1:100

GENERAL SHEET NOTES

- A. REFER TO SHEET A-001 FOR ARCHITECTURAL LEGEND AND ABBREVIATIONS.
- C. REFER TO SHEET A-101 FOR KEYNOTES THIS SHEET.
- D. REFER TO SHEET A-605 FOR PARTITION SCHEDULE.
- E. REFER TO SHEET A-606 FOR LOUVER SCHEDULE
- F. SEE INTERIOR 'I' SERIES SHEETS FOR FINISH, FURNITURE AND EQUIPMENT LAYOUTS
- G. PROVIDE FIRE RETARDANT TREATED WOOD BLOCKING FOR MOUNTING OF GRAB BARS, WALL MOUNTED CABINETS, AV OR IT EQUIPMENT, AND WHERE NECESSARY FOR PROPER ATTACHMENT OF ITEMS. SECURELY ATTACH BLOCKING TO METAL STUD FLANGES. COORDINATE EXACT LOCATION AND QUANTITY OF BLOCKING WITH EQUIPMENT REQUIRING REINFORCEMENT.
- H. PROVIDE PLYWOOD BACKBOARDS WHERE REQUIRED FOR COMMUNICATIONS EQUIPMENT. ALL EXPOSED EDGES OF BACKBOARD SHALL BE PAINTED ON FIVE SIDES WITH A FIRE RETARDANT PAINT. SEE ELECTRICAL/TELECOM DRAWINGS FOR ADDITIONAL INFORMATION
- I. FINISH FLOOR ELEVATION IS 0.00mm UNLESS OTHERWISE NOTED.



APPROVED: Digitally signed by signature
Date: 2026.01.28
FOR COMMANDER NAVFAC: 165555-0500

ACTIVITY
D. EVANS VIA EMAIL

SATISFACTORY TO: DATE: 01/15/2026

DES: PAK DRW: PAK CHK: JHR

PM/DM: MJD/ATP

BRANCH MANAGER: KMC

DES PROD DIR: WBF

FIRE PROTECTION ENGINEER: DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PLANNING, DESIGN AND CONSTRUCTION
NAVFAC
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
FLOOR PLAN

SCALE: AS NOTED
EPROVECT NO.: 1838720
CONSTR. CONTR. NO.

NAVFAC DRAWING NO.: 14157120
SHEET 58 OF 170
A-105

GRAPHIC SCALE

